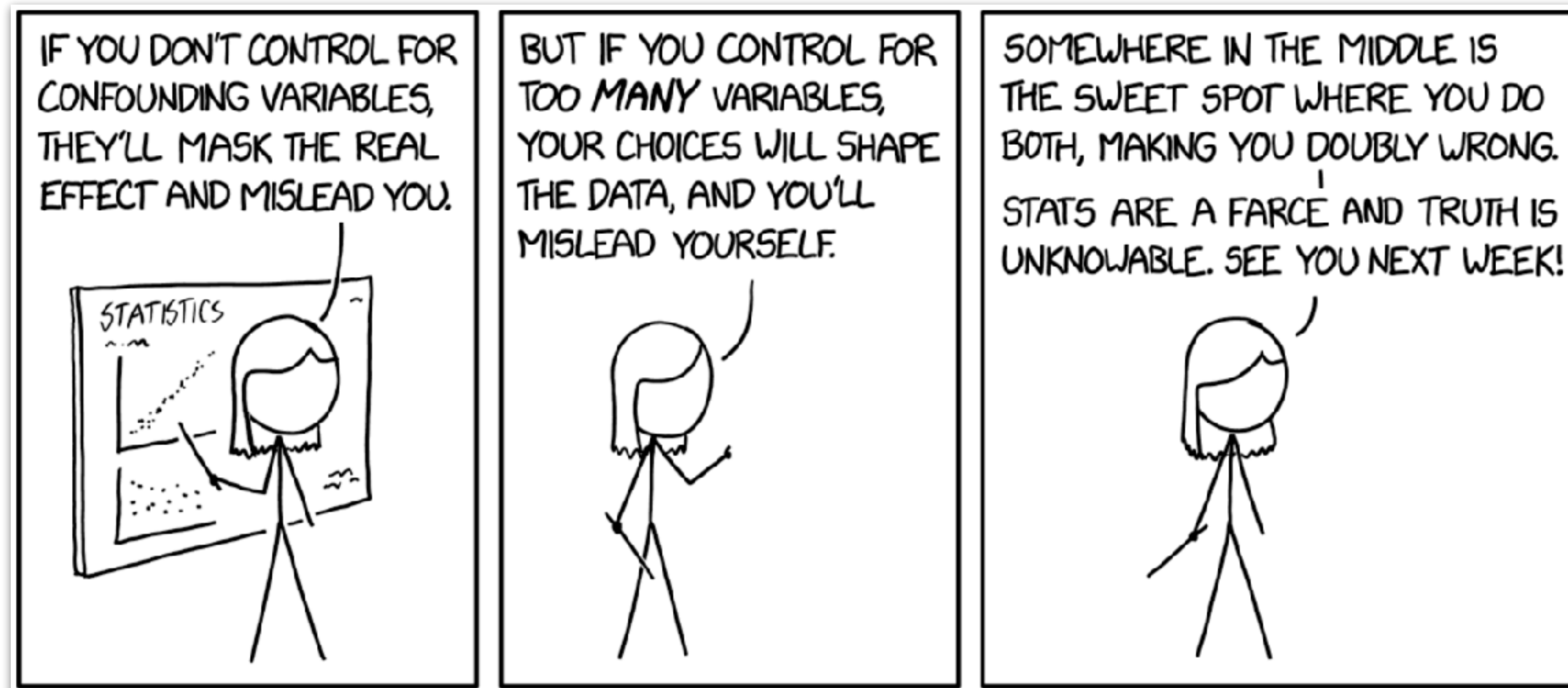


Causation



02/25/2026

Plan for today

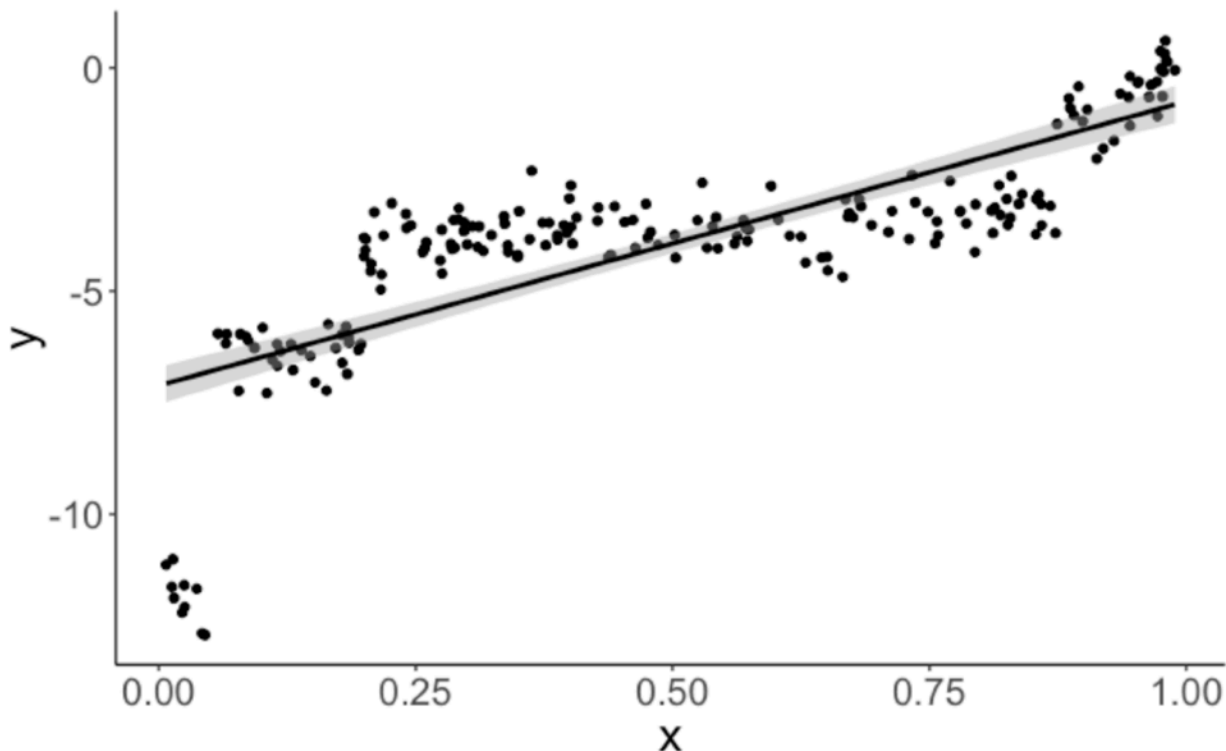
- Quick recap
- Causation
 - Causation vs. correlation
 - Controlling for variables
 - Mediation
 - Moderation

Quick recap

Quick recap: Simpson's paradox

Simpson's paradox

```
1 lm(formula = y ~ 1 + x,  
2   data = df.simpson) %>%  
3   summary()
```



```
Call:  
lm(formula = y ~ x, data = df.simpson)  
  
Residuals:  
    Min       1Q   Median       3Q      Max   
-5.8731 -0.6362  0.2272  1.0051  2.6410  
  
Coefficients:  
            Estimate Std. Error t value Pr(>|t|)      
(Intercept)  -7.1151     0.2107  -33.76  <2e-16 ***  
x              6.3671     0.3631   17.54  <2e-16 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 1.55 on 198 degrees of freedom  
Multiple R-squared:  0.6083,    Adjusted R-squared:  0.6064   
F-statistic: 307.5 on 1 and 198 DF,  p-value: < 2.2e-16
```

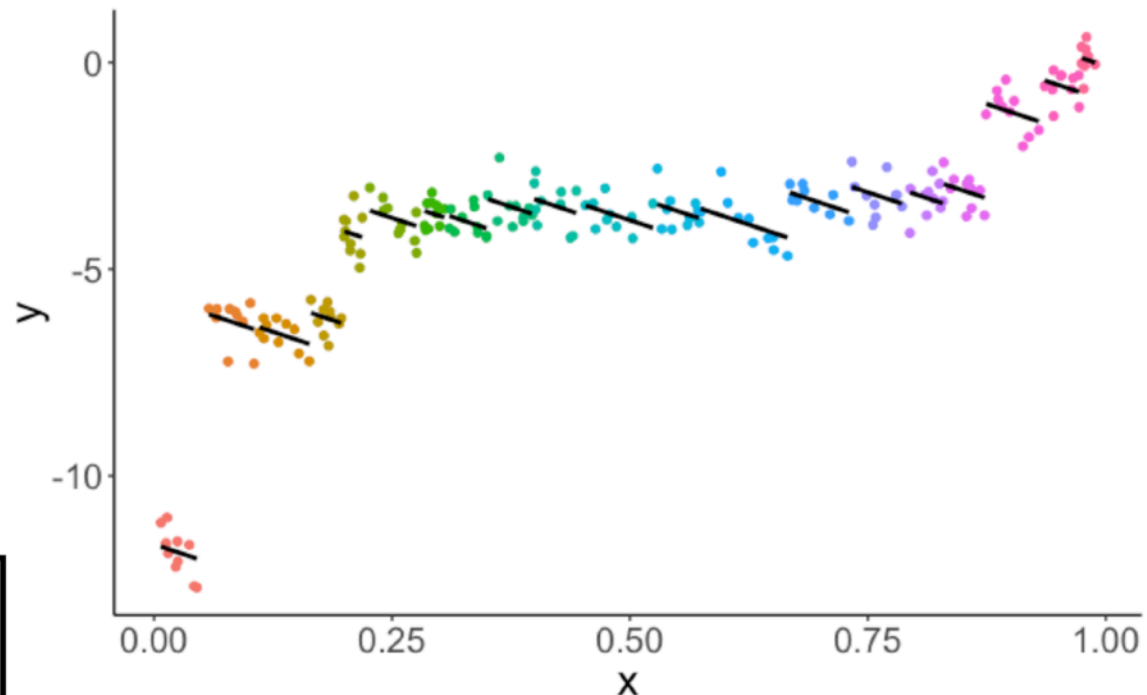
positive relationship
between x and y

7

Simpson's paradox

```
1 lmer(formula = y ~ 1 + x + (1 | participant),  
2   data = df.simpson) %>%  
3   summary()
```

```
Linear mixed model fit by REML ['lmerMod']  
Formula: y ~ 1 + x + (1 | participant)  
Data: df.simpson  
  
REML criterion at convergence: 345.1  
  
Scaled residuals:  
    Min       1Q   Median       3Q      Max   
-2.43394 -0.59687  0.04493  0.62694  2.68828  
  
Random effects:  
Groups      Name      Variance Std.Dev.  
participant (Intercept) 21.4898  4.6357  
Residual          0.1661  0.4075  
Number of obs: 200, groups: participant, 20  
  
Fixed effects:  
            Estimate Std. Error t value      
(Intercept)  -0.1577     1.3230  -0.119  
x             -7.6678     1.6572  -4.627  
---  
Correlation of Fixed Effects:  
 (Intr)  
x -0.621
```



negative (!)
relationship between
x and y

(once we take into
account individual
differences)

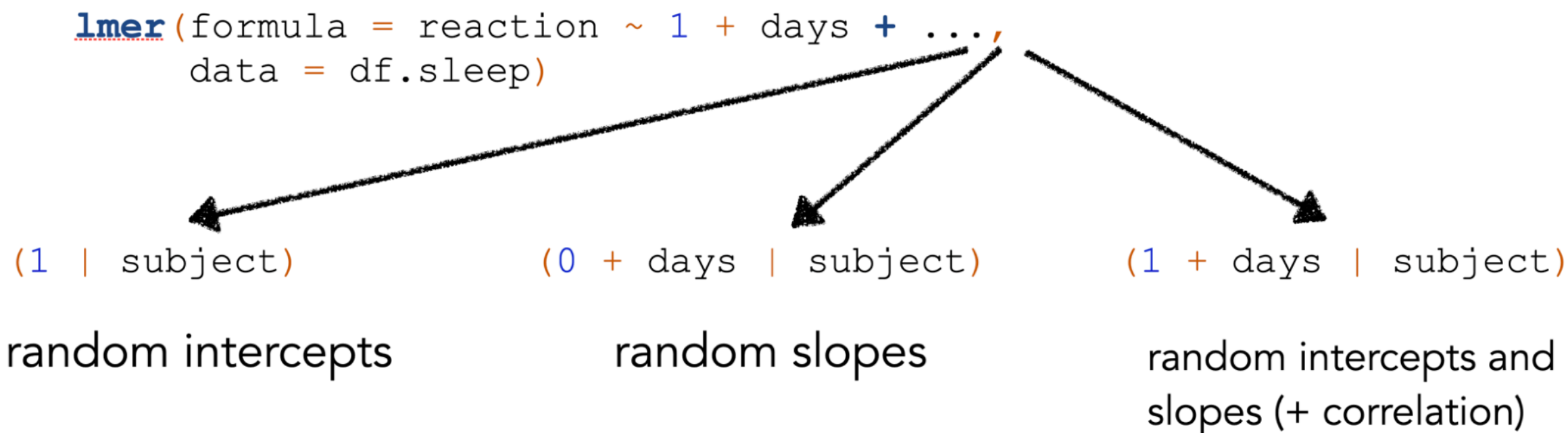
9

Quick recap: Understanding lmer () syntax

lmer () syntax summary

formula	description
<code>dv ~ 1 + x1 + (1 g)</code>	Random intercept for each level of `g`
<code>dv ~ 1 + x1 + (0 + x1 g)</code>	Random slope for each level of `g`
<code>dv ~ 1 + x1 + (1 + x1 g)</code>	Correlated random slope and intercept for each level of `g`
<code>dv ~ 1 + x1 + (1 + x1 g)</code>	Uncorrelated random slope and intercept for each level of `g`
<code>dv ~ 1 + x1 + (1 part) + (1 item)</code>	Random intercept for each level of `participant` and for each level of `item` (crossed)
<code>dv ~ 1 + x1 + (1 school) + (1 school:class)</code>	Random intercept for each level of `school` and for each level of `class` in `school` (nested)

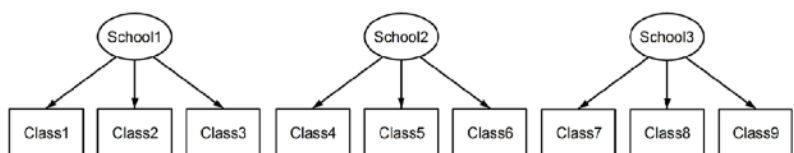
Coefficients



Multi-level models

nested

`(1|School) + (1|School:Class)`



random intercepts
of class within
each school

random intercepts of
schools

random intercepts

\$`class:school`				
(Intercept)				
a:I	53.77106	0.006106514	-0.007665927	0.0005404069
a:II	58.23842	0.006106514	-0.007665927	0.0005404069
a:III	58.71819	0.006106514	-0.007665927	0.0005404069
a:IV	58.68813	0.006106514	-0.007665927	0.0005404069
a:V	58.68268	0.006106514	-0.007665927	0.0005404069
a:VI	56.23088	0.006106514	-0.007665927	0.0005404069
b:I	59.54852	0.006106514	-0.007665927	0.0005404069
b:II	59.62643	0.006106514	-0.007665927	0.0005404069
b:III	59.70219	0.006106514	-0.007665927	0.0005404069
b:IV	59.73276	0.006106514	-0.007665927	0.0005404069
b:V	59.87036	0.006106514	-0.007665927	0.0005404069
b:VI	58.14865	0.006106514	-0.007665927	0.0005404069
c:I	62.28460	0.006106514	-0.007665927	0.0005404069
c:II	60.74743	0.006106514	-0.007665927	0.0005404069
c:III	60.70970	0.006106514	-0.007665927	0.0005404069
c:IV	60.86062	0.006106514	-0.007665927	0.0005404069
c:V	60.80225	0.006106514	-0.007665927	0.0005404069
c:VI	61.10164	0.006106514	-0.007665927	0.0005404069
d:I	64.14113	0.006106514	-0.007665927	0.0005404069
d:II	61.81189	0.006106514	-0.007665927	0.0005404069
d:III	61.65165	0.006106514	-0.007665927	0.0005404069
d:IV	61.83703	0.006106514	-0.007665927	0.0005404069
d:V	62.13593	0.006106514	-0.007665927	0.0005404069
d:VI	66.66561	0.006106514	-0.007665927	0.0005404069
\$school				
(Intercept)				
I	46.44407	0.006106514	-0.007665927	0.0005404069
II	54.20862	0.006106514	-0.007665927	0.0005404069
III	58.29847	0.006106514	-0.007665927	0.0005404069
IV	62.15074	0.006106514	-0.007665927	0.0005404069
V	66.41348	0.006106514	-0.007665927	0.0005404069
VI	73.91156	0.006106514	-0.007665927	0.0005404069

model coefficients
`fit.nested %>% coef()`

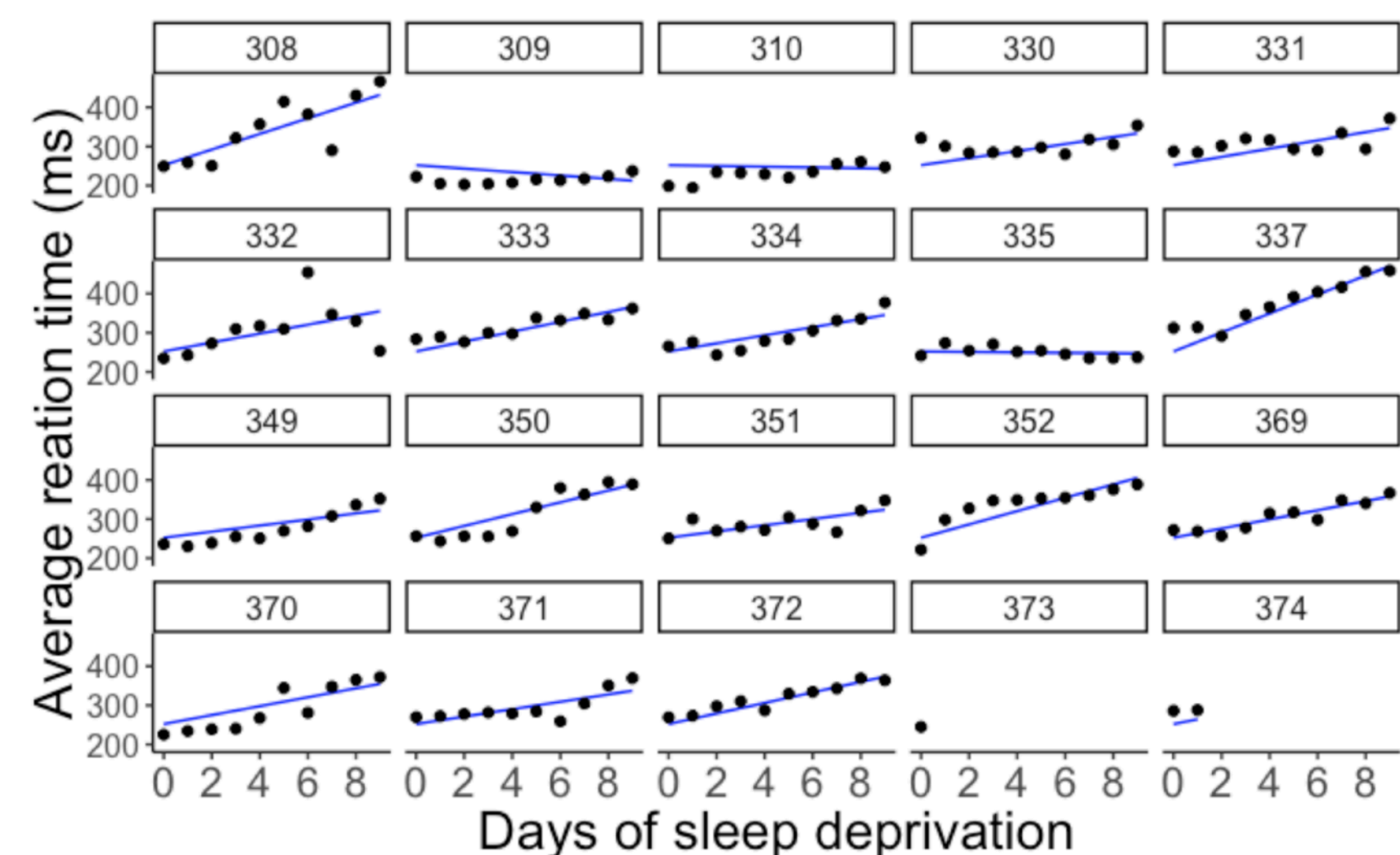
\$subject		
(Intercept)		
308	292.2749	10.43191
309	174.0559	10.43191
310	188.7454	10.43191
330	256.0247	10.43191
331	261.8141	10.43191
332	259.8262	10.43191
333	268.0765	10.43191
334	248.6471	10.43191
335	206.5096	10.43191
337	323.5643	10.43191
349	230.5114	10.43191
350	265.6957	10.43191
351	243.7988	10.43191
352	287.8850	10.43191
369	258.6454	10.43191
370	245.2931	10.43191
371	248.3508	10.43191
372	269.6861	10.43191
373	248.2086	10.43191
374	273.9400	10.43191

\$subject		
(Intercept)		
308	252.2965	19.9526801
309	252.2965	-4.3719650
310	252.2965	-0.9574726
330	252.2965	8.9909957
331	252.2965	10.5394285
332	252.2965	11.3994289
333	252.2965	12.6074020
334	252.2965	10.3413879
335	252.2965	-0.5722073
337	252.2965	24.2246485
349	252.2965	7.7702676
350	252.2965	15.0661415
351	252.2965	7.9675415
352	252.2965	17.0002999
369	252.2965	11.6982767
370	252.2965	11.3939807
371	252.2965	9.4535879
372	252.2965	13.4569059
373	252.2965	10.4142695
374	252.2965	11.9097917

\$subject		
(Intercept)		
308	253.9479	19.6264139
309	211.7328	1.7319567
310	213.1579	4.9061843
330	275.1425	5.6435987
331	273.7286	7.3862680
332	260.6504	10.1632535
333	268.3684	10.2245979
334	244.5523	11.4837825
335	251.3700	-0.3355554
337	286.2321	19.1090061
349	226.7662	11.5531963
350	238.7807	17.0156766
351	256.2344	7.4119501
352	272.3512	13.9920698
369	254.9484	11.2985741
370	226.3701	15.2027922
371	252.5051	9.4335432
372	263.8916	11.7253342
373	248.9752	10.3915245
374	271.1451	11.0782697

Quick recap: Reporting results

Visualization



show the data together with the model predictions

some text ...

Reporting results

Including Random Effects
in the Table

Table 2
Results From Final Models Examining How Familiarity of the Social Context Influences Momentary Affect Valence and Affect Arousal, and the Moderating Role of Age Therein

Parameters	Parameter estimates (SE)	
	Affect valence	Affect arousal
Fixed effects		
Intercept, γ_{00}	77.5231* (.836)	62.8809* (.953)
Familiarity ecology, γ_{01}	.2571* (.077)	-.1005 (.092)
Familiarity situation, γ_{10}	.0805* (.007)	-.0199* (.008)
Familiarity situation ² , γ_{20}	.0004* (.0001)	-.0007* (.0001)
Age, γ_{02}	-.5752* (.234)	.2413 (.265)
Age ² , γ_{03}	.0064* (.002)	-.0014 (.003)
Age $\times$ familiarity ecology, γ_{04}	.0027 (.004)	.0033 (.005)
Age $\times$ familiarity situation, γ_{11}	.0003 (.0004)	.0005 (.0004)
Health, γ_{05}	.5364* (.2021)	.3569 (.230)
Familiarity ecology $\times$ familiarity situation, γ_{12}	—	-.0018* (.001)
Valence trait, γ_{06}	—	.6008* (.094)
Valence momentary, γ_{13}	—	.2797* (.006)
Familiarity ecology $\times$ valence momentary, γ_{14}	—	-.0044* (.0005)
Familiarity situation $\times$ valence trait, γ_{15}	—	.0012 (.001)
Familiarity situation $\times$ valence momentary, γ_{16}	—	-.0002 (.000)
Familiarity situation ² $\times$ valence momentary, γ_{17}	—	-.0000* (.0000)
Age $\times$ Familiarity ecology $\times$ valence momentary, γ_{18}	—	-.0003* (.000)
Age $\times$ Familiarity situation $\times$ valence momentary, γ_{19}	—	-.00002* (.000)
Random effects		
Variance intercept, σ^2_{u0}	103.50* (12.30)	134.02* (16.01)
Variance familiarity situation, σ^2_{u1}	.007* (.001)	.008* (.001)
Covariance intercept, familiarity situation, $\sigma_{u0, u1}$	-.096 (.074)	.29* (.097)
Autocorrelation, ar(1)	.21* (.004)	.29* (.004)
Residual variance, σ^2_e	190.16* (1.117)	272.08* (1.671)
-2LL	514,339	534,375

Note. SE = standard error. Data are reports on 64,213 social interactions nested within 150 persons. Bold lines indicate tests for the main hypotheses of interest.
* $p < .05$.

Quick recap: Simulation

Let's simulate an `lmer()`

```
1 # make example reproducible
2 set.seed(1)
3
4 # parameters
5 sample_size = 100
6 b0 = 1
7 b1 = 2
8 sd_residual = 1
9 sd_participant_intercept = 0.5
10
11 # generate the data
12 df.mixed = tibble(participant = rep(1:sample_size, 2),
13                   condition = rep(0:1, each = sample_size)) %>%
14   group_by(participant) %>%
15   mutate(participant_intercepts = rnorm(n = 1, sd = sd_participant_intercept)) %>%
16   ungroup() %>%
17   mutate(value = b0 + b1 * condition + participant_intercept + rnorm(n(), sd = sd_residual)) %>%
18   arrange(participant, condition)
```

participant	condition	participant_intercept	value
1	0	-0.31	0.07
1	1	-0.31	3.10
2	0	0.09	1.13
2	1	0.09	4.78
3	0	-0.42	-0.33
3	1	-0.42	4.17
4	0	0.80	1.96
4	1	0.80	3.47
5	0	0.16	0.51
5	1	0.16	0.88

$$\text{value}_{ij} = b_0 + b_1 \cdot \text{condition}_{ij} + U_i + e_{ij}$$

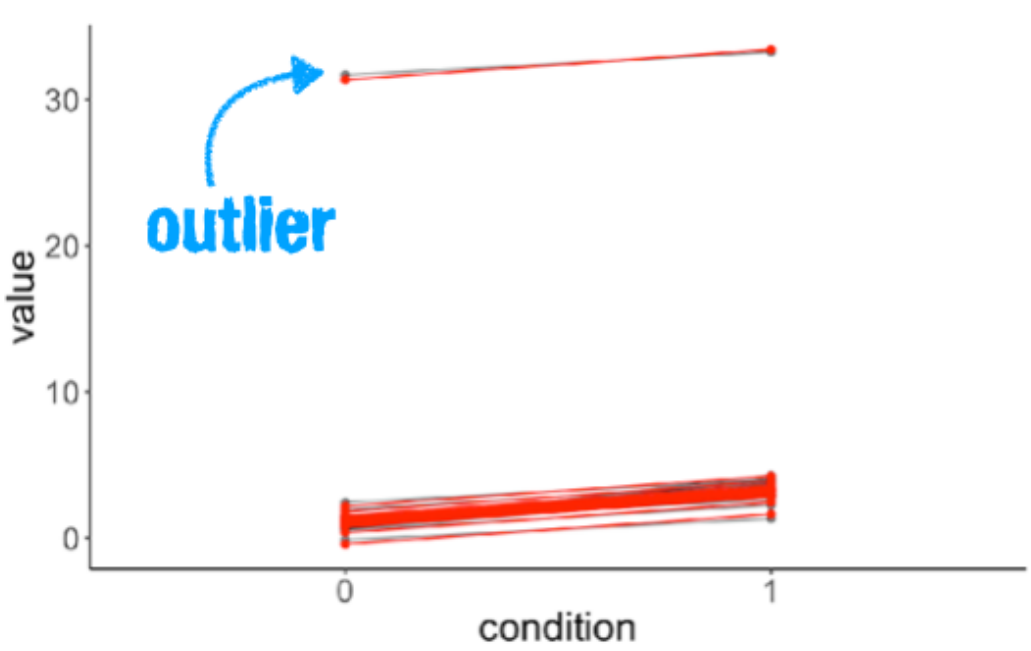
$$e_{ij} \sim \mathcal{N}(\text{mean} = 0, \text{sd} = s_{\text{error}})$$

$$U_i \sim \mathcal{N}(\text{mean} = 0, \text{sd} = s_U)$$

simulating data from a model and trying to recover the parameters is a great way to check one's understanding of what the model does

15

With outlier



```
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
Data: df.test

REML criterion at convergence: 171.7

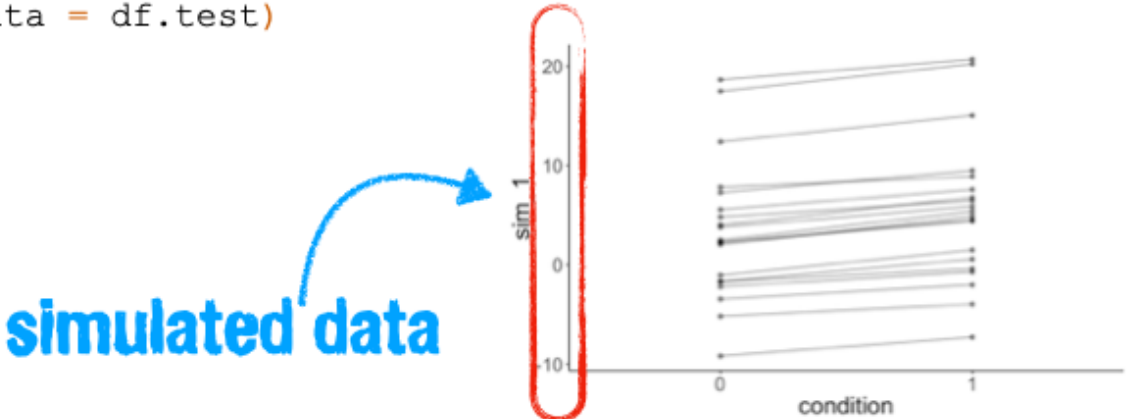
Scaled residuals:
   Min       1Q   Median       3Q      Max
-1.4038 -0.4678 -0.0094  0.5800  1.3930

Random effects:
 Groups      Name      Variance Std.Dev.
participant (Intercept) 46.198   6.7969
Residual      0.227    0.4764
Number of obs: 40, groups: participant, 20

Fixed effects:
              Estimate Std. Error t value
(Intercept)  2.5920    1.5236   1.701
condition1   2.0726    0.1507  13.758

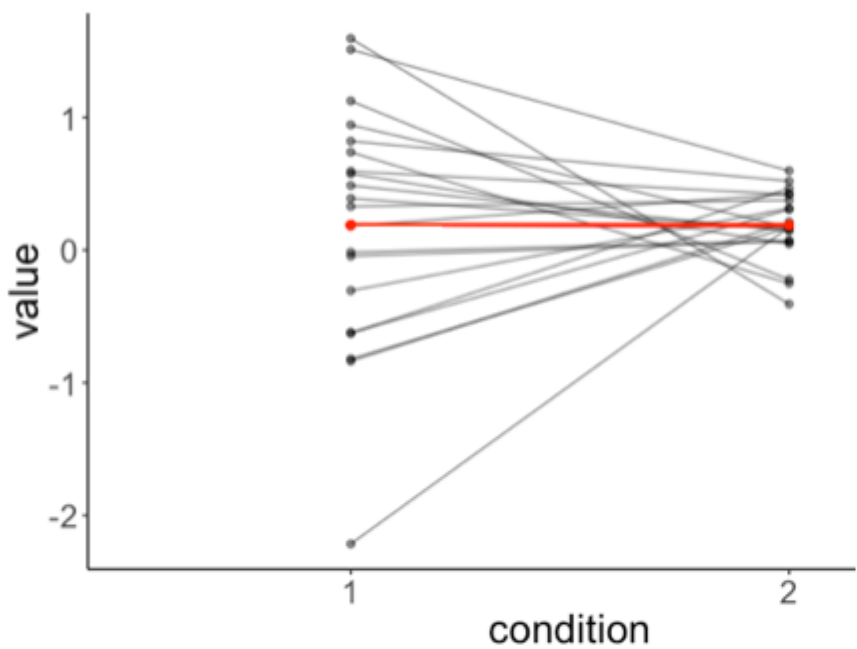
Correlation of Fixed Effects:
          (Intr)
condition1 -0.049
```

```
1 # fit model
2 fit.test = lmer(formula = value ~ 1 + condition + (1 | participant),
3                 data = df.test)
4
5 # simulate data
6 fit.test %>%
7   simulate()
```



18

Non-equal variance



```
singular fit
Linear mixed model fit by REML ['lmerMod']
Formula: value ~ 1 + condition + (1 | participant)
Data: df.test

REML criterion at convergence: 83.6

Scaled residuals:
   Min       1Q   Median       3Q      Max
-3.5808 -0.3184  0.0130  0.4551  2.0913

Random effects:
 Groups      Name      Variance Std.Dev.
participant (Intercept) 0.0000   0.0000
Residual      0.4512   0.6717
Number of obs: 40, groups: participant, 20

Fixed effects:
              Estimate Std. Error t value
(Intercept)  0.190524    0.150197   1.268
condition2  -0.001941    0.212411  -0.009

Correlation of Fixed Effects:
          (Intr)
condition2 -0.707
convergence code: 0
singular fit
```

clearly there are interindividual differences though?!

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Quick recap: Worked examples

Sleep data

```
1 fit = lmer(formula = reaction ~ 1 + days + (1 | subject),
2           data = df.sleep %>%
3             mutate(days = as.factor(days)))
4
5 contrast = list(first_vs_second = c(-1, 1, rep(0, 8)),
6               early_vs_late = c(rep(-1, 5)/5, rep(1, 5)/5))
7
8 fit %>%
9   emmeans(specs = "days",
10          contr = contrast) %>%
11   pluck("contrasts")
```

fit the model

define the contrasts

test the contrasts

contrast	estimate	SE	df	t.ratio	p.value
first_vs_second	7.82	10.10	156	0.775	0.4398
early_vs_late	53.66	4.65	155	11.534	<.0001

days	reaction
0	257.54
1	265.73

index	reaction
early	271.67
late	325.39

Degrees-of-freedom method: kenward-roger

<https://aosmith.rbind.io/2019/03/25/getting-started-with-emmeans/>

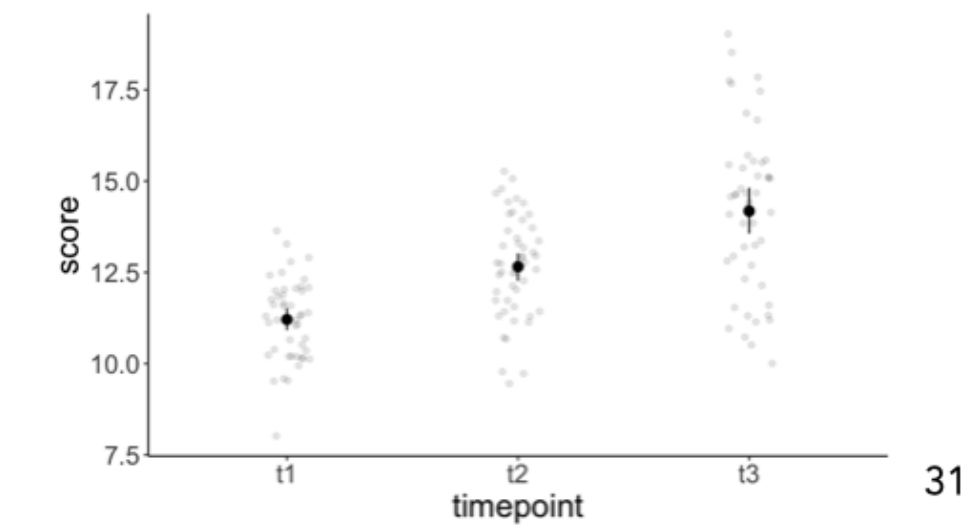
Weight loss data

1. Is the score at the third time point different from the other two time points?
2. Is there a linear increase across time points?

```
1 fit = aov_ez(id = "id",
2             dv = "score",
3             between = "exercises",
4             within = c("diet", "timepoint"),
5             data = df.weightloss)
6
7 contrasts = list(first_two_vs_last = c(-0.5, -0.5, 1),
8               linear_increase = c(-1, 0, 1))
9
10 fit %>%
11   emmeans(spec = "timepoint",
12          contr = contrasts)
```

df.weightloss

id	diet	exercises	timepoint	score
1	no	no	t1	10.43
1	no	no	t2	13.21
1	no	no	t3	11.59
1	yes	no	t1	10.20
1	yes	no	t2	12.51
1	yes	no	t3	14.60
2	no	no	t1	11.59

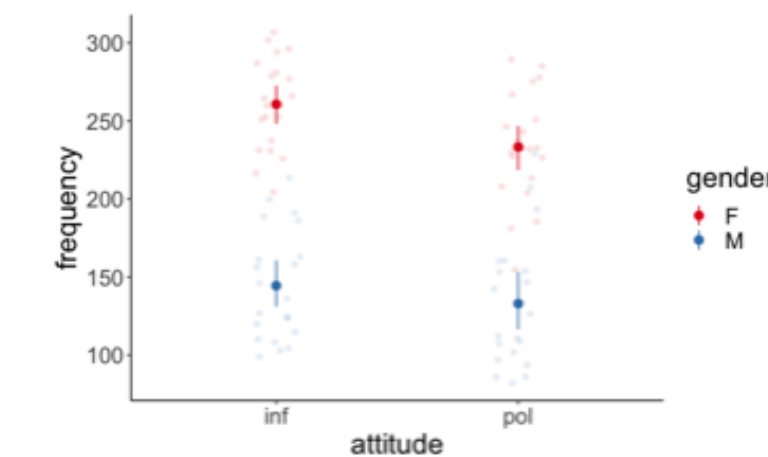


Politeness

Was there an effect of gender and attitude on pitch?

```
1 lmer(formula = frequency ~ 1 + attitude * gender + (1 + attitude | subject) + (1 | scenario),
2     data = df.politeness) %>%
3   joint_tests()
```

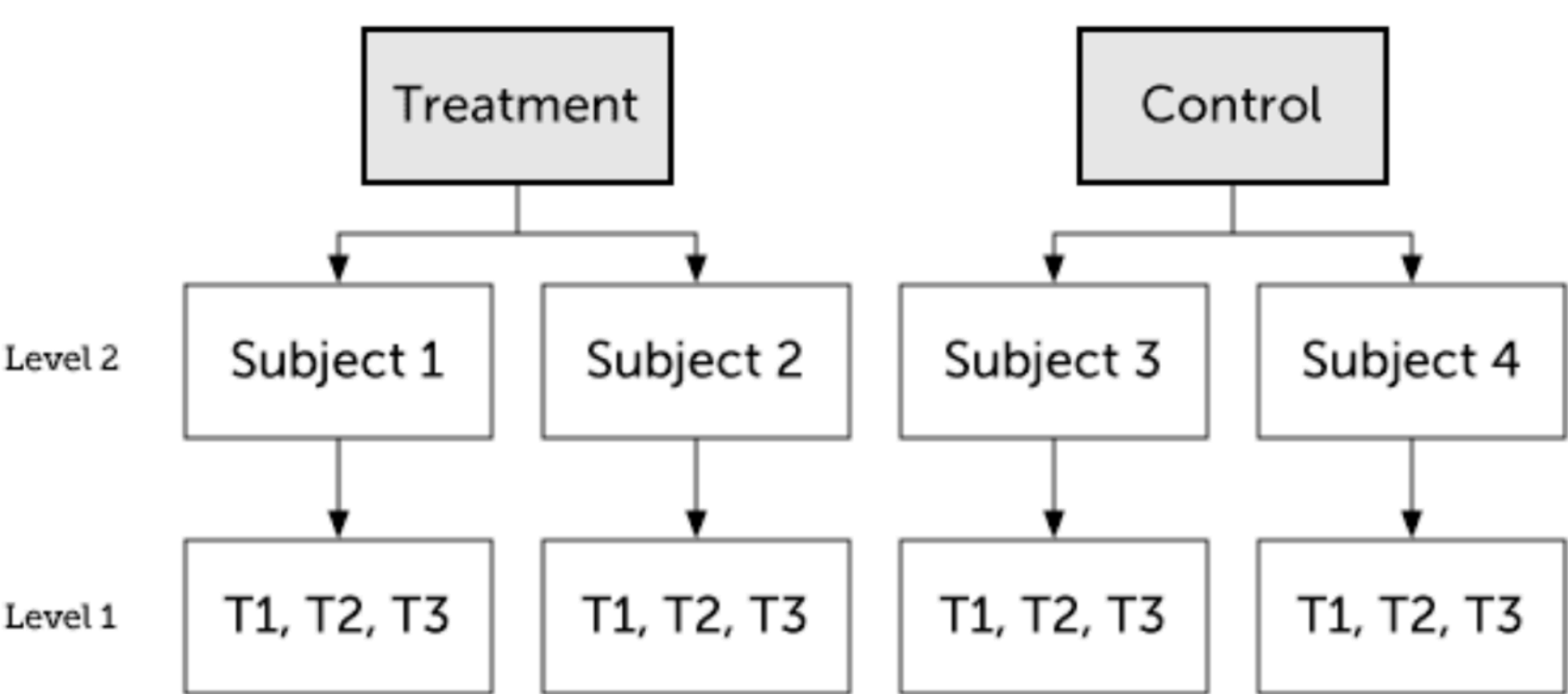
model term	df1	df2	F.ratio	p.value
attitude	1	3.99	12.411	0.0244
gender	1	4.00	26.570	0.0067
attitude:gender	1	3.99	1.959	0.2342



main effect of attitude, main effect of gender, no significant interaction effect

Quick recap: Growth models

Graphical representation



assume a linear effect of time

Simulate data

```
1 set.seed(1)
2
3 n_participants = 100
4 n_timepoints = 3
5 n_conditions = 2
6 p_condition = 0.5
7 b0 = 10 # intercept
8 b1 = 10 # condition
9 b2 = 2 # time
10 b3 = 3 # interaction
11 sd_intercept_participant = 2
12 sd_time_participant = 2
13 sd_residual = 1
14
15 df.data = tibble(participant = rep(1:n_participants, each = n_timepoints),
16                  timepoint = rep(1:n_timepoints, times = n_participants),
17                  intercept_participant = rep(rnorm(n_participants, sd = sd_intercept_participant),
18                  each = n_timepoints),
19                  time_participant = rep(rnorm(n_participants, sd = sd_time_participant),
20                  each = n_timepoints)) %>%
21 group_by(participant) %>%
22 mutate(condition = rbinom(n = 1, size = 1, prob = p_condition)) %>%
23 ungroup() %>%
24 mutate(value = b0 + intercept_participant +
25          b1 * condition +
26          (b2 + time_participant) * timepoint +
27          b3 * condition * timepoint +
28          rnorm(n_participants * n_timepoints, sd = sd_residual))
```

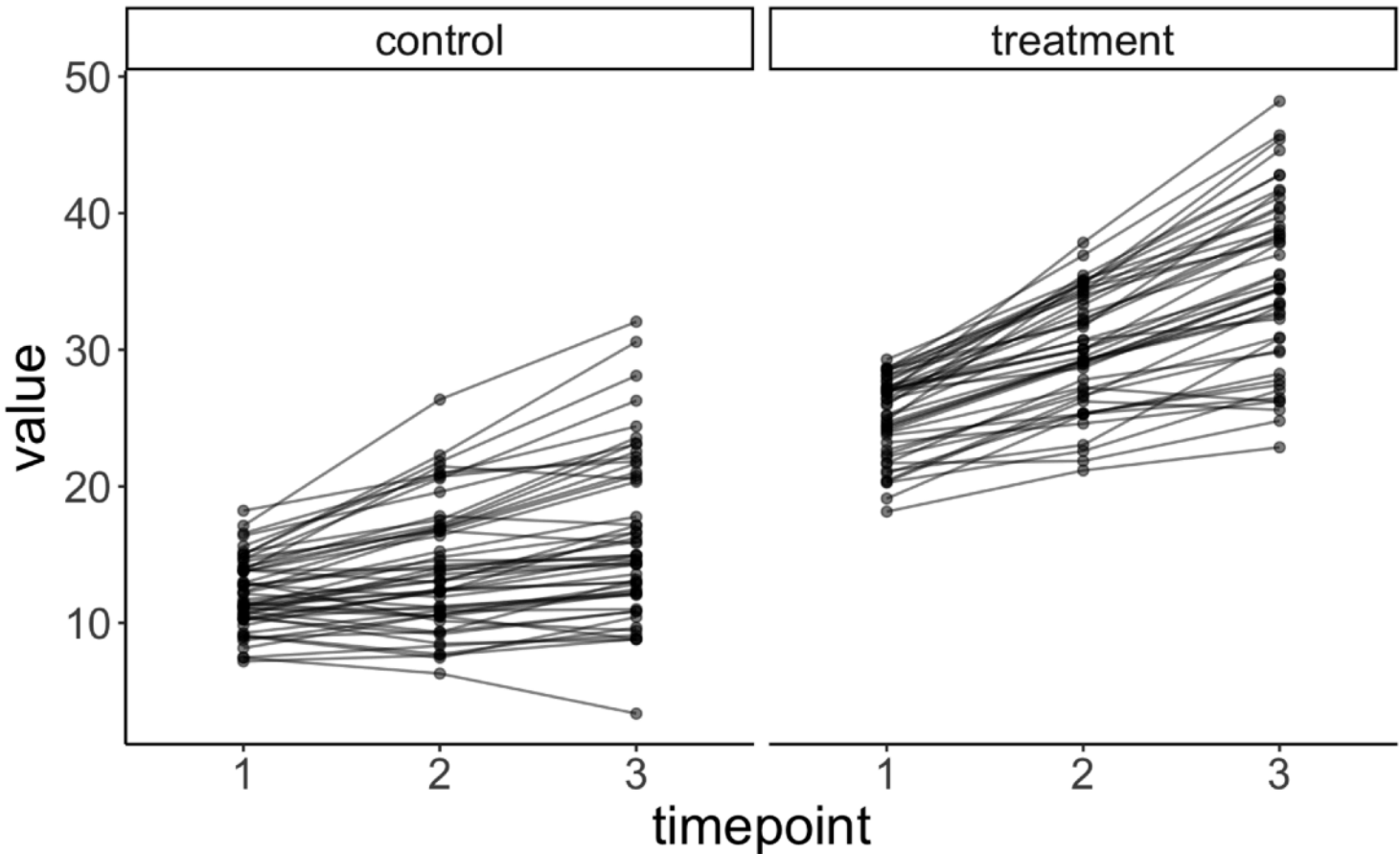
	participant	timepoint	intercept_participant	time_participant	condition	value
1	1	1	-1.252907621	-0.310183339	1	20.573131
2	1	2	-1.252907621	-0.310183339	1	22.844077
3	1	3	-1.252907621	-0.310183339	1	24.367254
4	2	1	0.367286648	0.021057937	0	12.119622
5	2	2	0.367286648	0.021057937	0	15.062837
6	2	3	0.367286648	0.021057937	0	17.513515
7	3	1	-1.671257225	-0.455460824	1	17.470186
8	3	2	-1.671257225	-0.455460824	1	22.446022
9	3	3	-1.671257225	-0.455460824	1	24.248006
10	4	1	3.190561604	0.079014386	1	24.844308
11	4	2	3.190561604	0.079014386	1	28.220589
12	4	3	3.190561604	0.079014386	1	28.880339
13	5	1	0.659015544	-0.327292322	1	22.047393
14	5	2	0.659015544	-0.327292322	1	25.189133
15	5	3	0.659015544	-0.327292322	1	28.051351

Fit the model

```
1 fit = lmer(formula = value ~ 1 + condition * timepoint + (1 + timepoint | participant),
2            data = df.data)
```

Linear mixed model fit by REML ['lmerMod']					
Formula: value ~ 1 + condition * timepoint + (1 + timepoint participant)					
Data: df.data					
REML criterion at convergence: 1360.3					
Scaled residuals:					
	Min	1Q	Median	3Q	Max
	-2.14633	-0.46360	0.03902	0.42302	2.82945
Random effects:					
Groups	Name	Variance	Std.Dev.	Corr	
participant	(Intercept)	3.190	1.786		
	timepoint	3.831	1.957	-0.06	
Residual		1.149	1.072		
Number of obs: 300, groups: participant, 100					
Fixed effects:					
	Estimate	Std. Error	t value		
(Intercept)	10.0101	0.3328	30.079		
condition	10.0684	0.4854	20.741		
timepoint	2.0595	0.2883	7.143		
condition:timepoint	2.9090	0.4205	6.917		
Correlation of Fixed Effects:					
	(Intr)	condtn	timptn		
condition	-0.686				
timepoint	-0.266	0.182			
condtn:timptn	0.182	-0.266	-0.686		

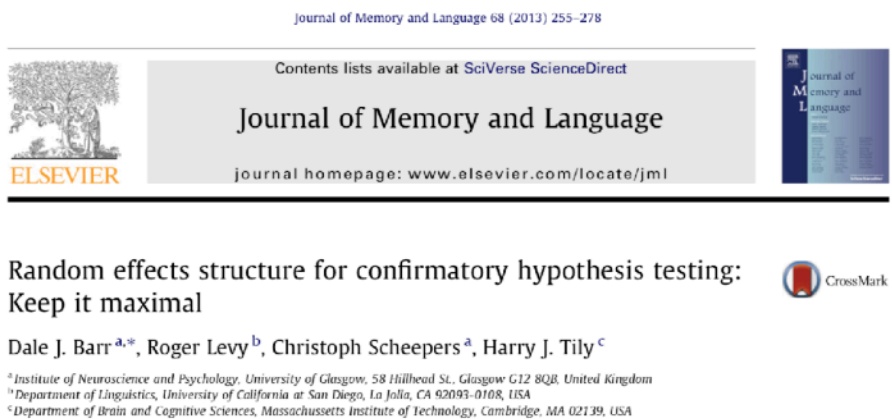
Plot data



Quick recap: standard operating procedures

What shall I include as random effects?

- mixed opinions on the topic
- go maximal!

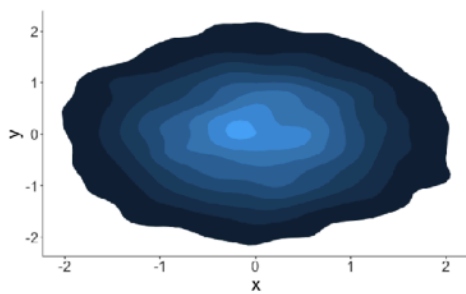


"Through theoretical arguments and Monte Carlo simulation, we show that LMEMs generalize best when they include the maximal random effects structure justified by the design. ...
Maximal LMEMs should be the 'gold standard' for confirmatory hypothesis testing in psycholinguistics and beyond."

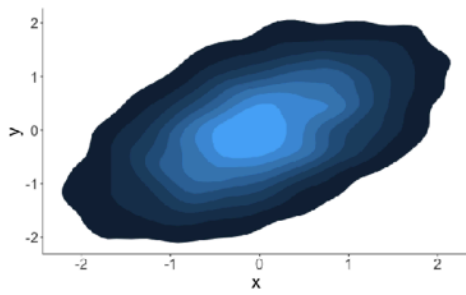
Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, 68(3), 255--278.

70

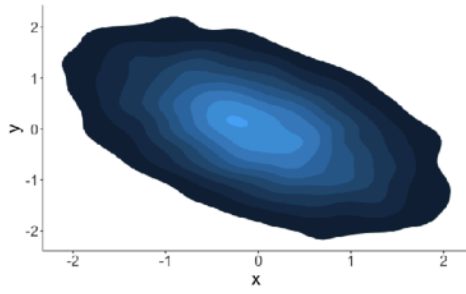
Remove the correlation component from your model



uncorrelated



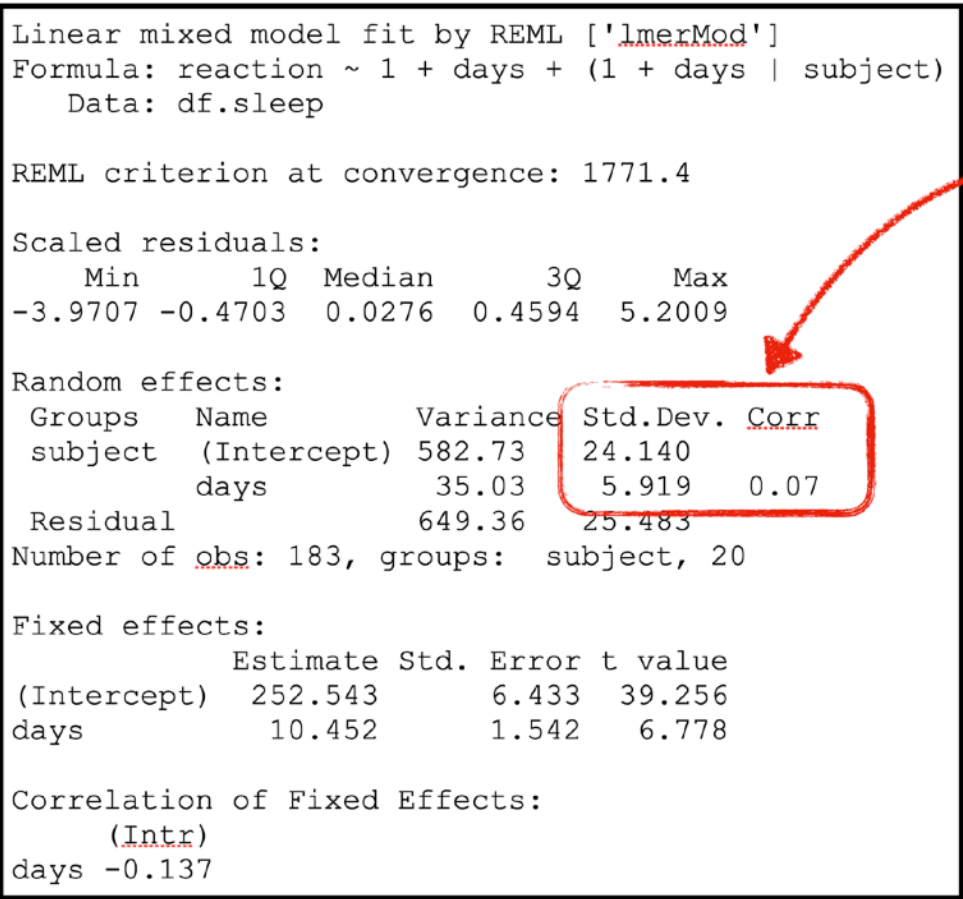
positively correlated



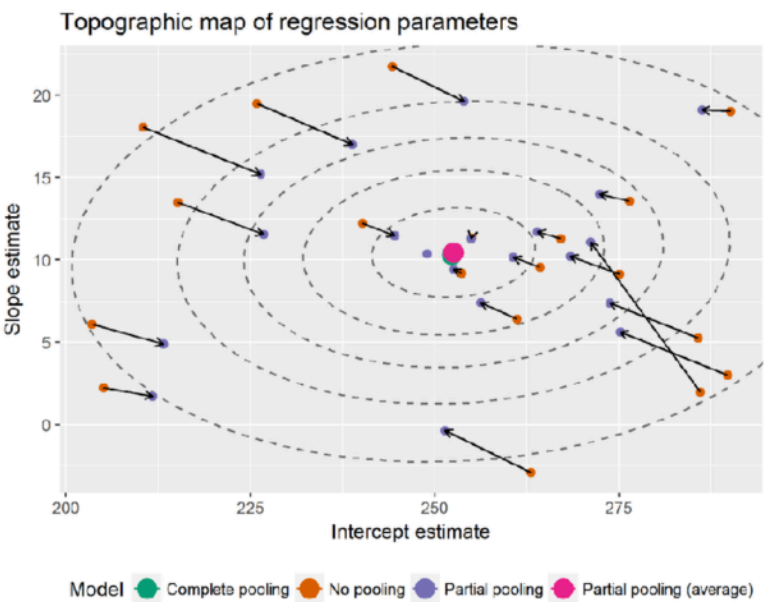
negatively correlated

Remove the correlation component from your model

```
1 # fit the model
2 fit_lmer = lmer(formula = reaction ~ 1 + days + (1 + days | subject),
3                 data = df.sleep)
4 # model summary
5 fit_lmer %>%
6   summary()
```



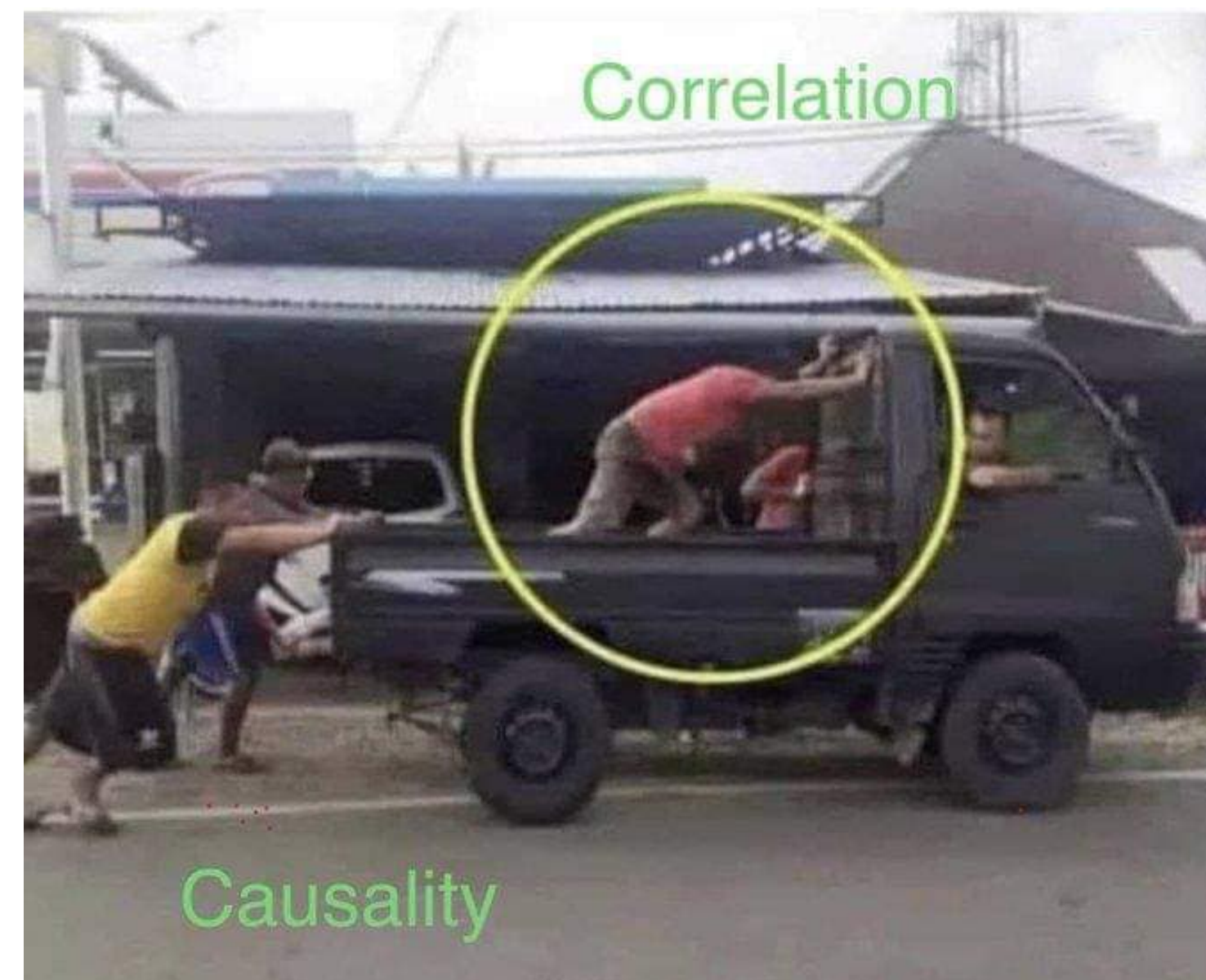
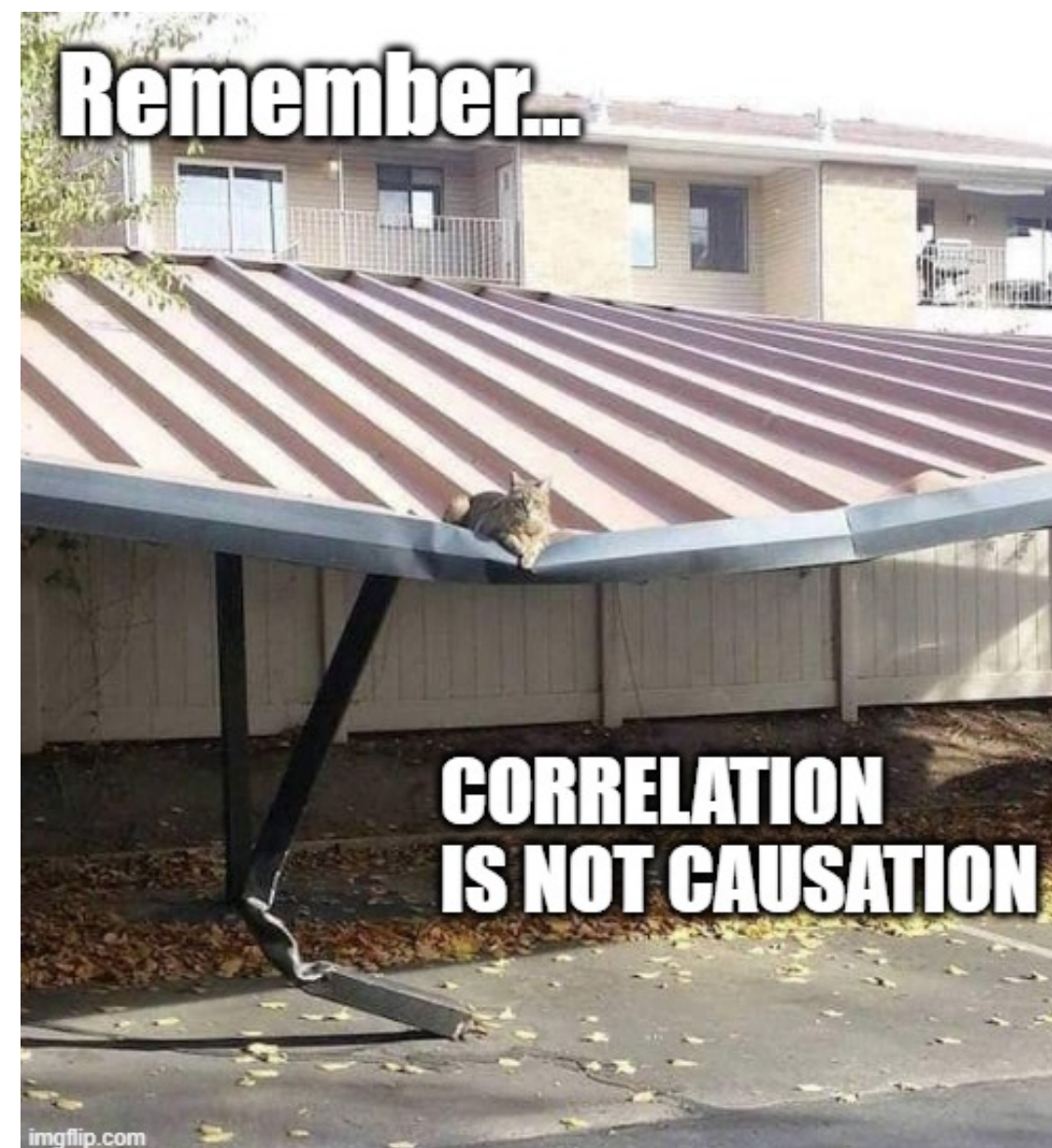
multivariate
Gaussian

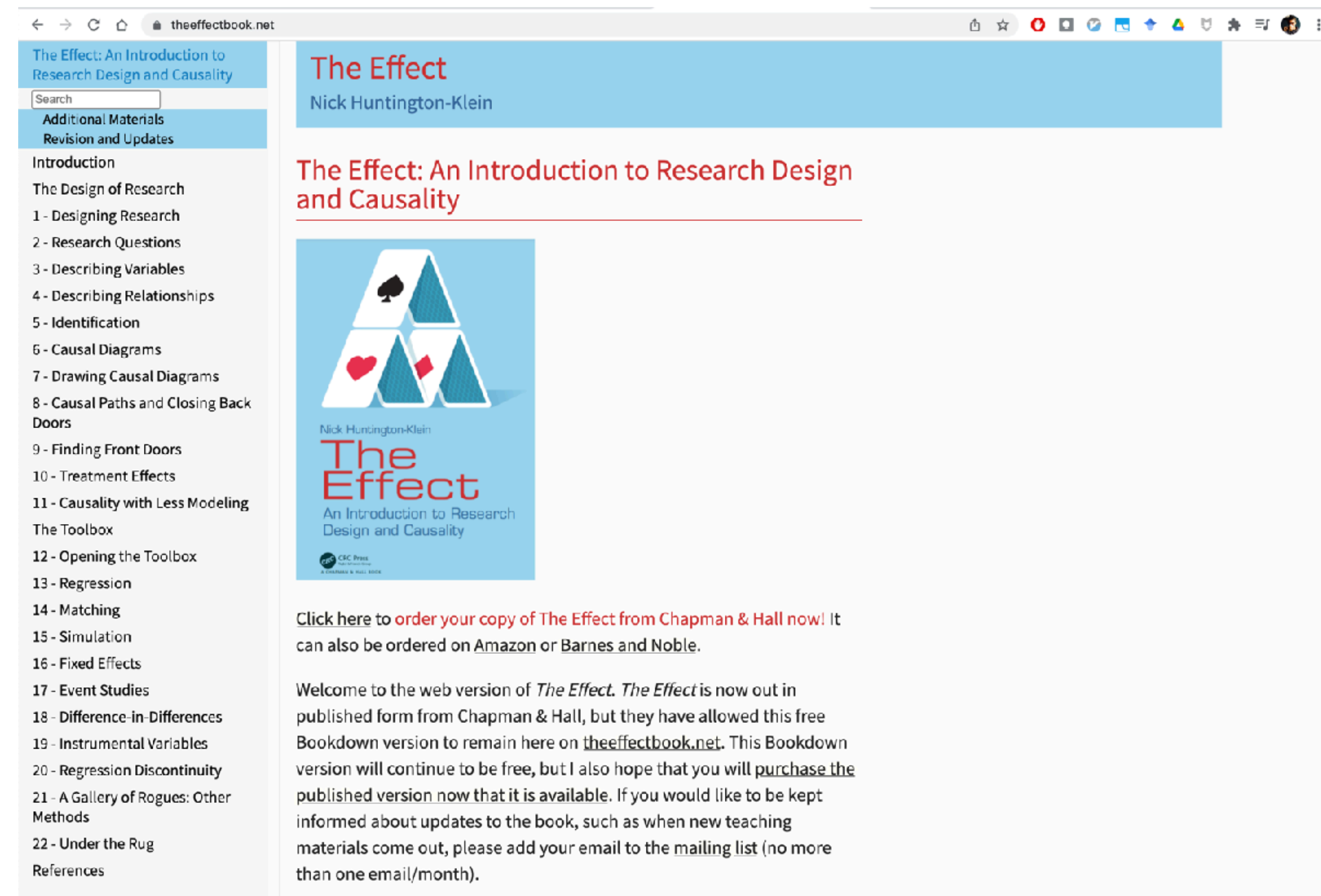
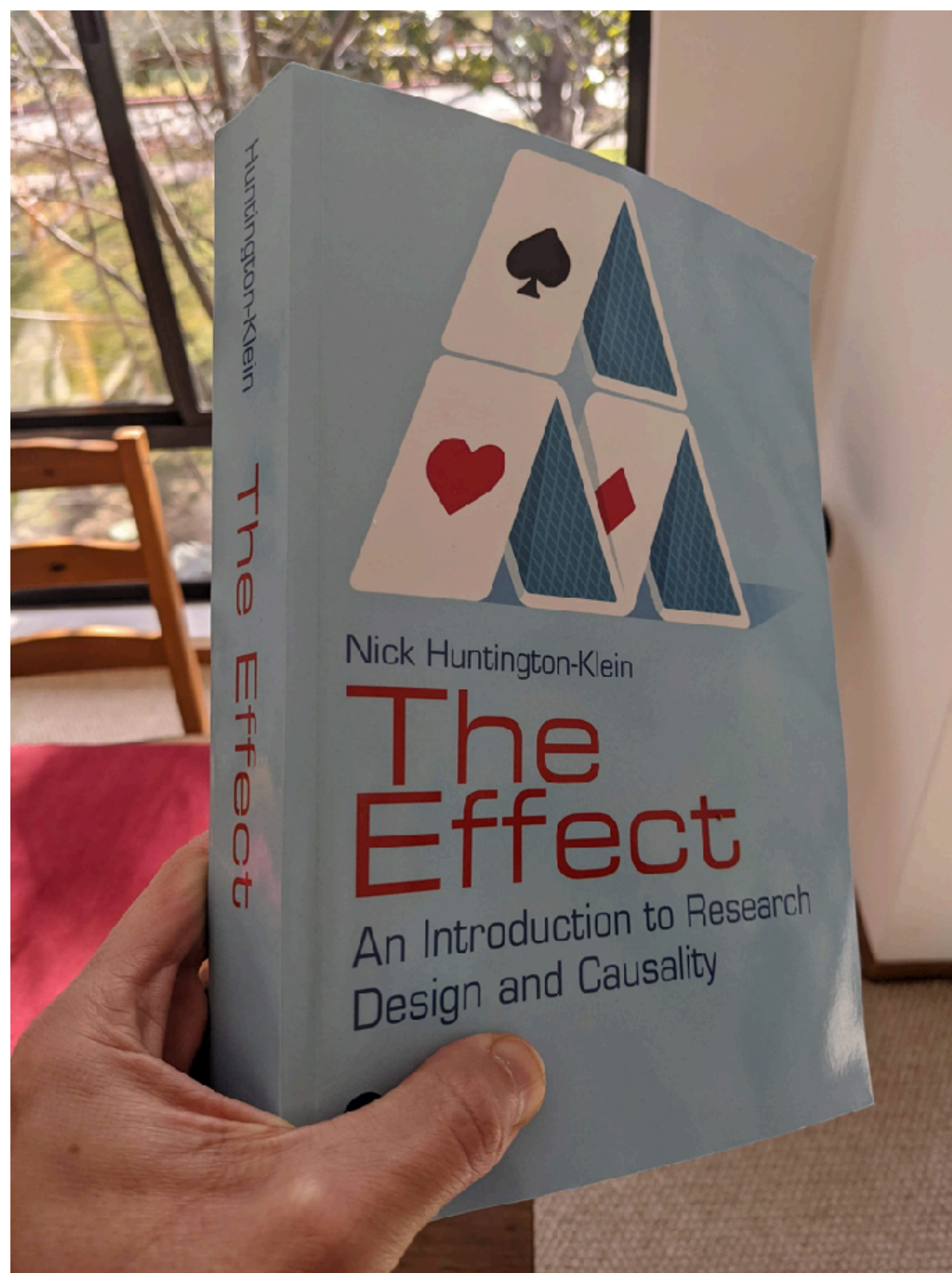


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Causation vs. correlation

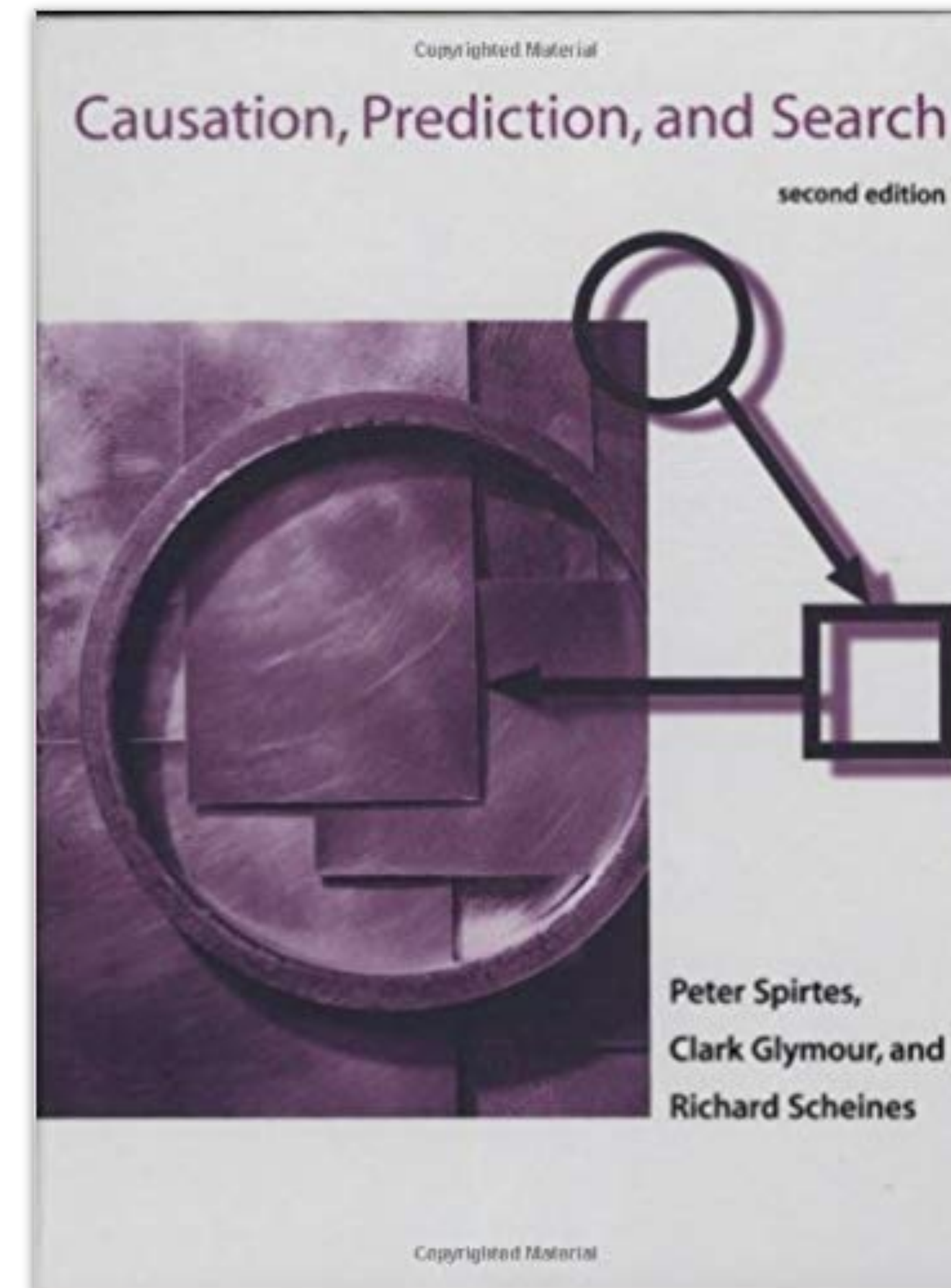
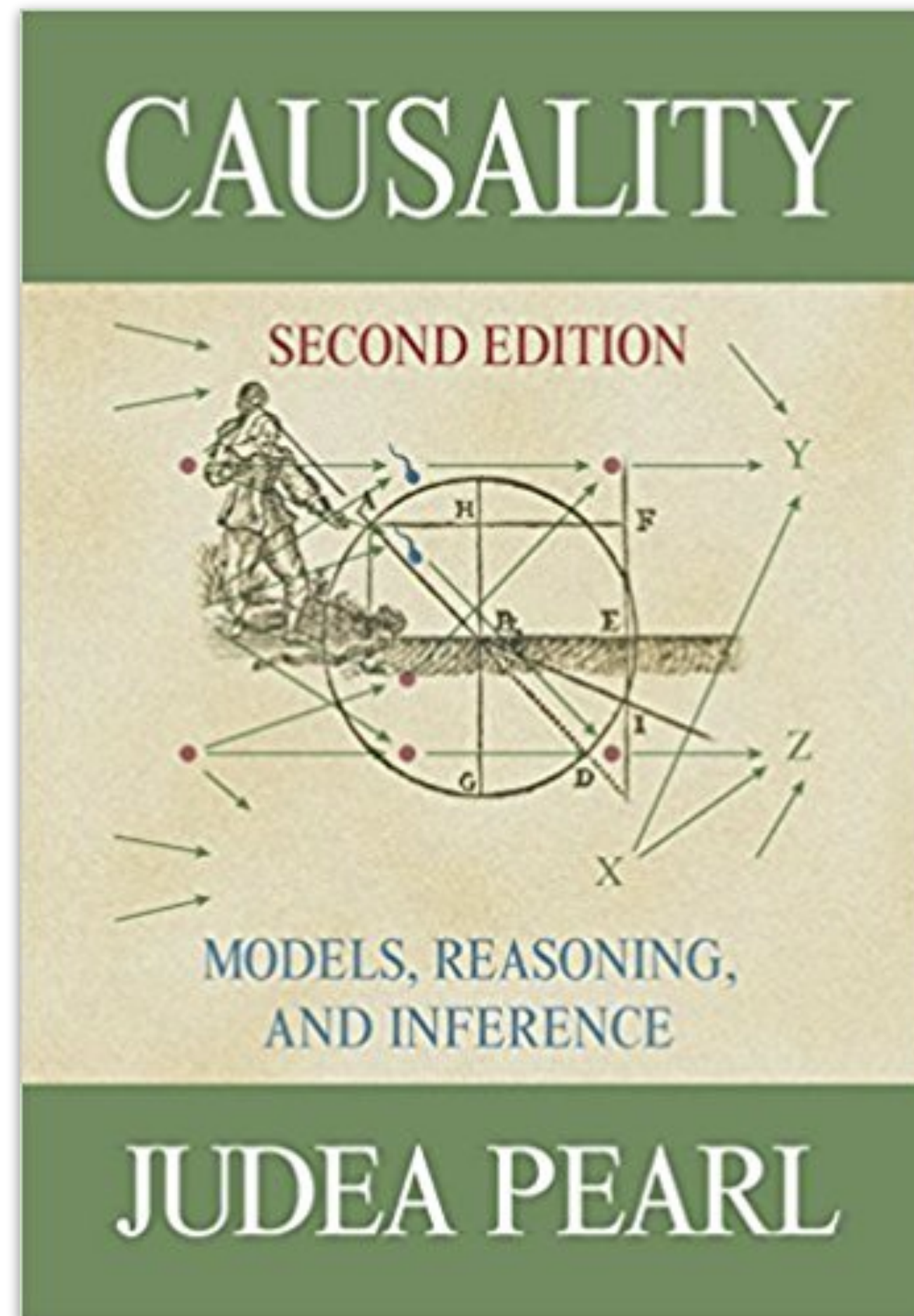
correlation does not imply causation





made with RMarkdown

<https://theeffectbook.net/>



Pearl, J. (2000). Causality: Models, reasoning and inference. Cambridge, England: Cambridge University Press.

Spirtes, P., Glymour, C. N., & Scheines, R. (2000). Causation, prediction, and search. The MIT Press.

What do you think?

Suppose there is a robust, statistically significant, and long-term correlation between the color of cars and the annual rate at which they are involved in accidents.

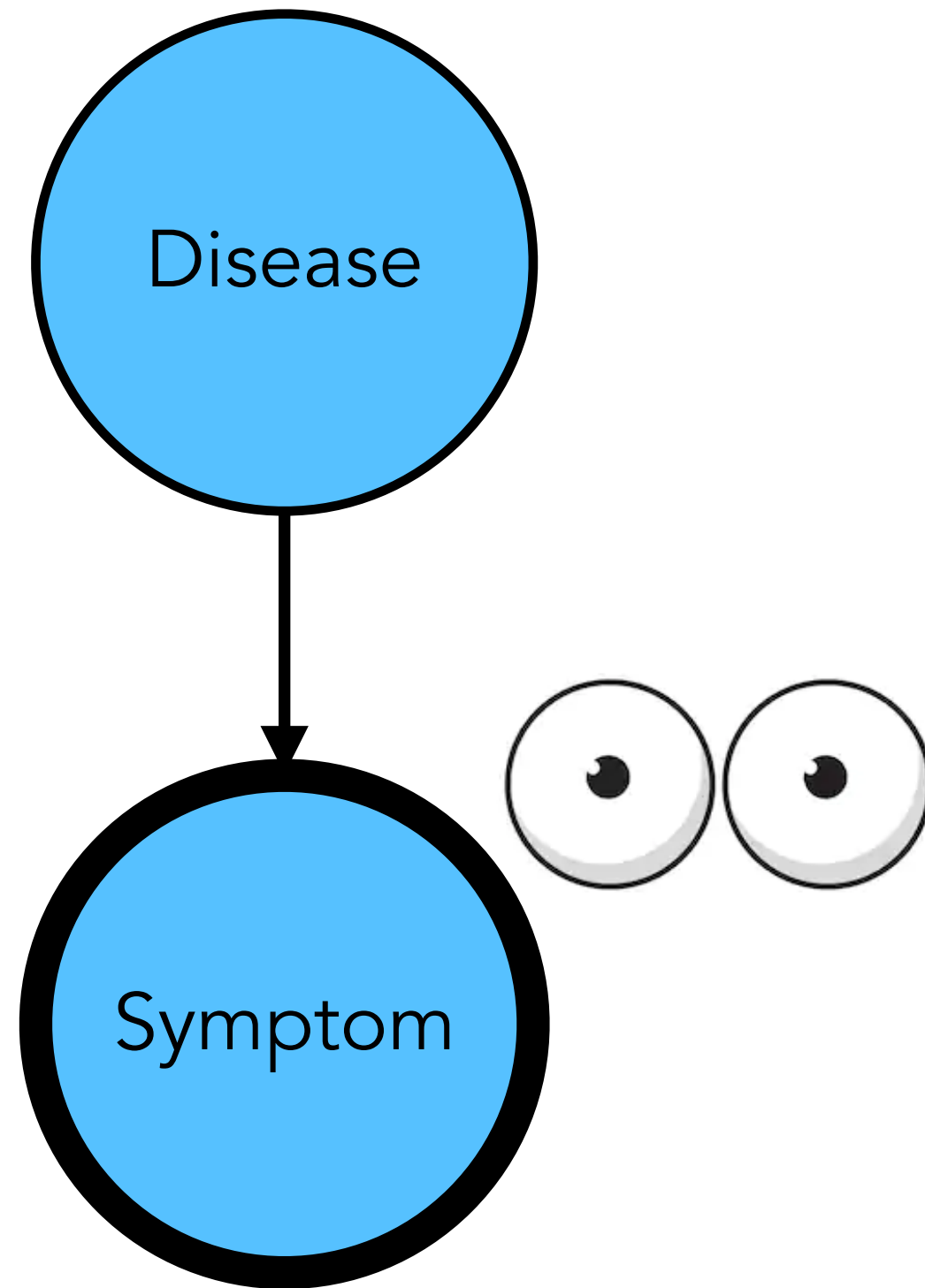
To be concrete, assume that red cars, in particular, are involved in accidents year after year at a higher rate than cars of any other color. When you go to buy a new car, should you avoid the color red in your quest to remain safe on the road?

XXX

Seeing vs. doing

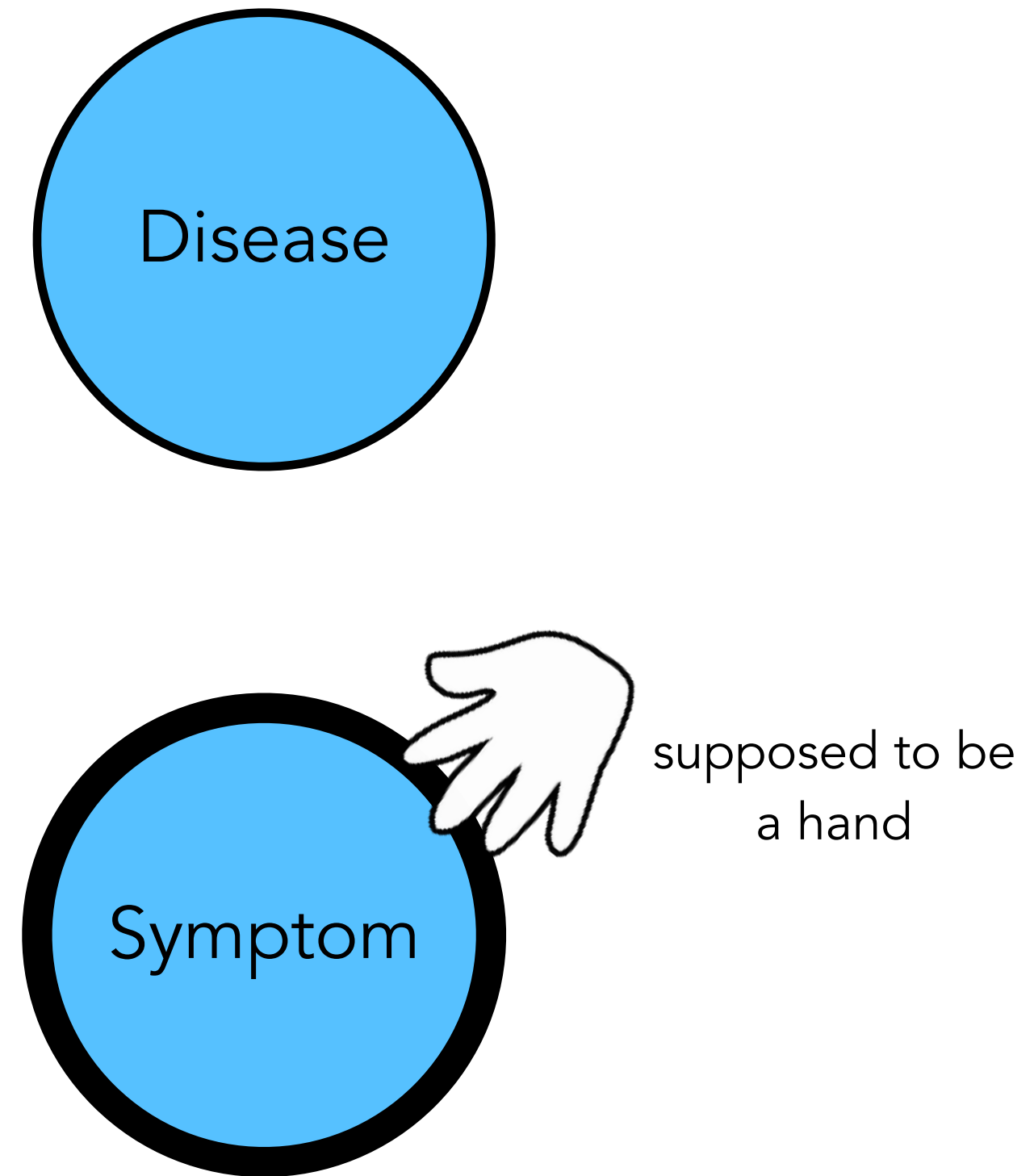
Observation vs. Intervention

seeing



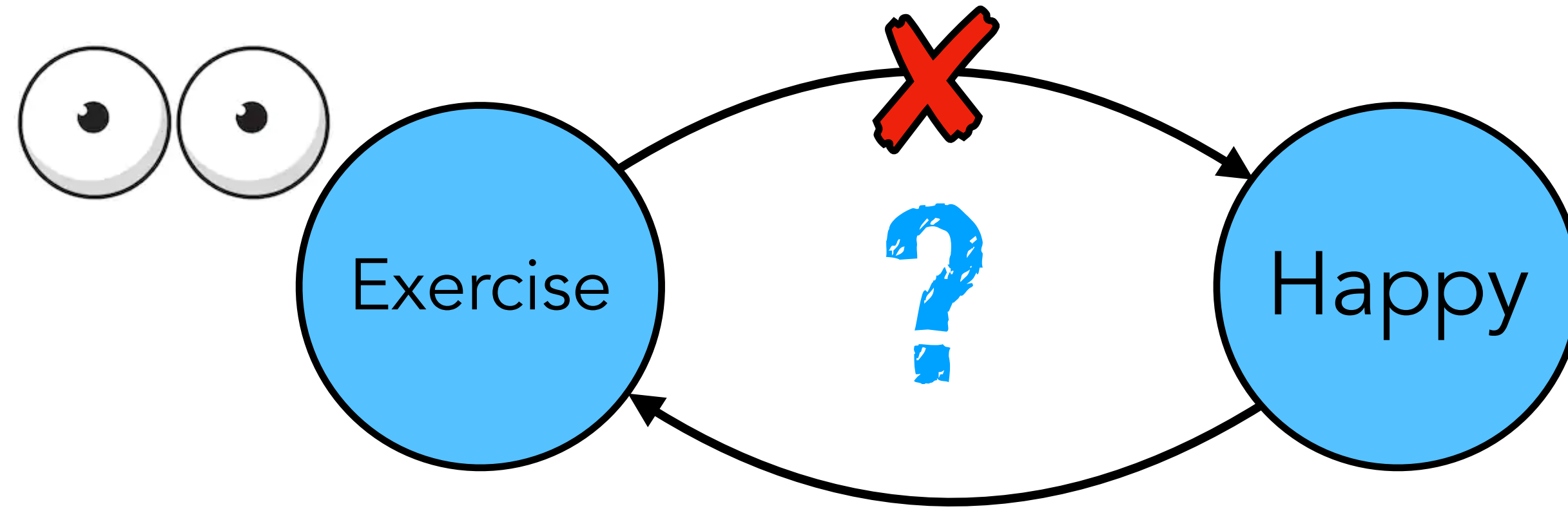
$$p(D | S) > p(D)$$

doing

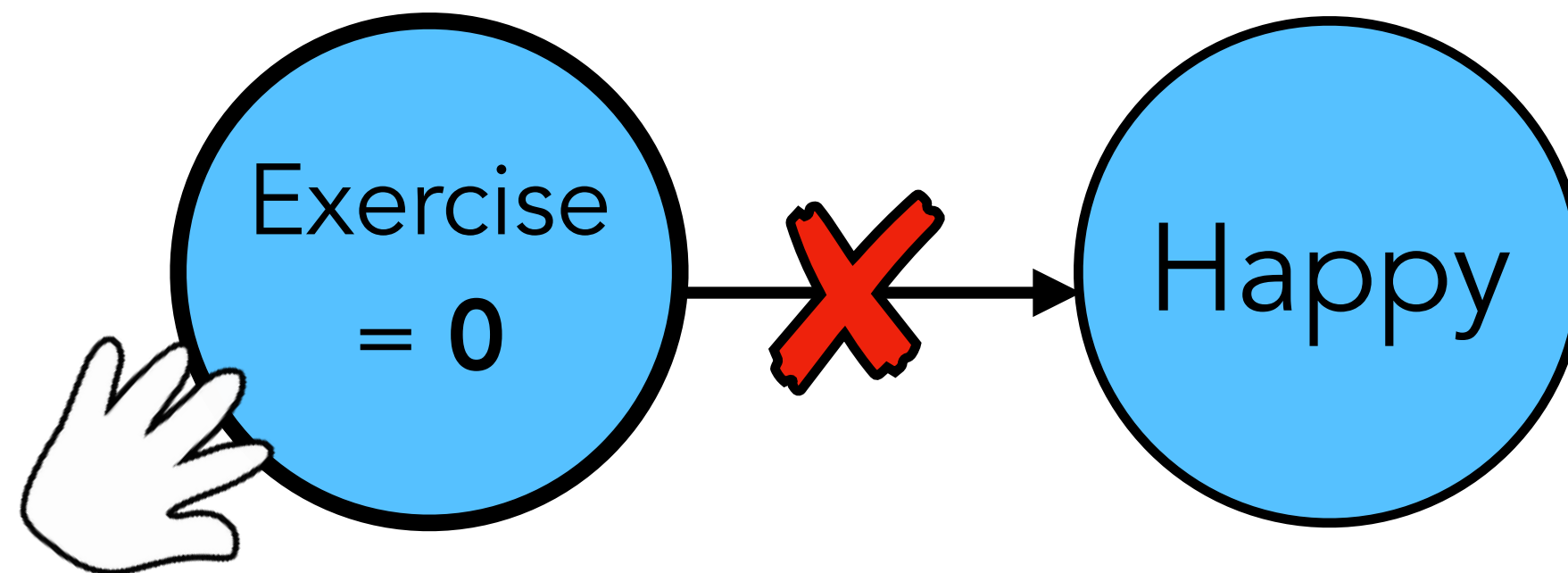


$$p(D | \text{do}(S)) = p(D)$$

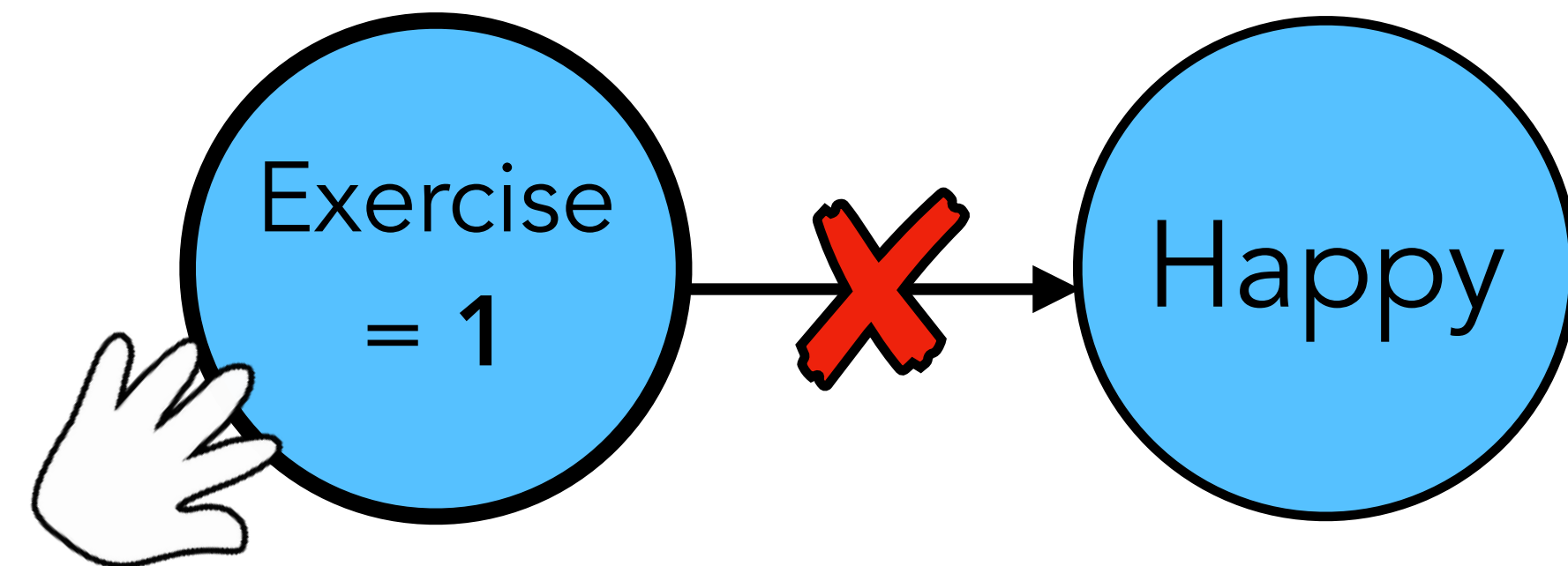
Inferring causal structure through intervention



Experiment 1

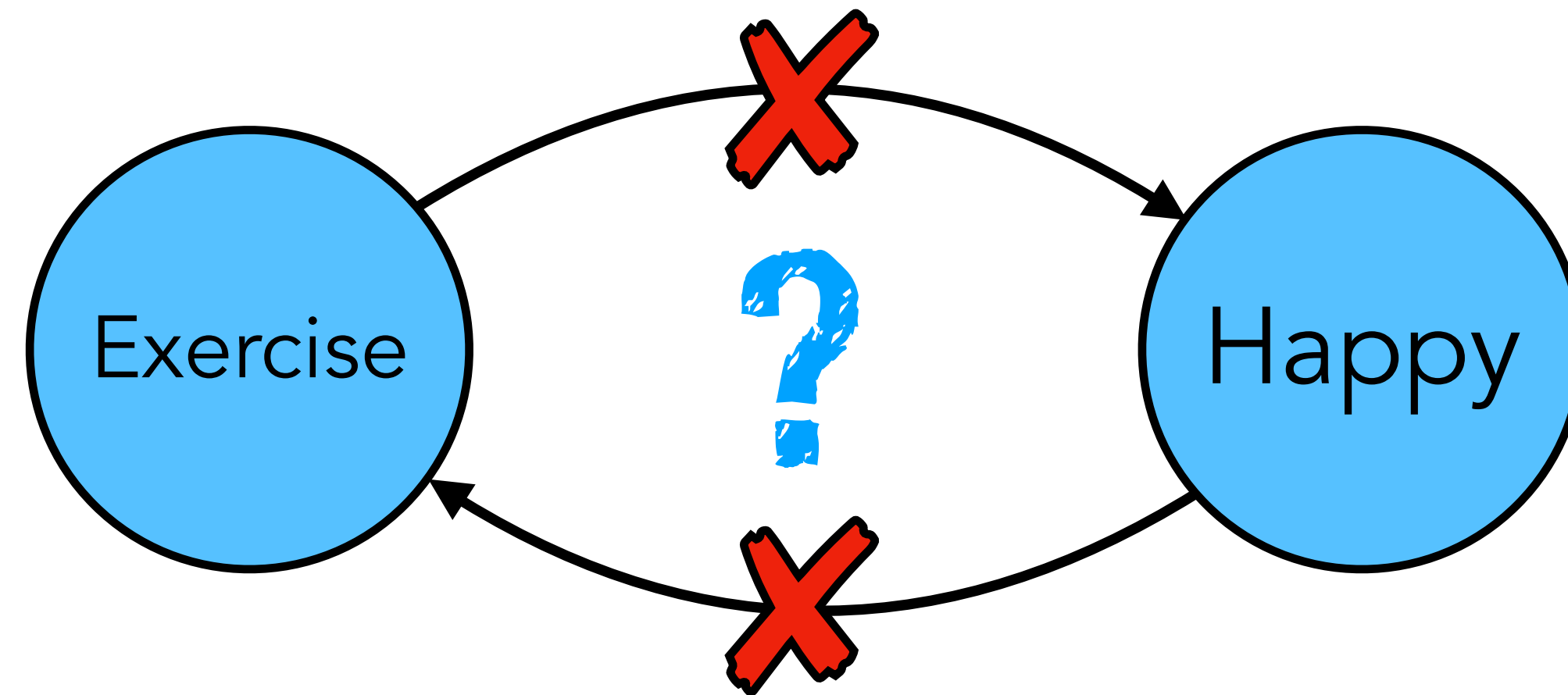


$$p(\text{Happy} \mid \text{do}(\text{Exercise} = 0)) = 0.3$$

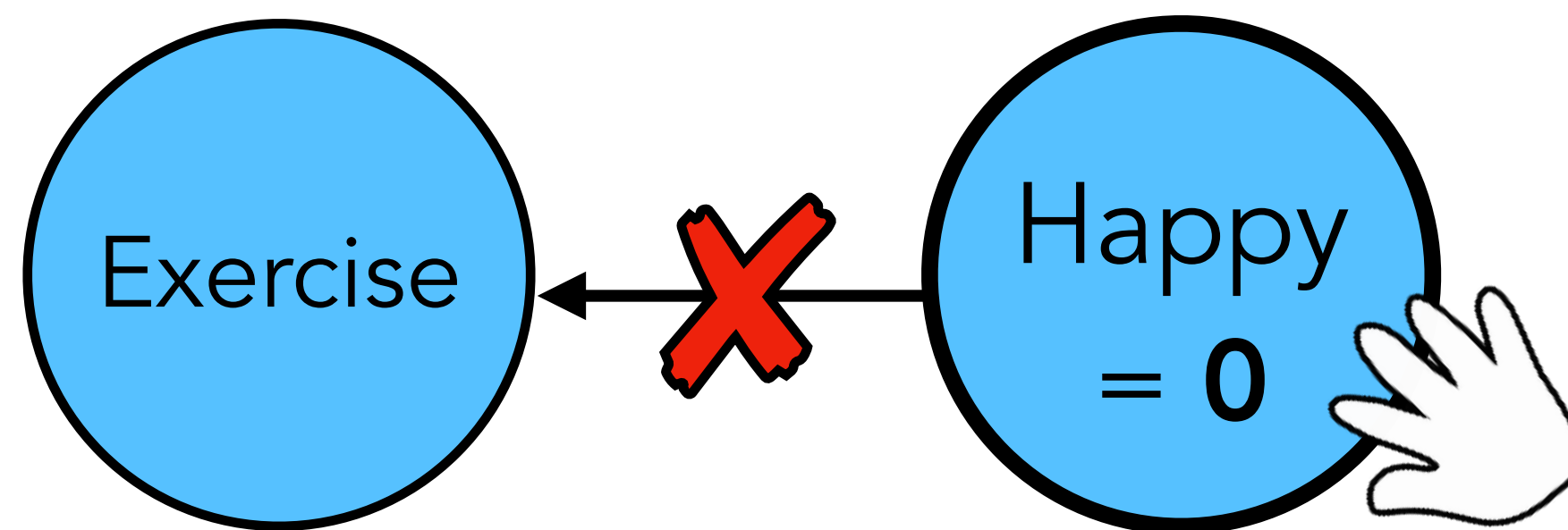


$$p(\text{Happy} \mid \text{do}(\text{Exercise} = 1)) = 0.3$$

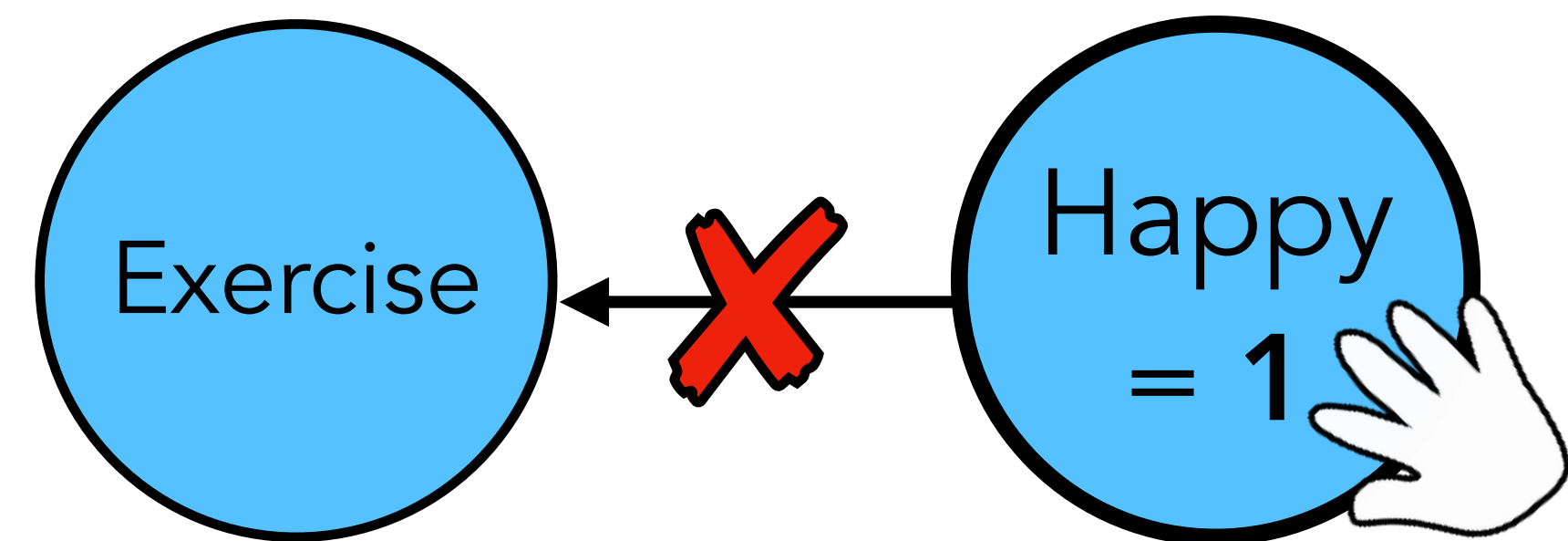
Inferring causal structure through intervention



Experiment 2

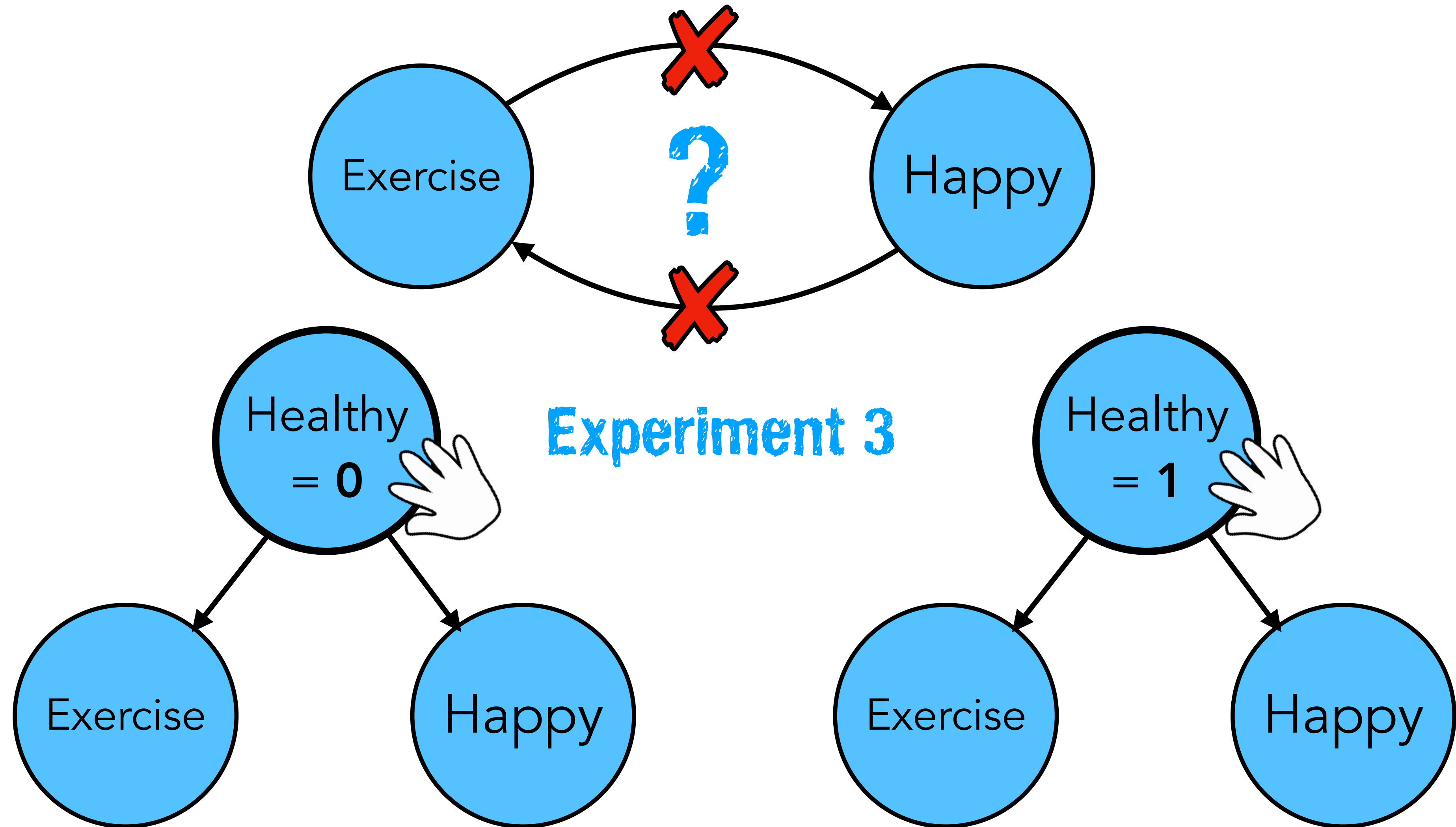


$$p(\text{Exercise} \mid \text{do}(\text{Happy} = 0)) = 0.1$$



$$p(\text{Exercise} \mid \text{do}(\text{Happy} = 1)) = 0.1$$

Inferring causal structure through intervention



$$p(\text{Happy} \mid \text{do}(\text{Healthy} = 0)) = 0.05$$

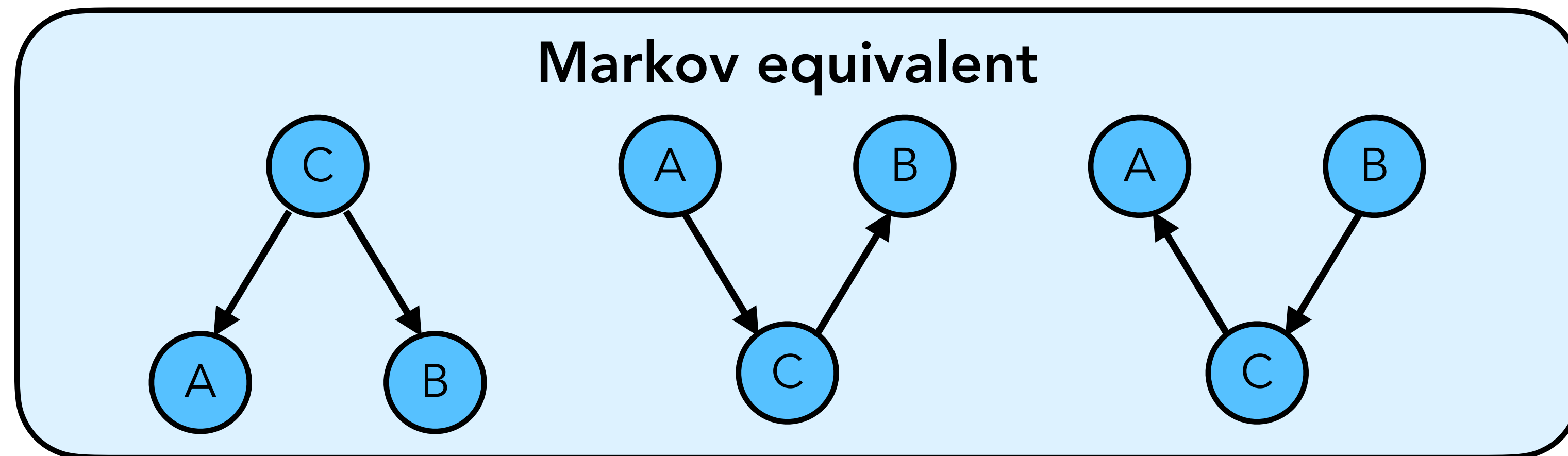
$$p(\text{Exercise} \mid \text{do}(\text{Healthy} = 0)) = 0.1$$

$$p(\text{Happy} \mid \text{do}(\text{Healthy} = 1)) = 0.5$$

$$p(\text{Exercise} \mid \text{do}(\text{Healthy} = 1)) = 0.75$$

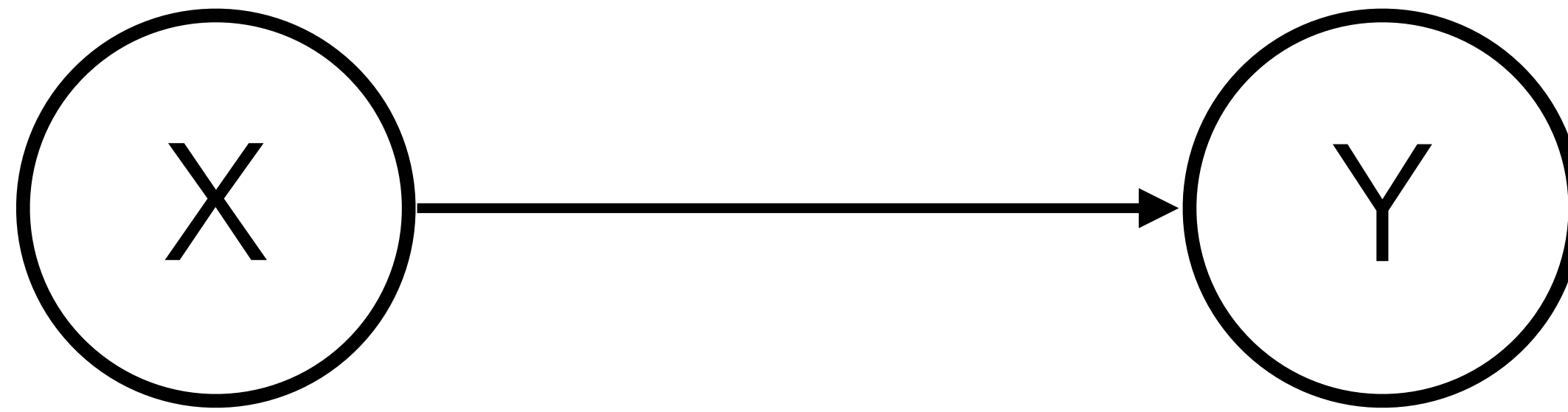
Important take home message

- correlation is not causation
- correlation (= probabilistic dependence) suggests that there is *some* causal relationship
- but we don't know which one it is

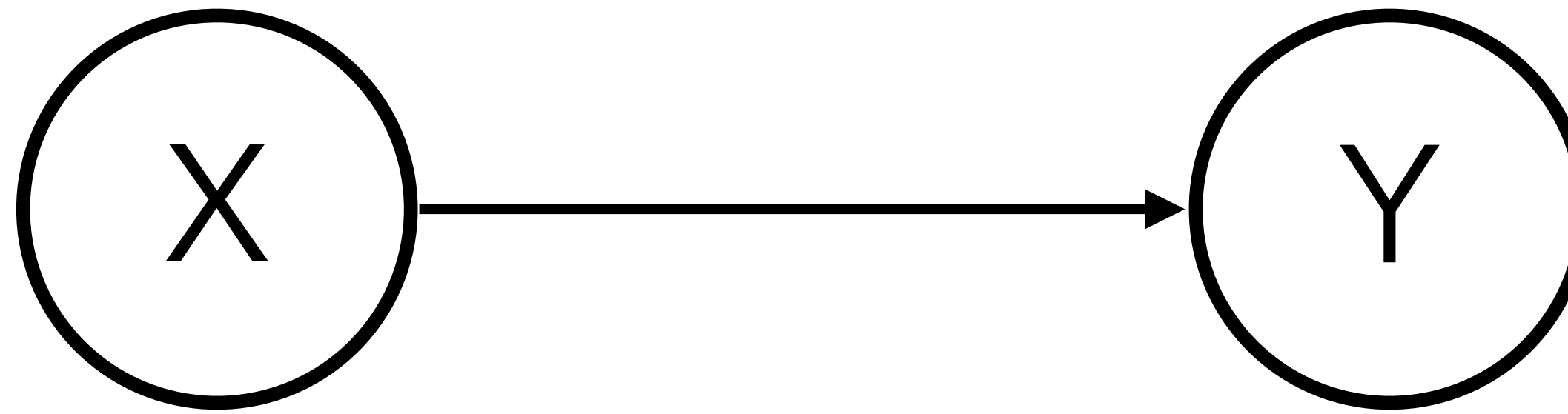


- **causal interventions** / experiments can reveal the underlying causal structure

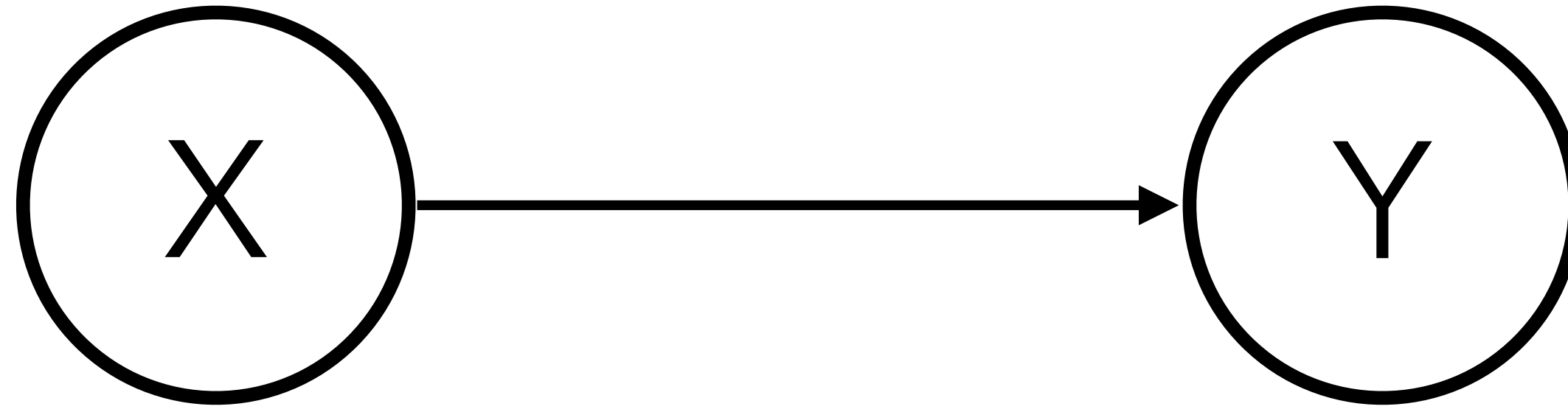
Observation, intervention, counterfactual



Level	Concept	Expression	Activity	Question	Example
I	Observation/ Prediction	$p(y x)$	Seeing	How does x change my belief in y ?	Would the grass be wet if we <i>found</i> the sprinkler off?

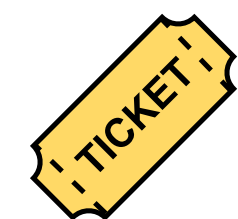
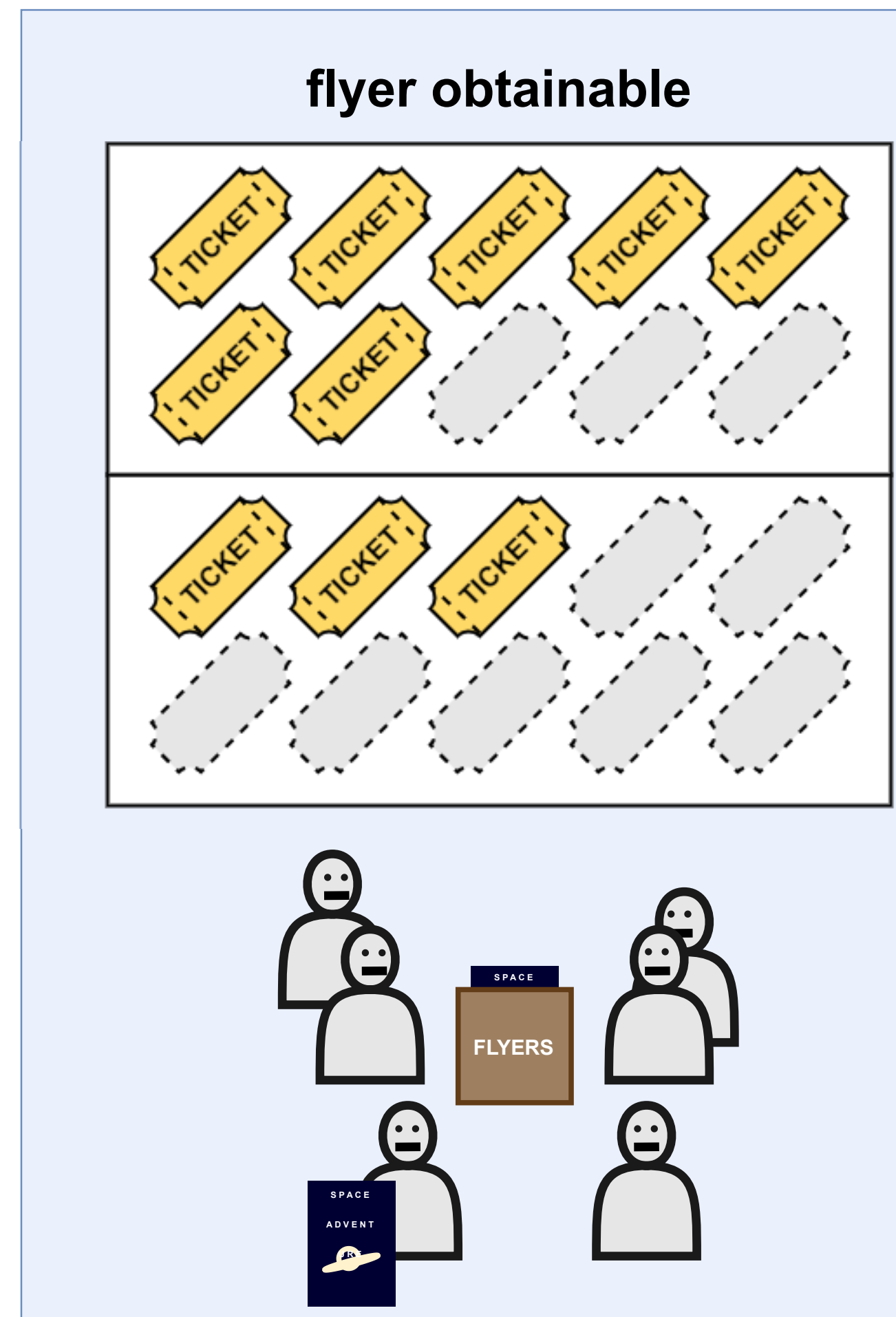
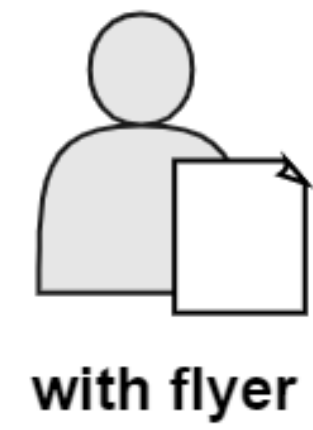


Level	Concept	Expression	Activity	Question	Example
I	Observation/ Prediction	$p(y x)$	Seeing	How does x change my belief in y ?	Would the grass be wet if we <i>found</i> the sprinkler off?
II	Intervention/ Hypothetical	$p(y \text{do}(x))$	Doing	Would y happen if I did x ?	Would the the grass be wet if <i>made sure</i> that the sprinkler was off?

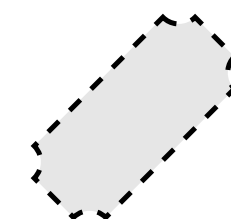


Level	Concept	Expression	Activity	Question	Example
I	Observation/ Prediction	$p(y x)$	Seeing	How does x change my belief in y ?	Would the grass be wet if we <i>found</i> the sprinkler off?
II	Intervention/ Hypothetical	$p(y \text{do}(x))$	Doing	Would y happen if I did x ?	Would the the grass be wet if <i>made sure</i> that the sprinkler was off?
III	Counterfactual	$p(y_x x', y')$	Explaining	Would y have happened instead of y' , if I had done x instead of x' ?	Would the grass have been wet if we <i>had</i> <i>made sure</i> that the sprinkler was off, given that the grass is wet and the sprinkler on?

Observation

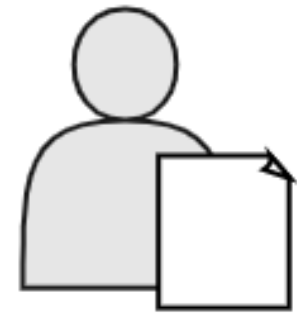


= returned



= didn't return

Intervention/Experiment

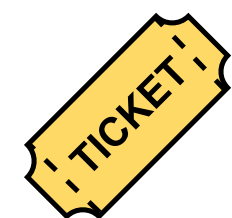
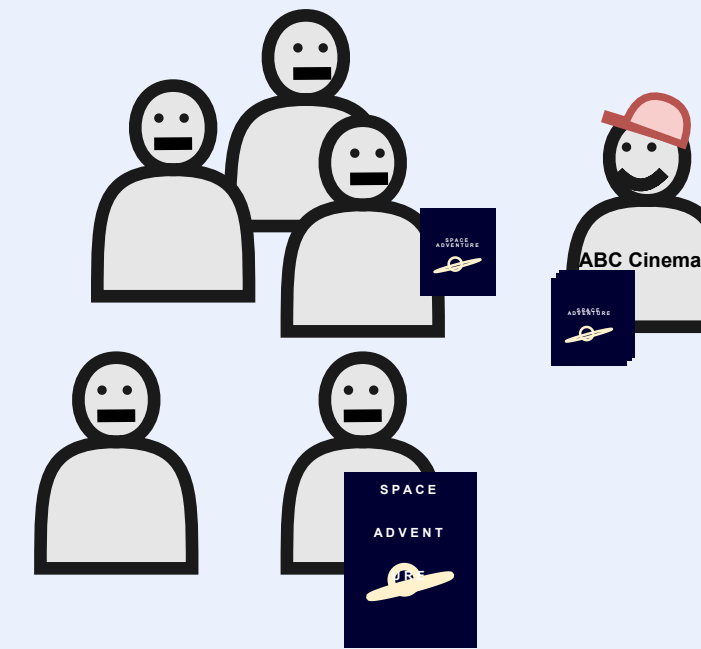
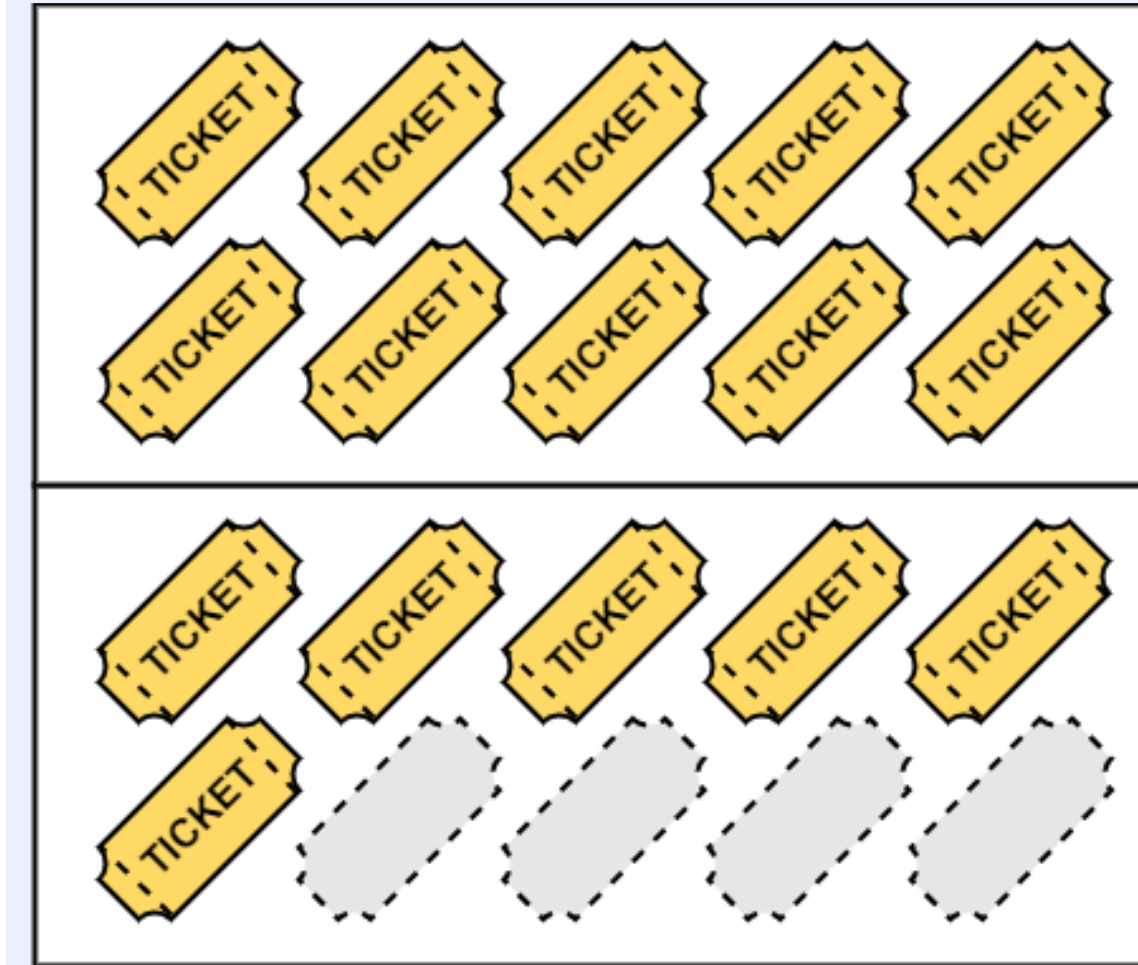


with flyer

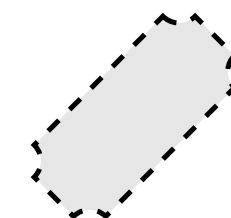


without flyer

flyer handed out



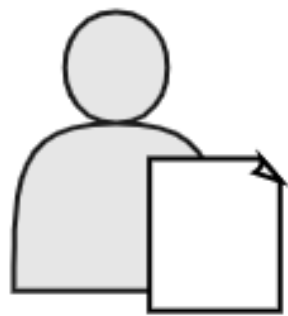
= returned



= didn't return

Counterfactual

Did this person go because they were handed a flyer?

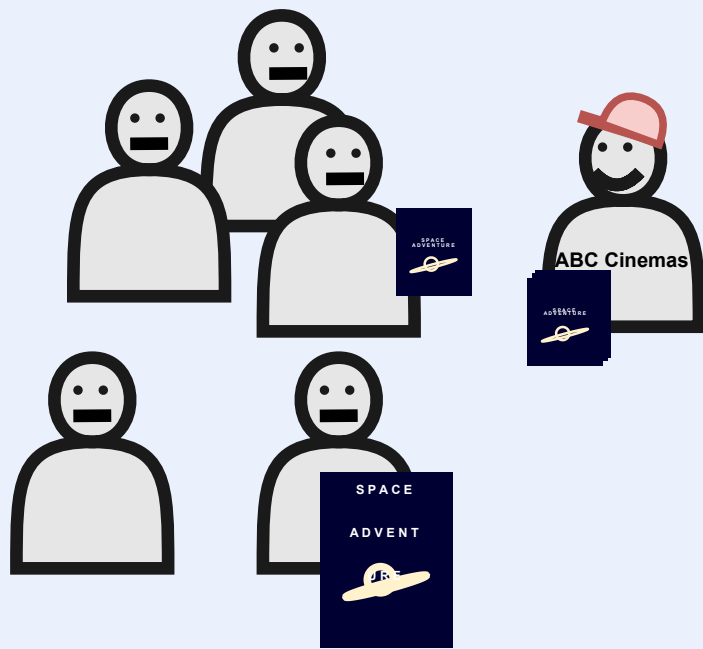
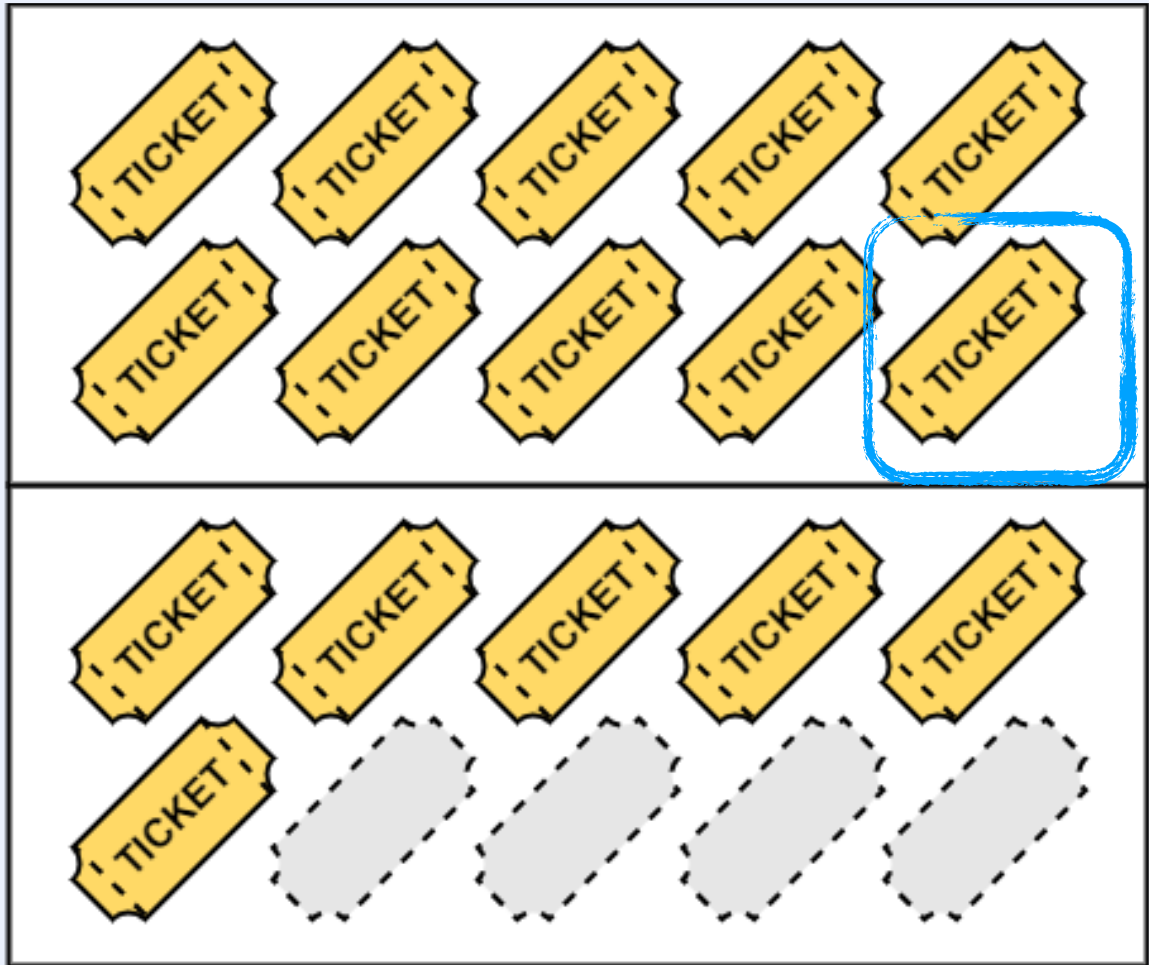


with flyer

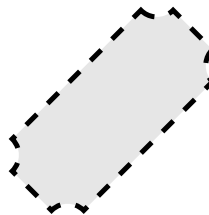


without flyer

flyer handed out

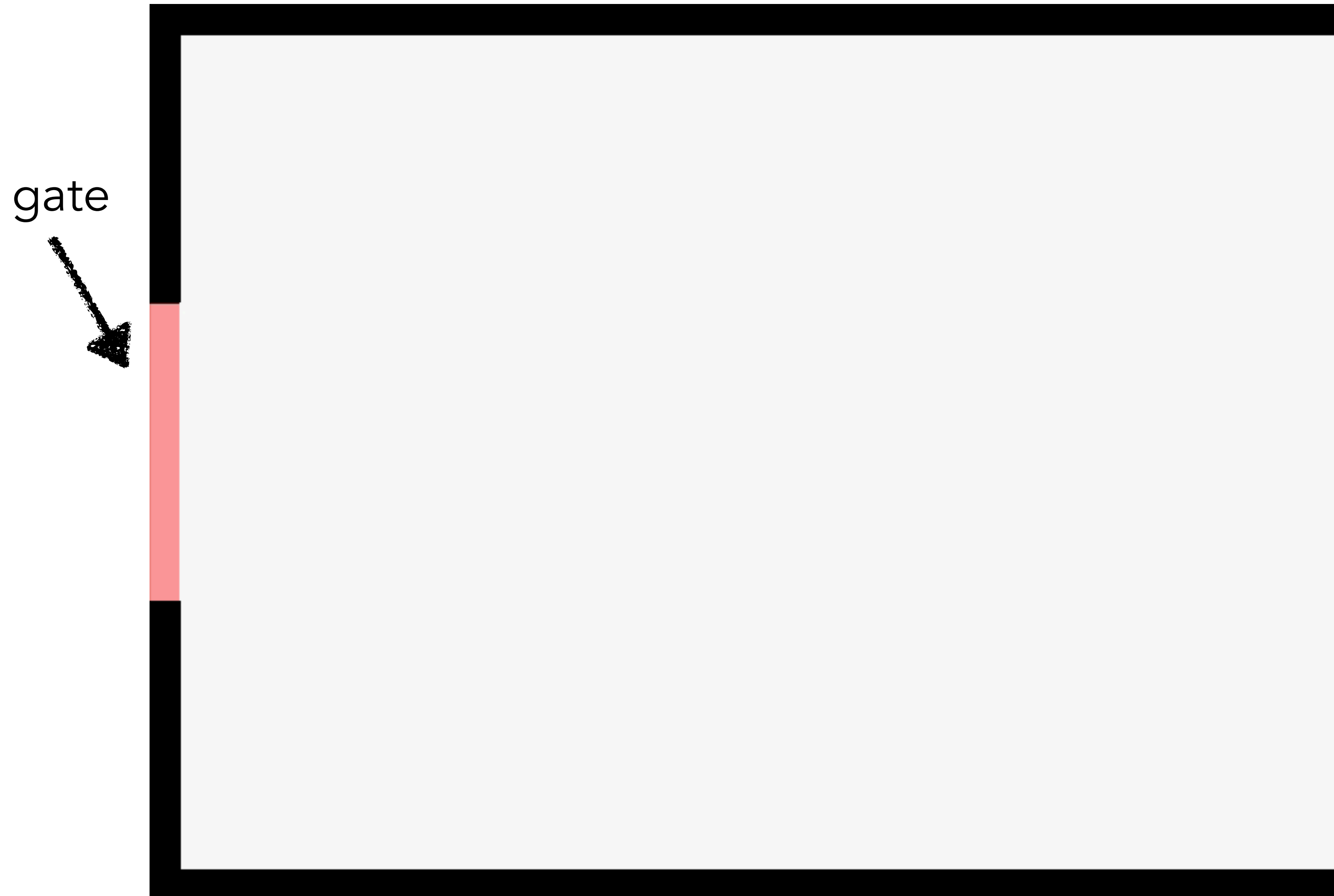


= returned



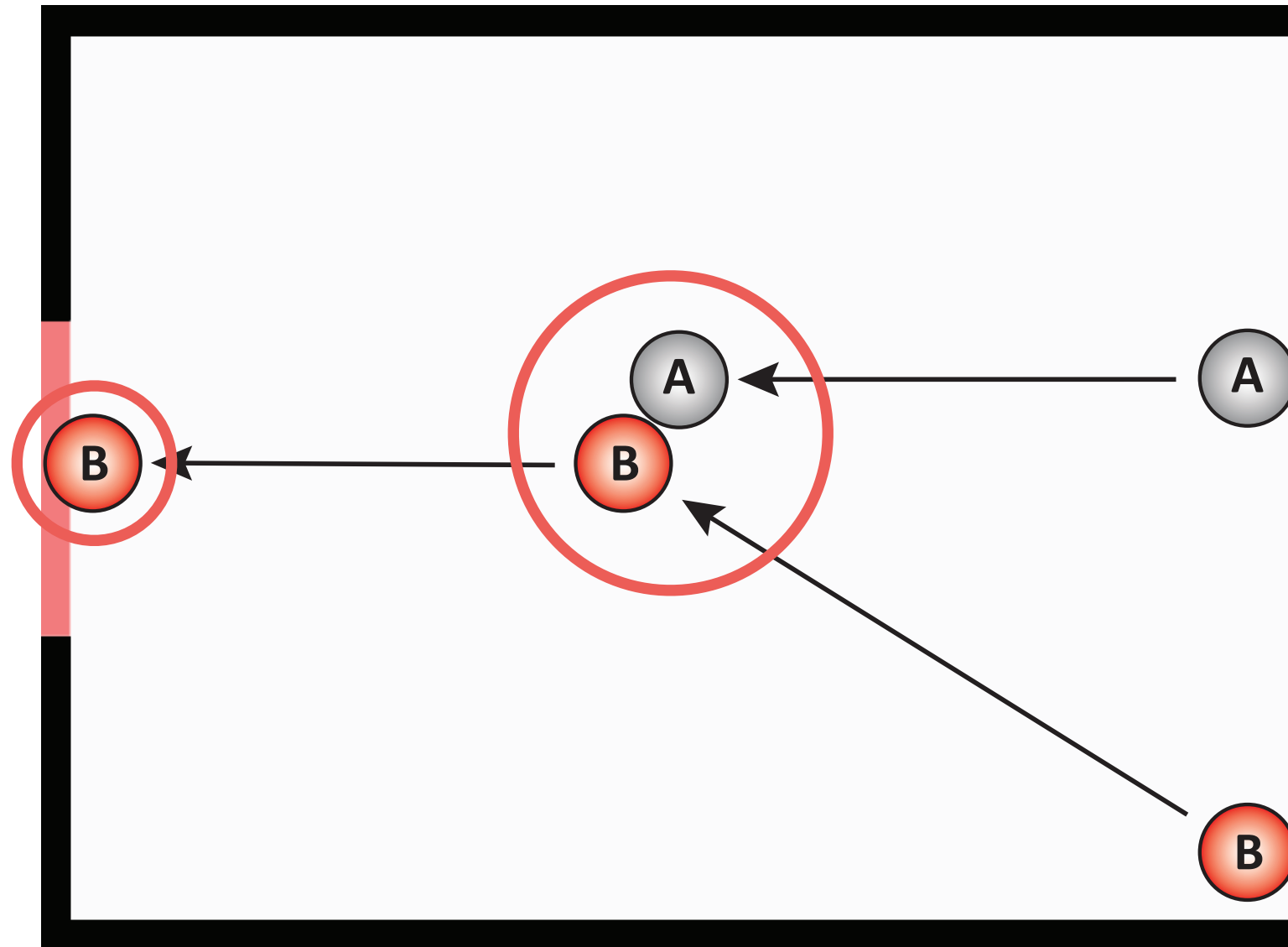
= didn't return

Did **A** cause **B** to go through the gate?



Counterfactual Simulation Model

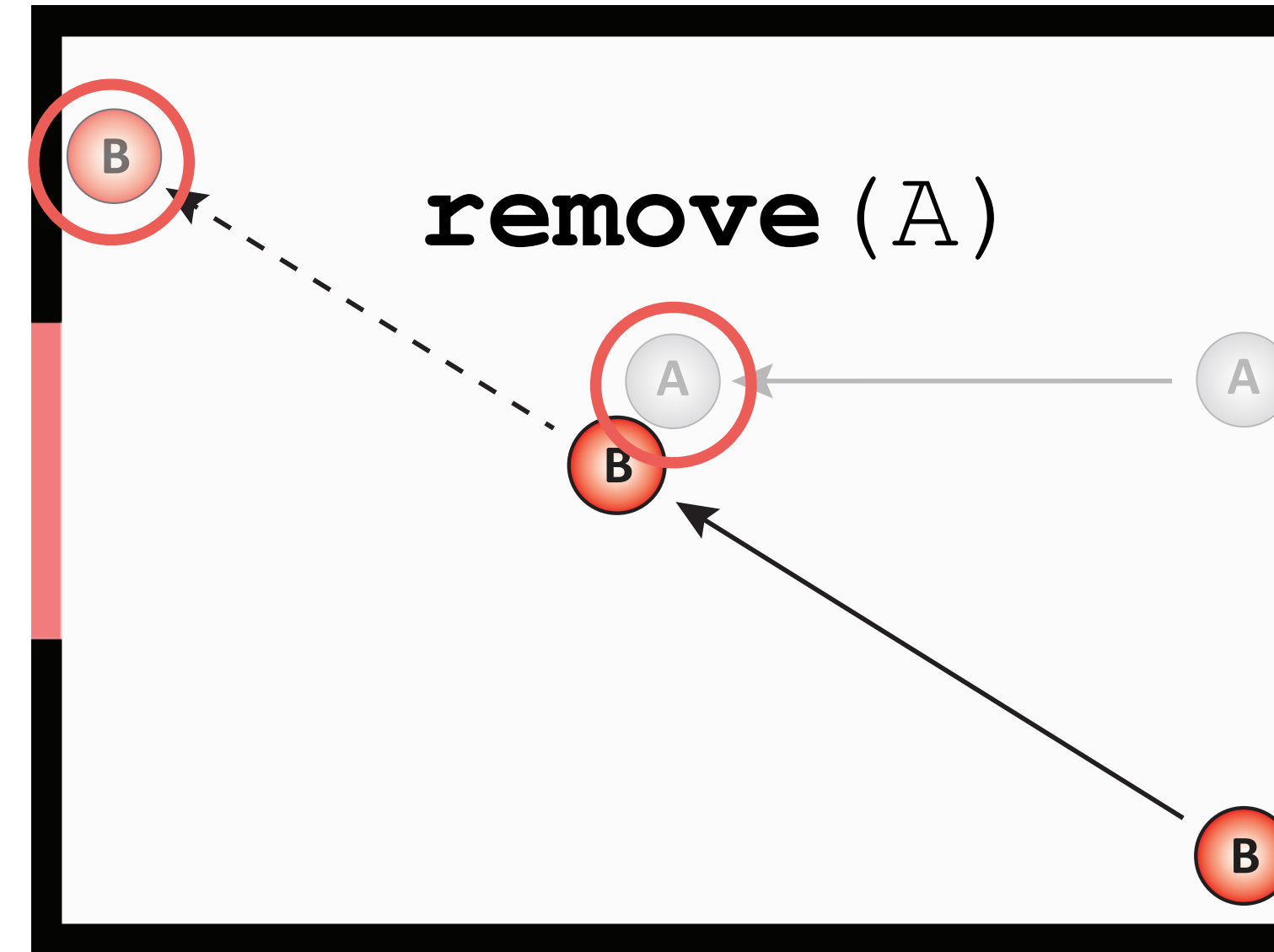
What happened?



Actual situation

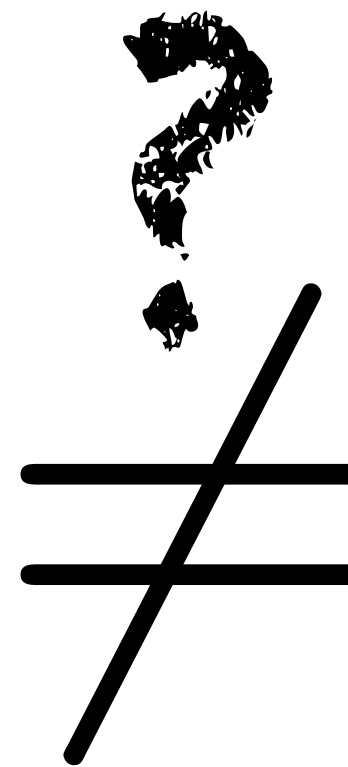
B went through the gate

What would have happened?



Counterfactual situation

B would have missed the gate

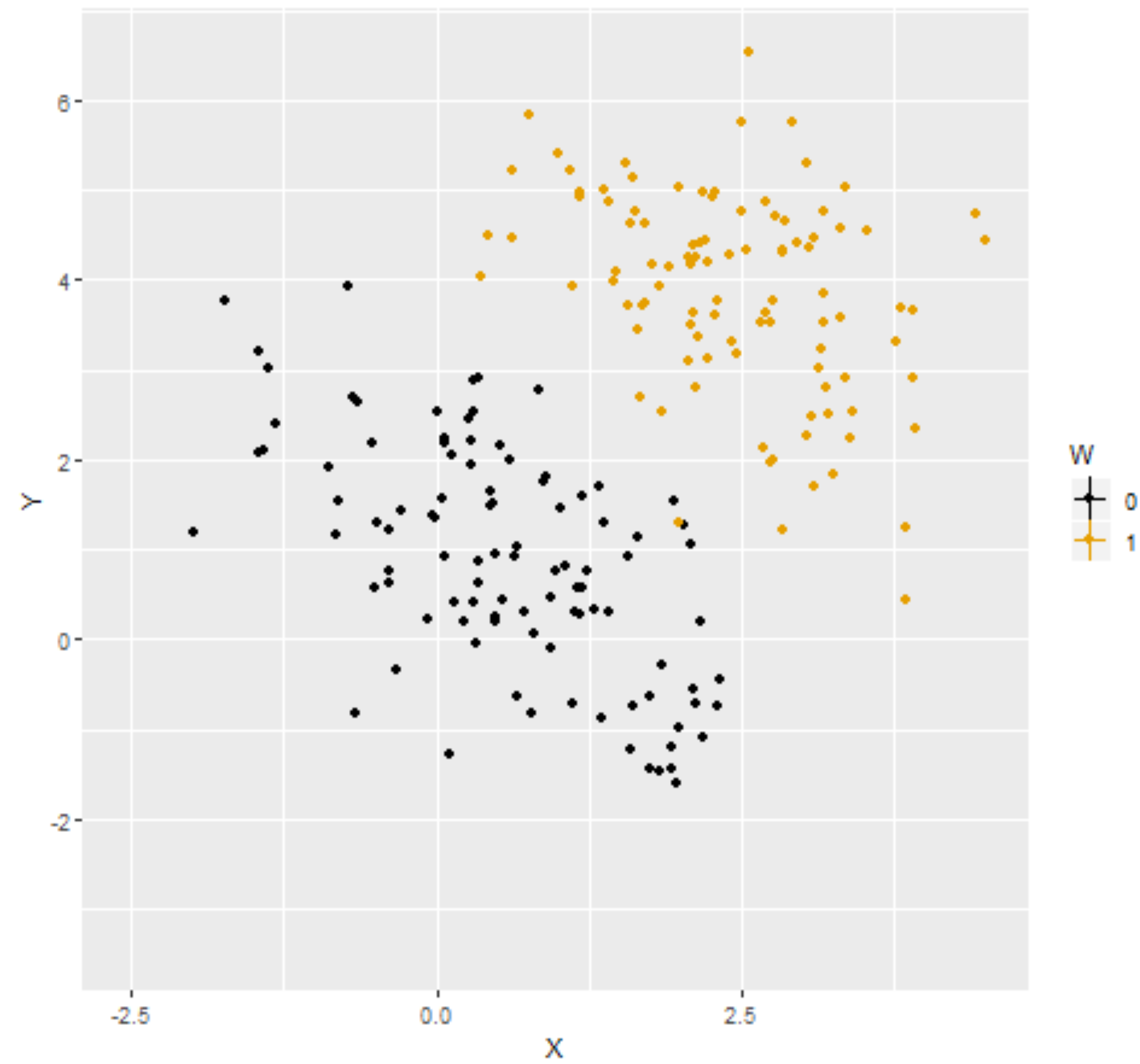


Did **A** prevent **B** from go through the gate?

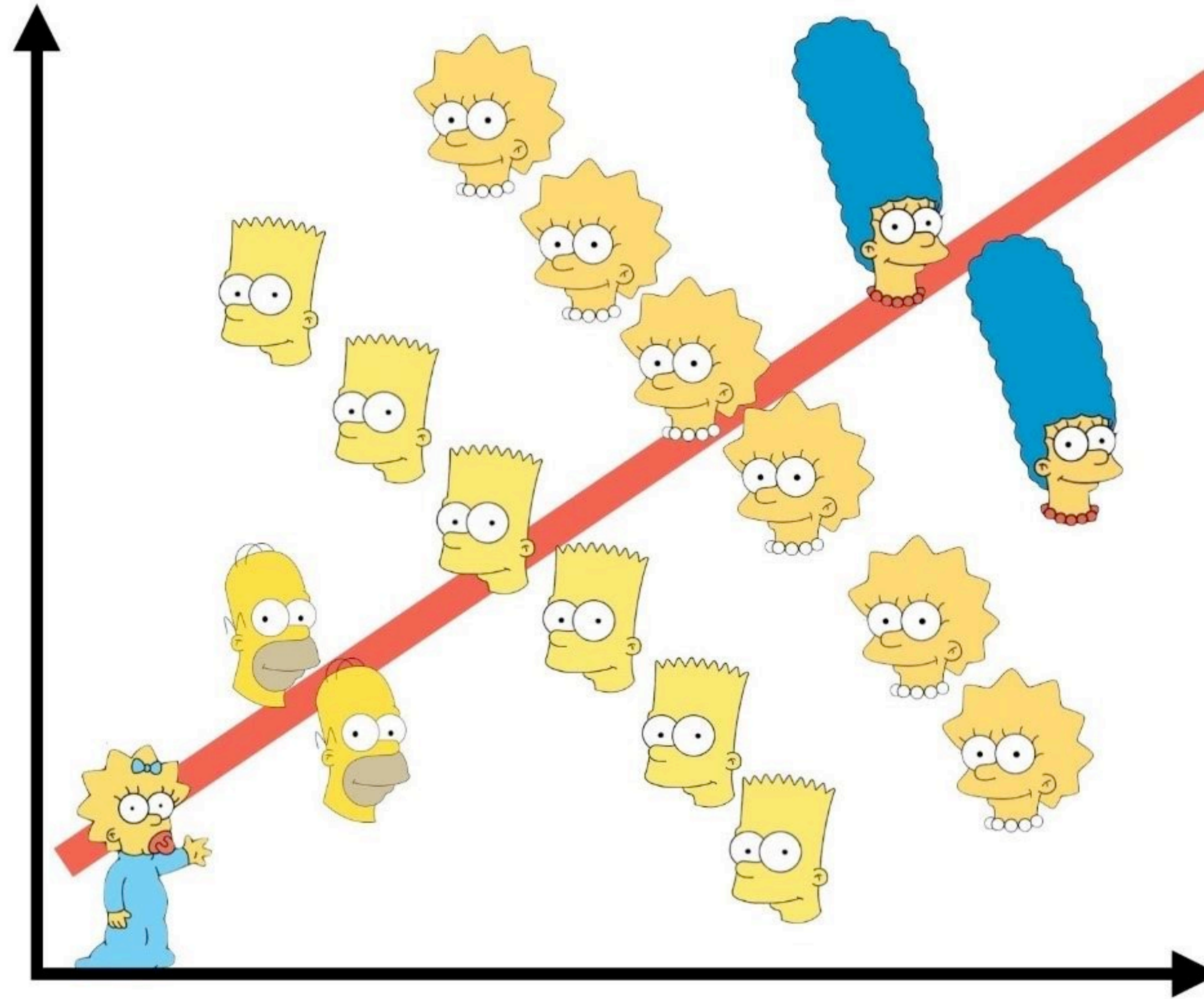
1/2 speed

Controlling for variables

The Relationship between Y and X, Controlling for a Binary Variable W
1. Start with raw data. Correlation between X and Y: 0.319



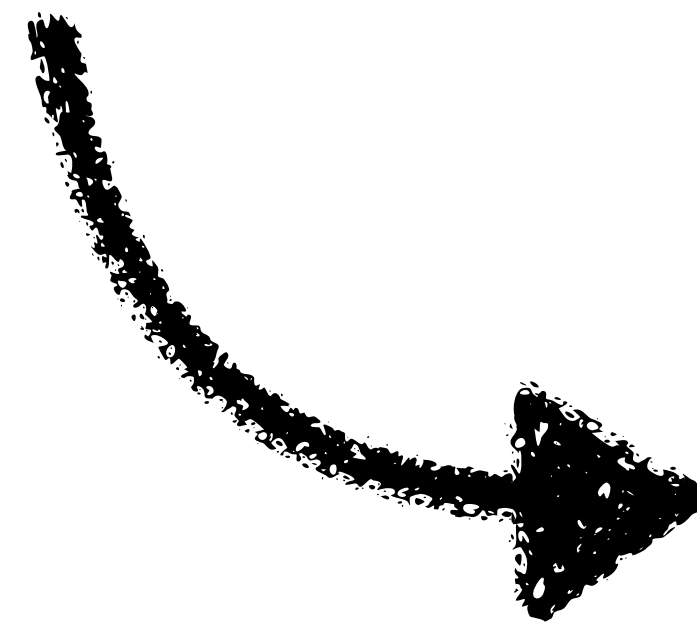
Simpson's paradox



What does controlling for variables mean?

we are not actually "**controlling**" the variable

instead, we are taking the variable into consideration when making predictions



the hope is that we get a better estimate of the parameter that we are interested in by taking into account other factors

Patterns of inference

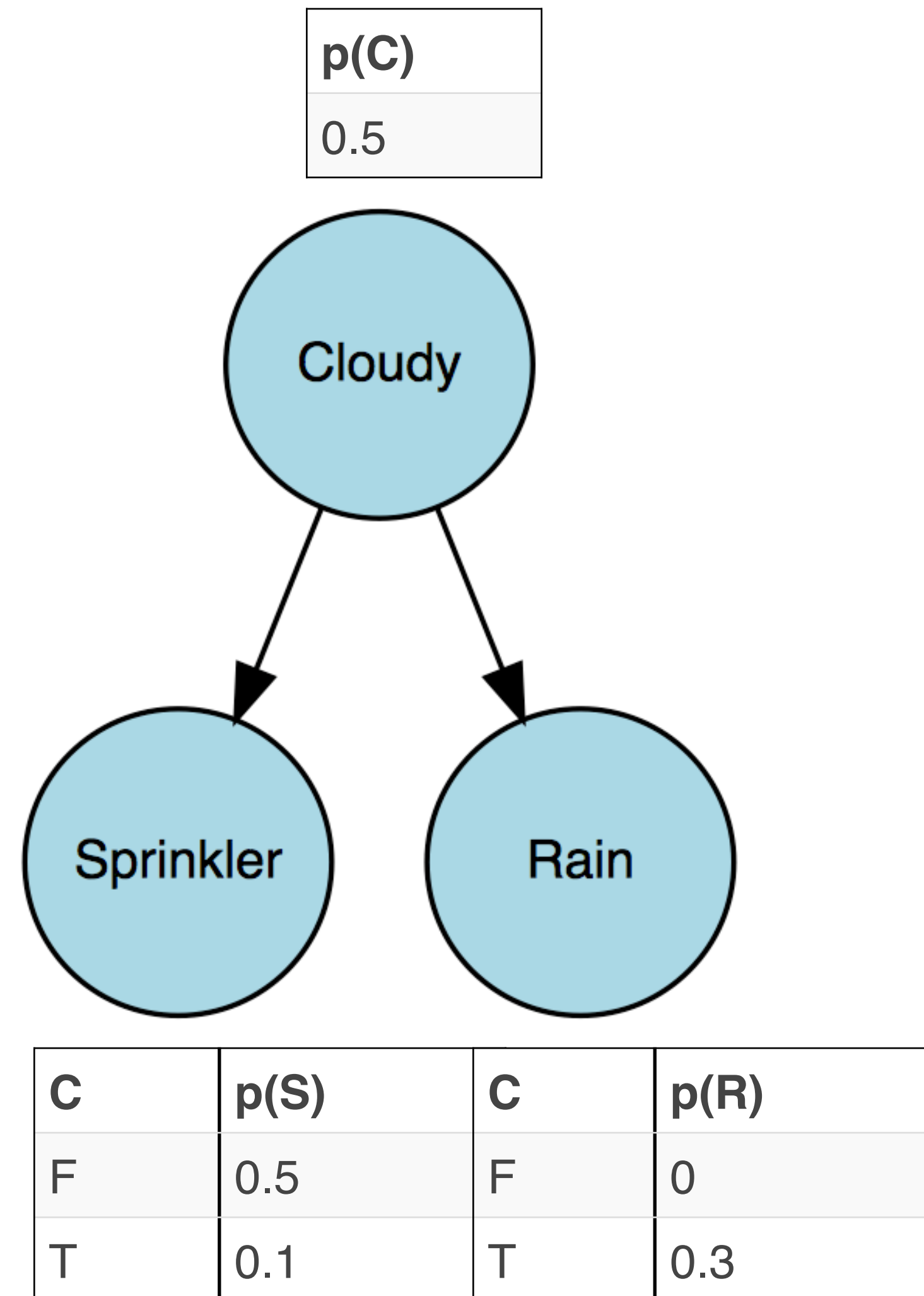
Patterns of inference: **Common cause**

$$p(S | R) = p(S)$$

or

$$p(S | R) \neq p(S)$$

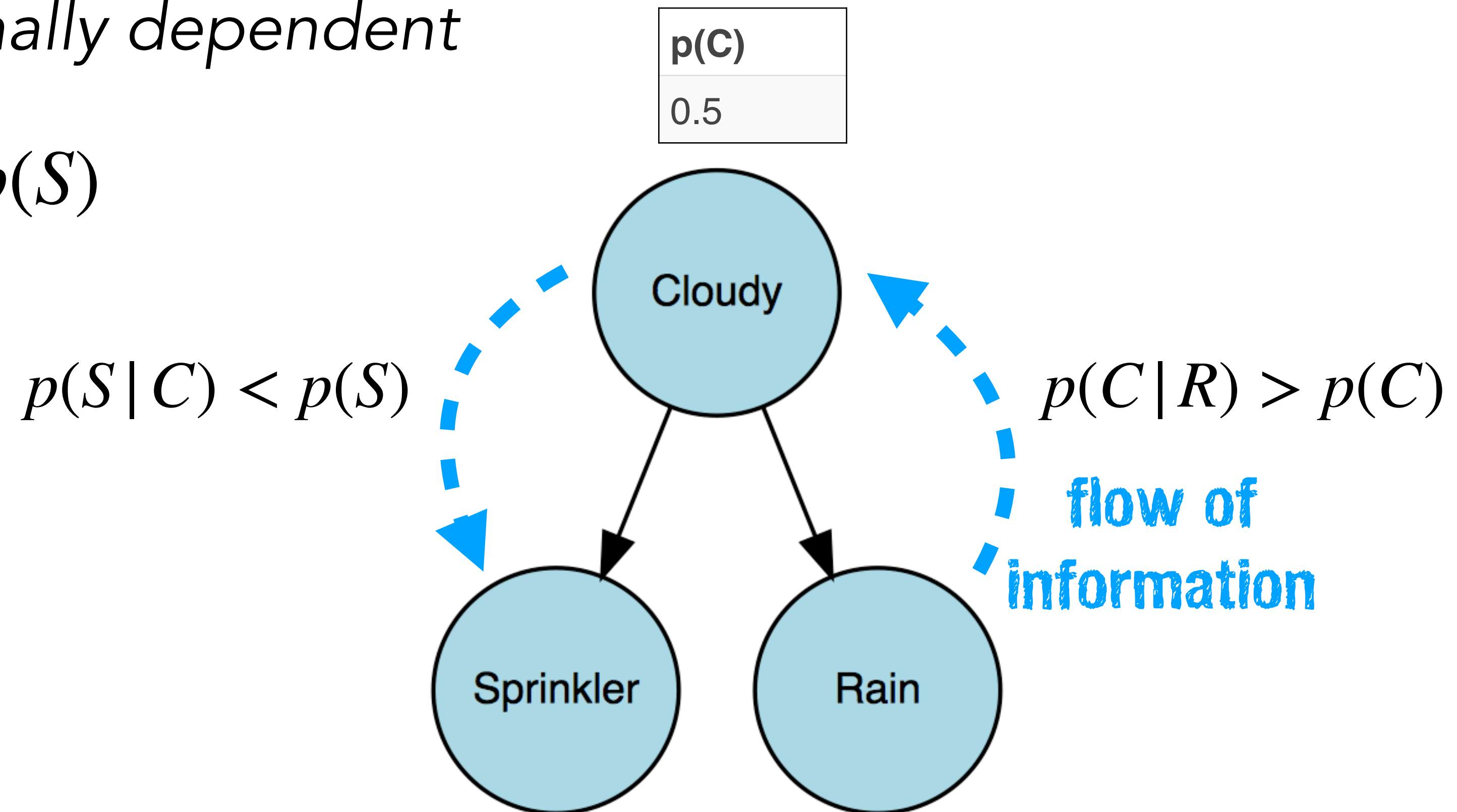
?



Patterns of inference: **Common cause**

- effects of a common cause are *unconditionally dependent*

$$p(S|R) \neq p(S)$$

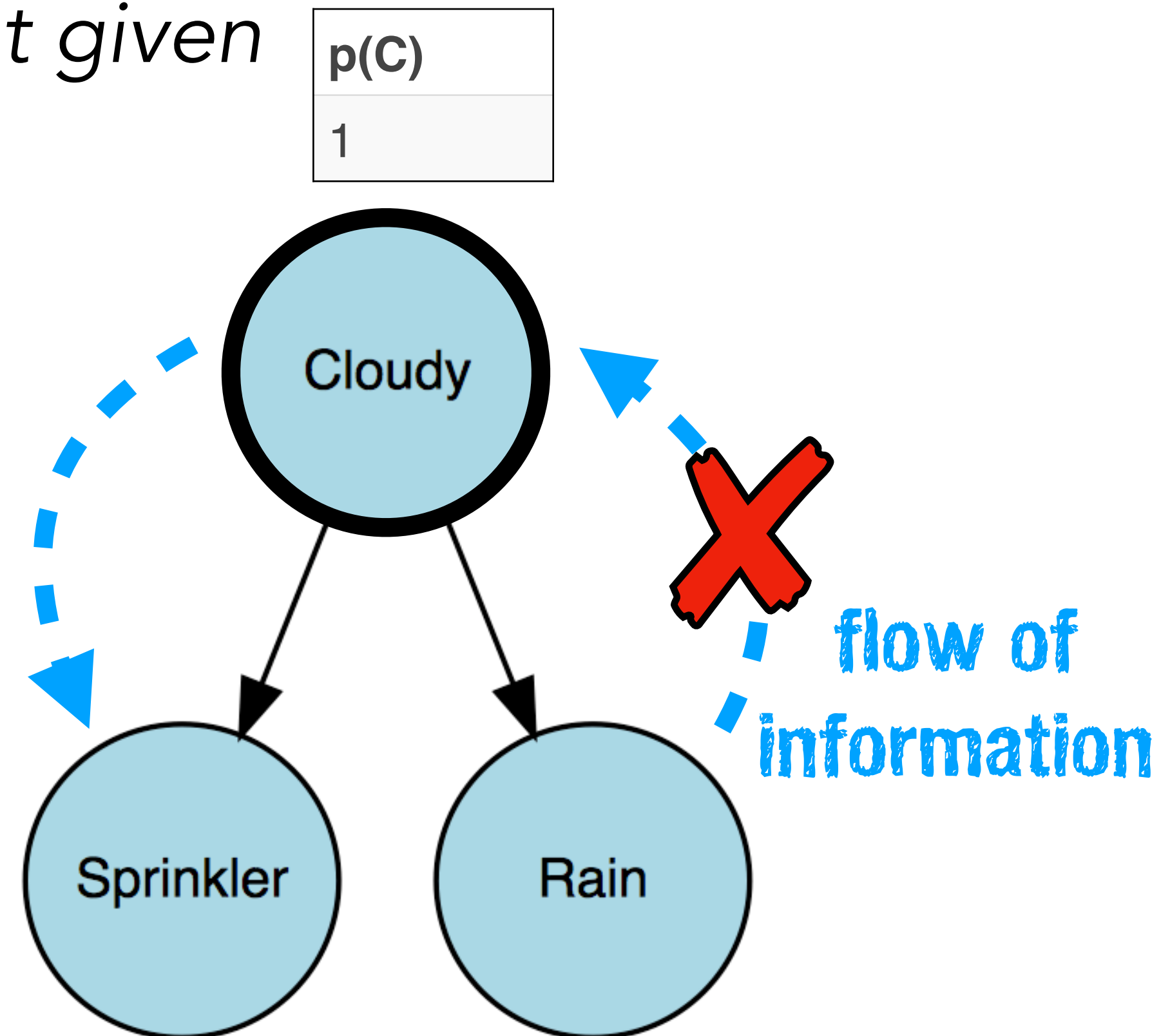


C	p(S)	C	p(R)
F	0.5	F	0
T	0.1	T	0.3

Patterns of inference: **Common cause**

- effects of a common cause are *conditionally independent given the cause*

$$p(S | R, C) = p(S | C)$$

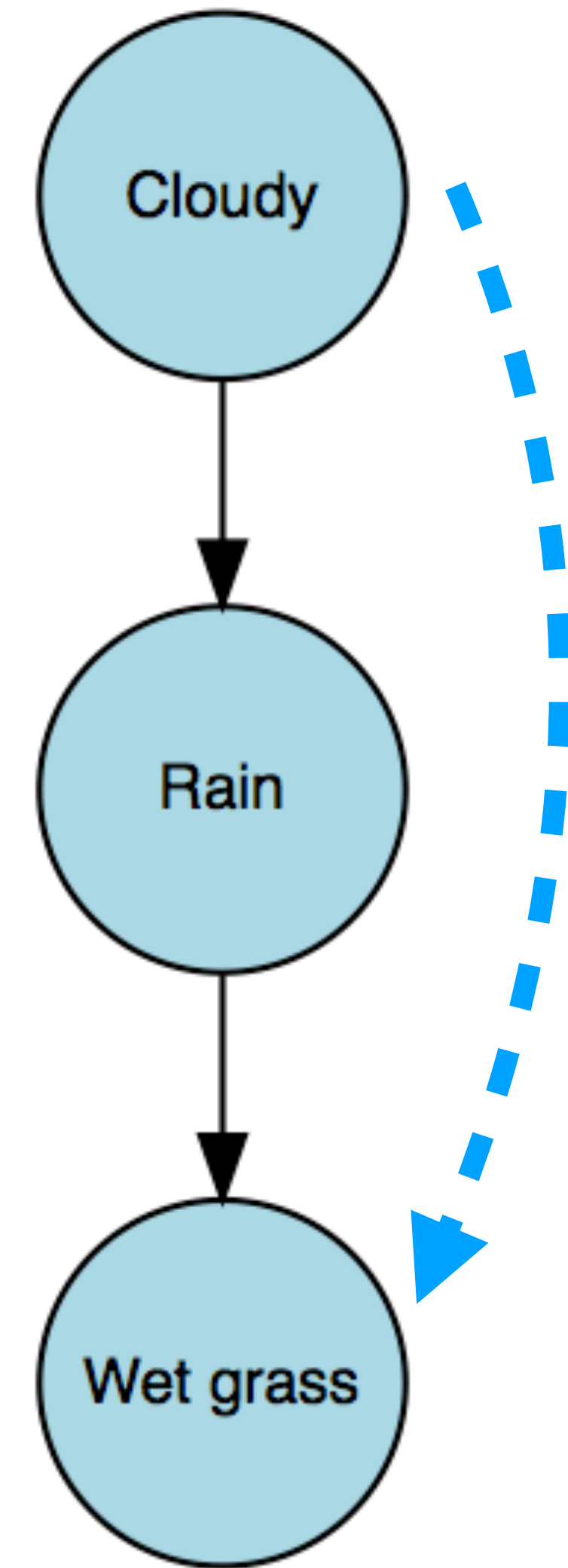


C	p(S)	C	p(R)
F	0.5	F	0
T	0.1	T	0.3

Patterns of inference: **Causal chain**

- cause and effect in a causal chain are *unconditionally dependent*

$$p(W|C) \neq p(W)$$

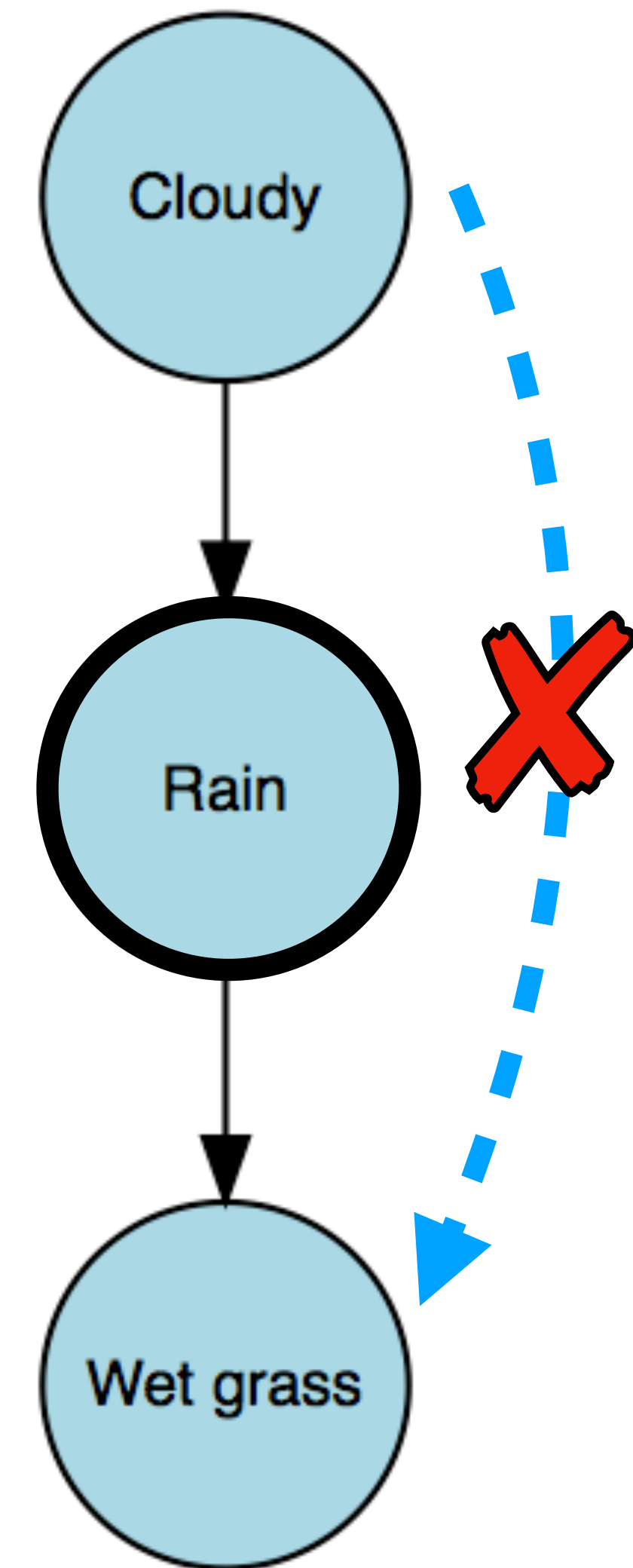


Patterns of inference: **Causal chain**

- cause and effect in a causal chain are *conditionally independent*

$$p(W | C, R) = p(W | R)$$

screening off

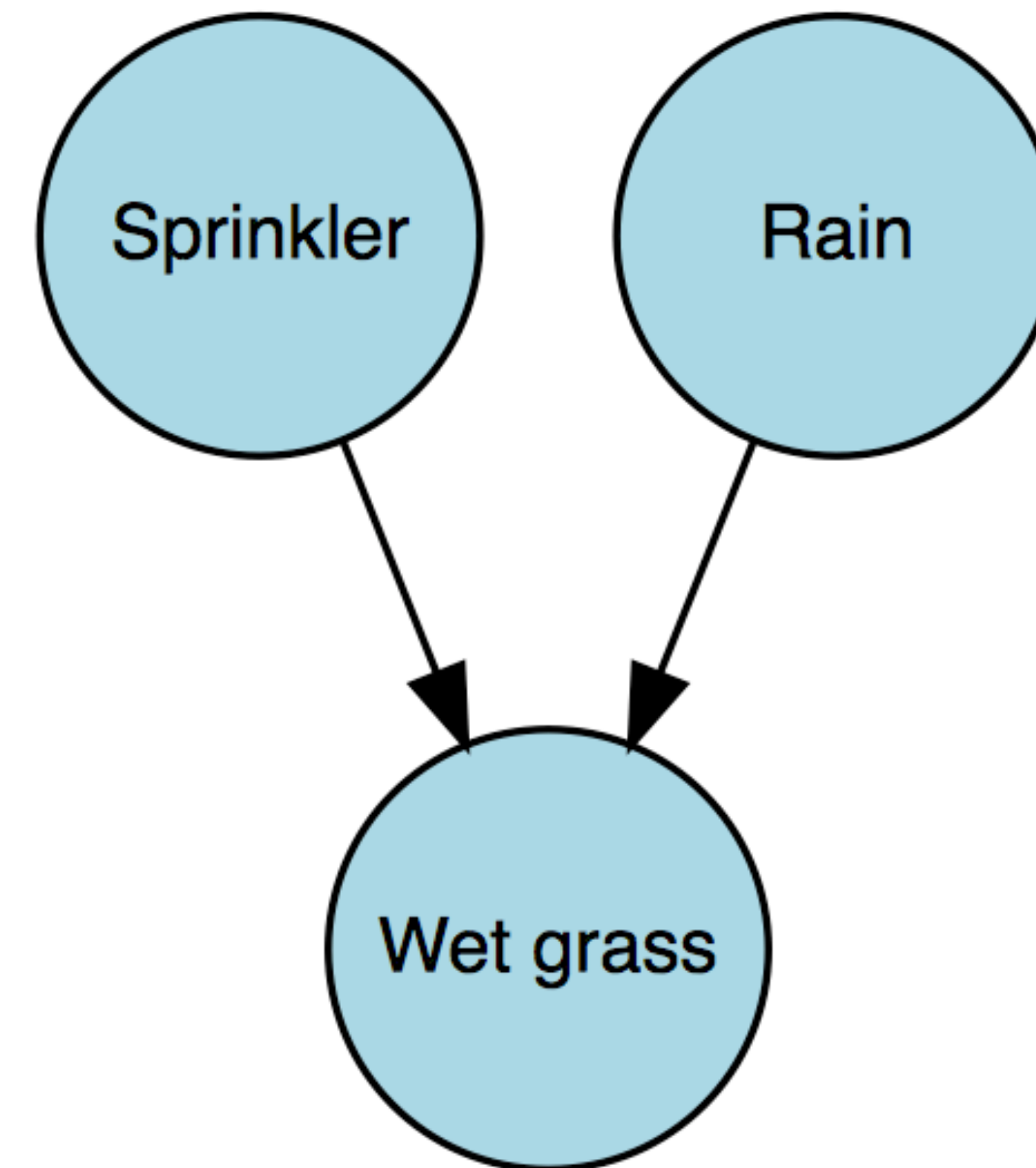


Patterns of inference: **Common effect**

- two causes of a common effect are *unconditionally independent*

$$p(S | R) = p(S)$$

(e.g. Sprinkler is set by a timer)

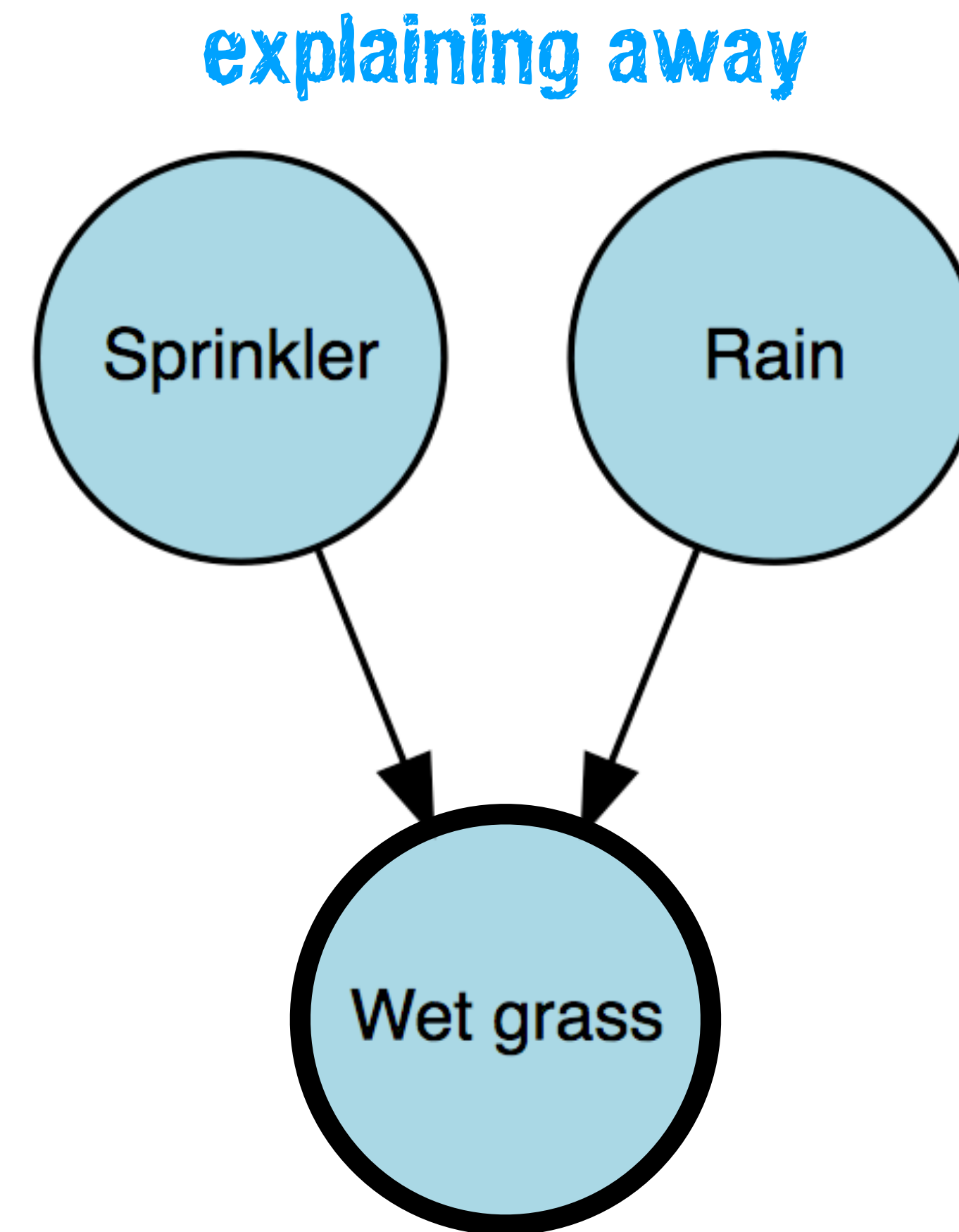


Patterns of inference: **Common effect**

- two causes of a common effect are *conditionally dependent given the effect*

$$p(S | R, W) \neq p(S | W)$$

- intuitively: both causes compete to explain the effect



Note: The pattern of inference depends on the structural form which captures how `Sprinkler` and `Rain` jointly affect `Wet grass`. Explaining away holds for the commonly used noisy-or integration function.

Should I control?

When should I control for variables?

recent advances in graphical models have produced a way to help distinguish good from bad controls

 **d-separation**
directional

decide from a causal graph whether a set of variables X is independent of another set Y , given a third set Z

Goal: we want a precise (and unbiased) estimate of the predictive relationship between X and Y

 **we want to block all other paths from X to Y**

When should I control for variables?

How can I tell whether two variables are independent?

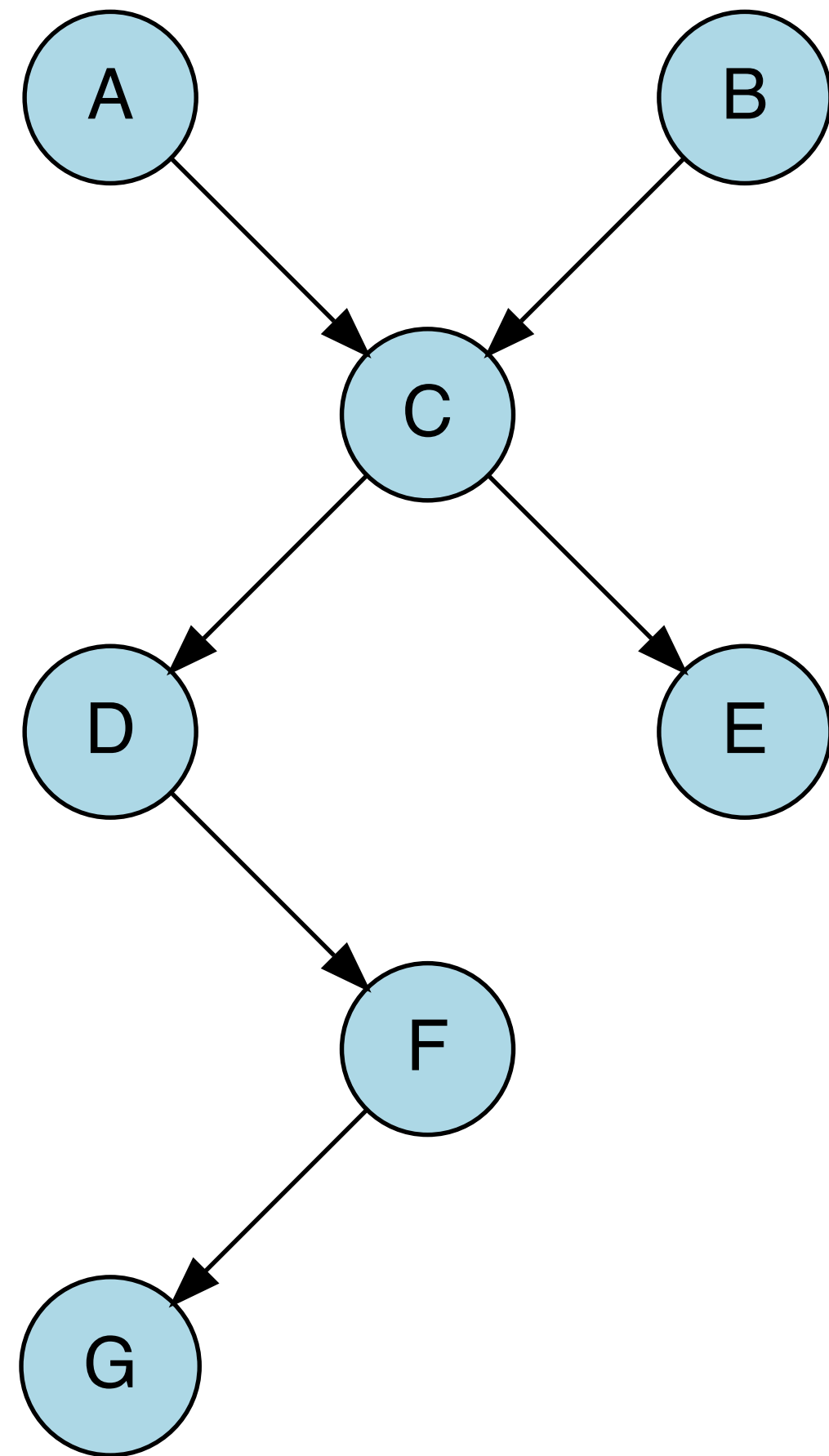
Recipe for independence

1. Draw the ancestral graph
2. "Moralize" the graph by "marrying" the parents
3. "Disorient" the graph by replacing arrows with edges
4. Delete the givens and their edges
5. Read the answer off the graph
 - if variables are **disconnected** they are independent
 - if variables are connected (have a path between them) they are not guaranteed to be independent

When should I control for variables?

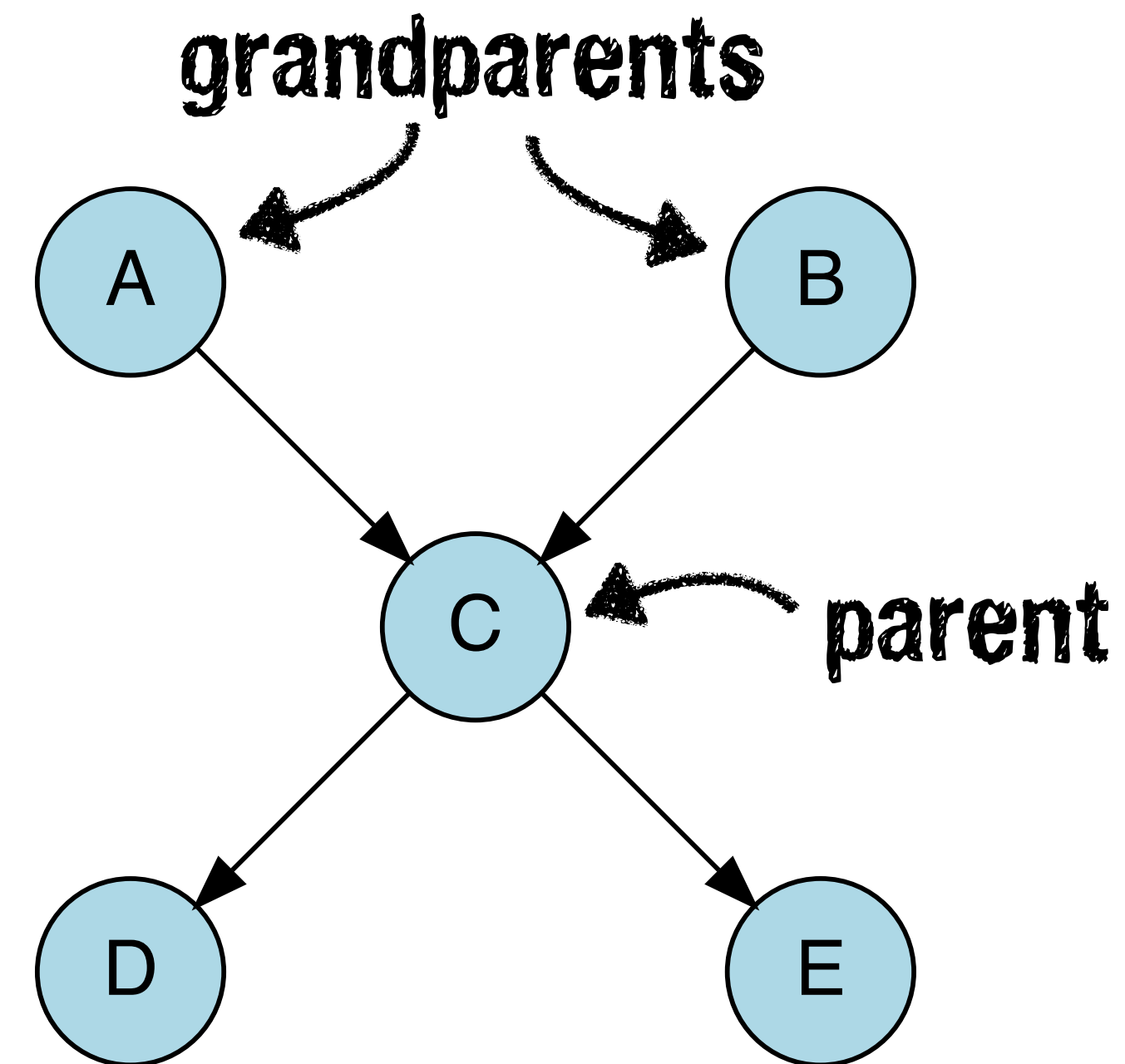
Are D and E independent?

$$p(D | E) = p(D) \text{ ?}$$



1. Draw the ancestral graph

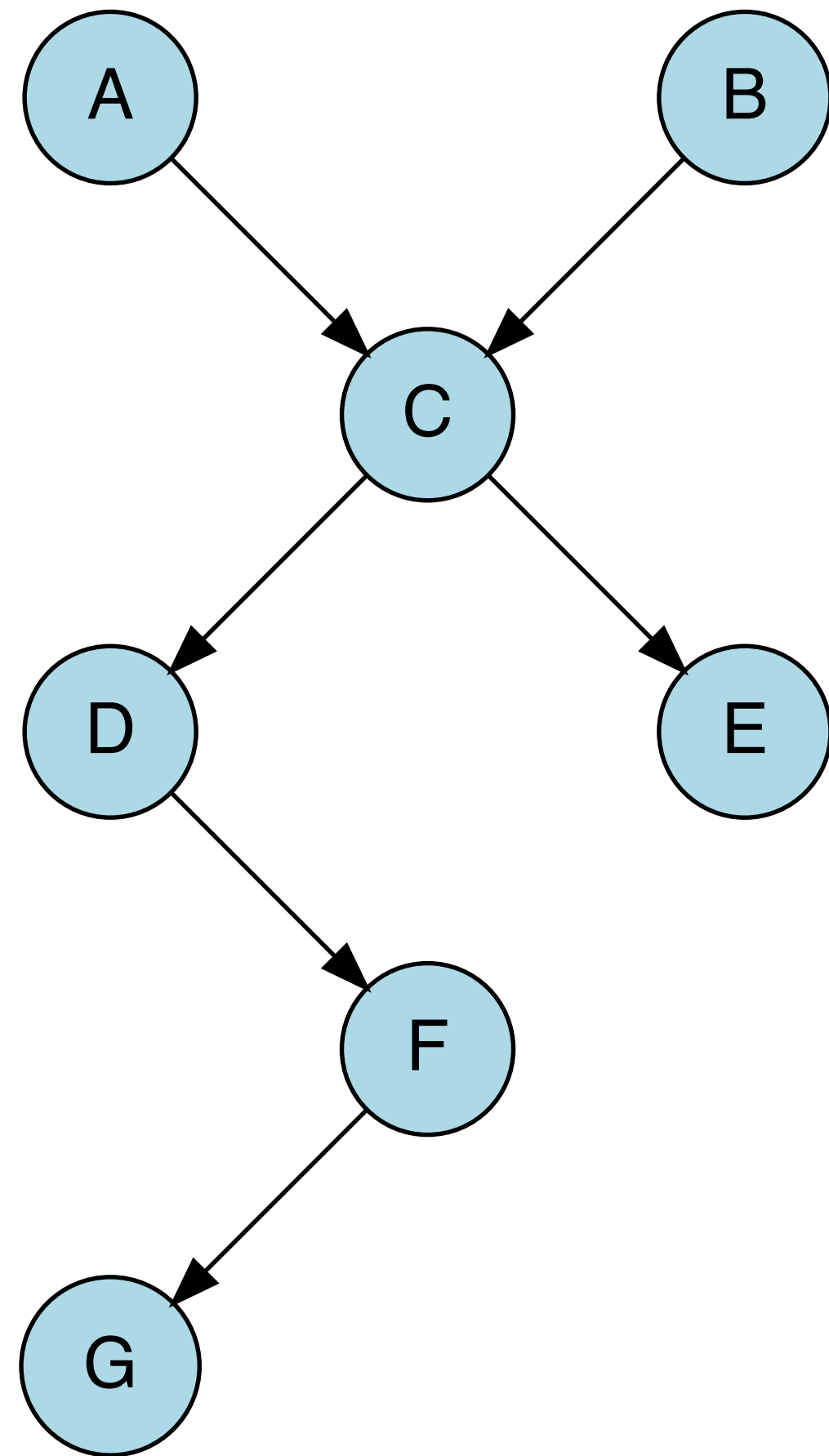
Construct the "ancestral graph" of all variables mentioned in the probability expression. This is a reduced version of the original net, consisting only of the variables mentioned and all of their ancestors (parents, parents' parents, etc.)



When should I control for variables?

Are D and E independent?

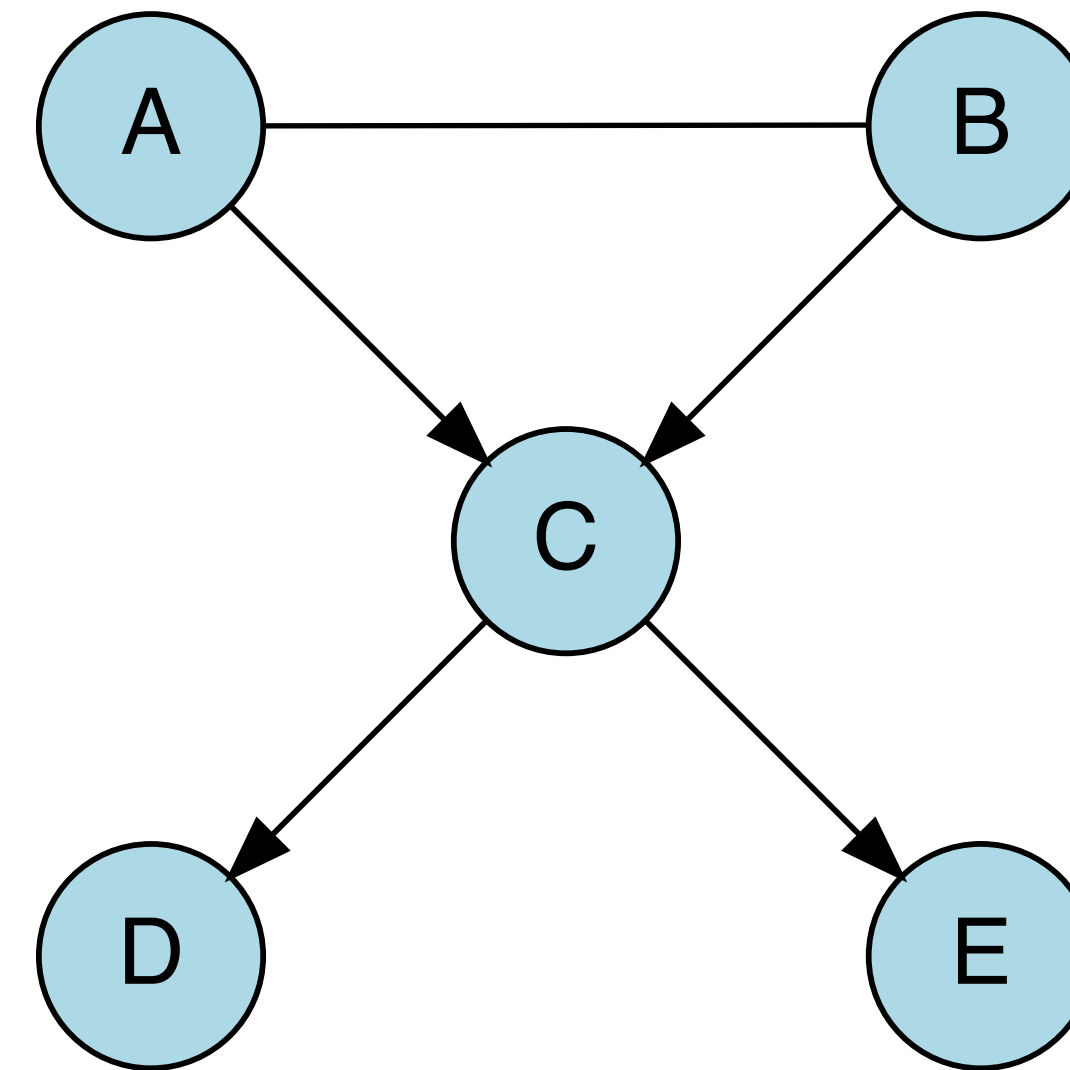
$$p(D | E) = p(D) \text{ ?}$$



2. "Moralize" the graph

let's get married!

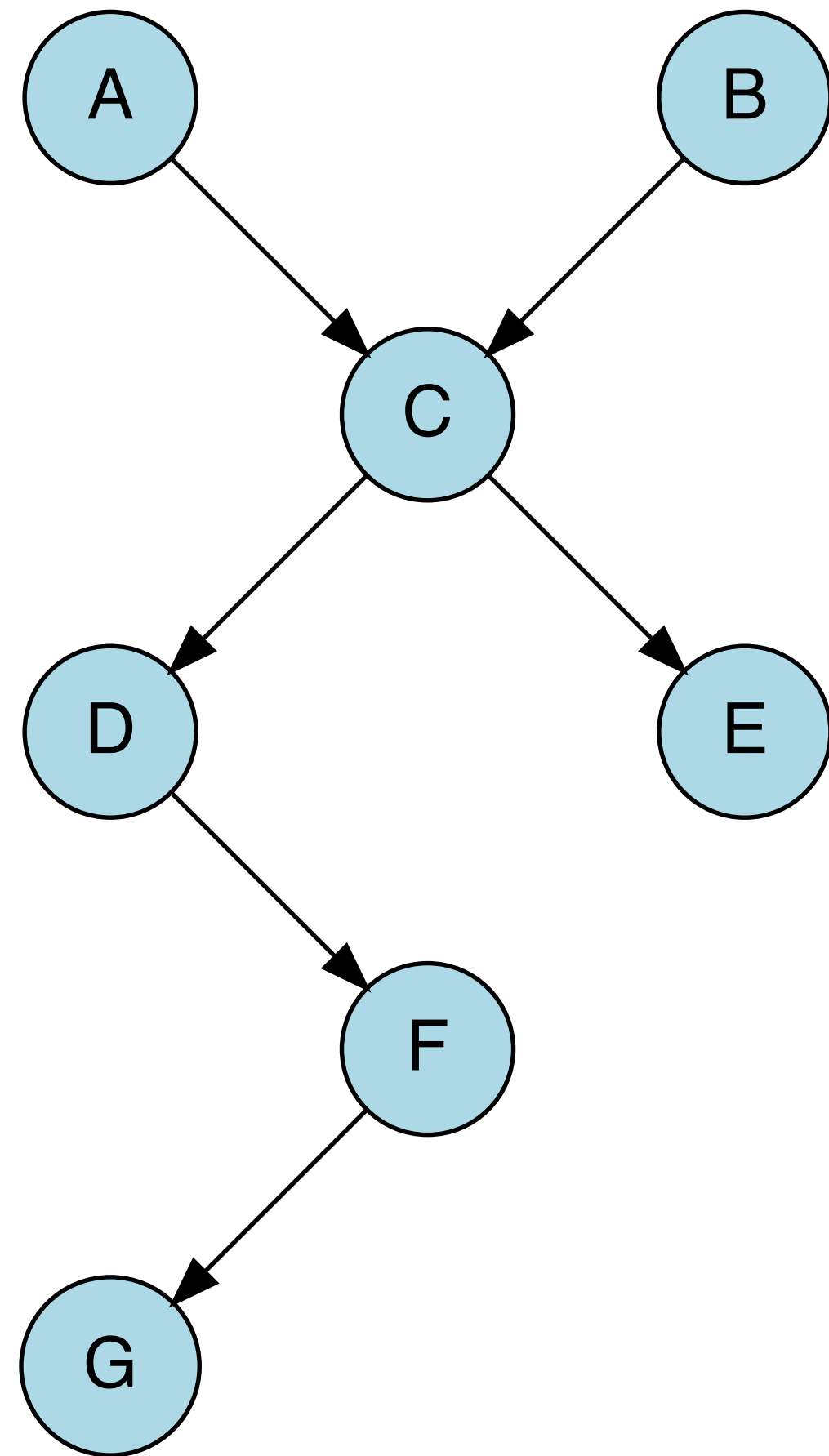
For each pair of variables with a common child, draw an undirected edge (line) between them. (If a variable has more than two parents, draw lines between every pair of parents.)



When should I control for variables?

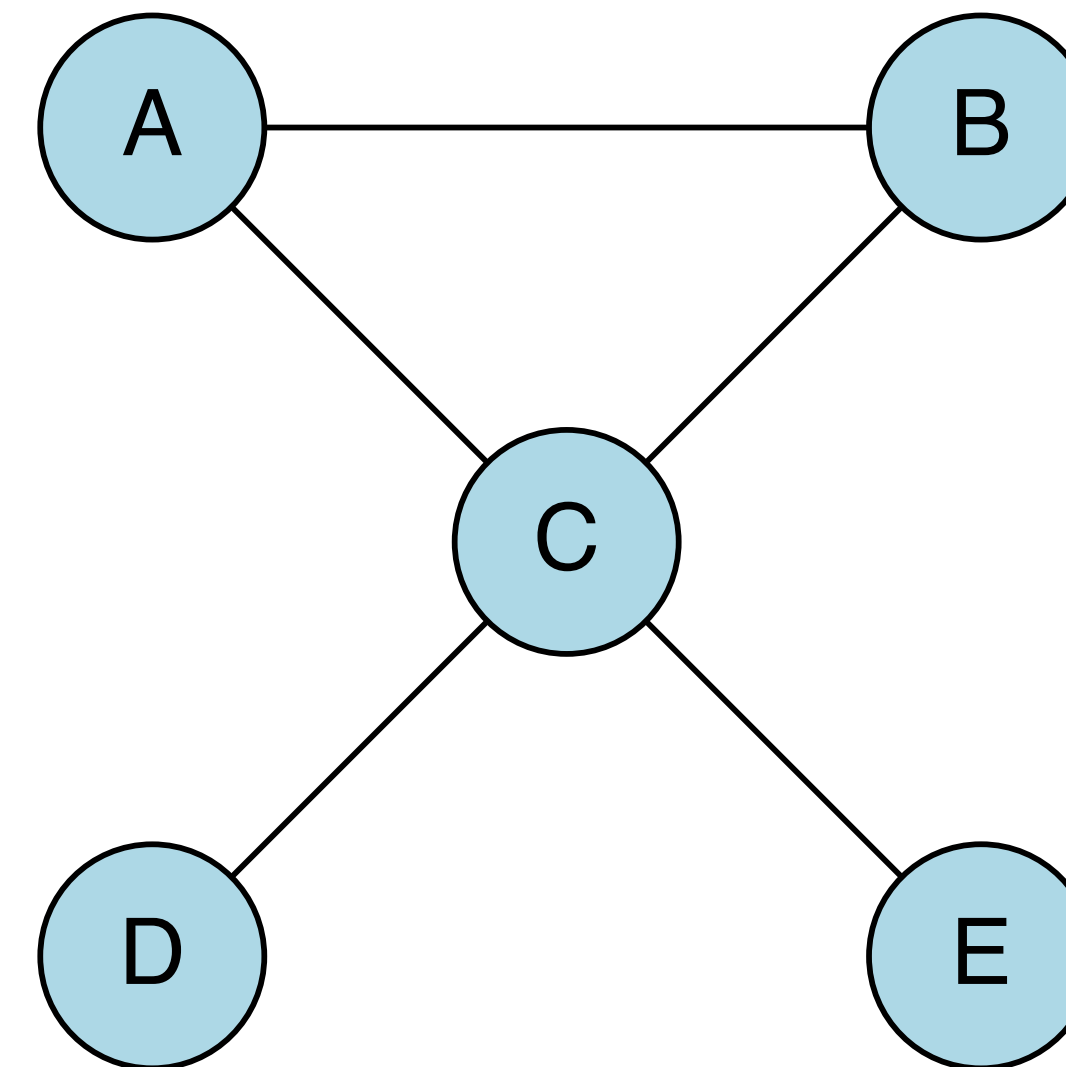
Are D and E independent?

$$p(D | E) = p(D) ?$$



3. "Disorient" the graph

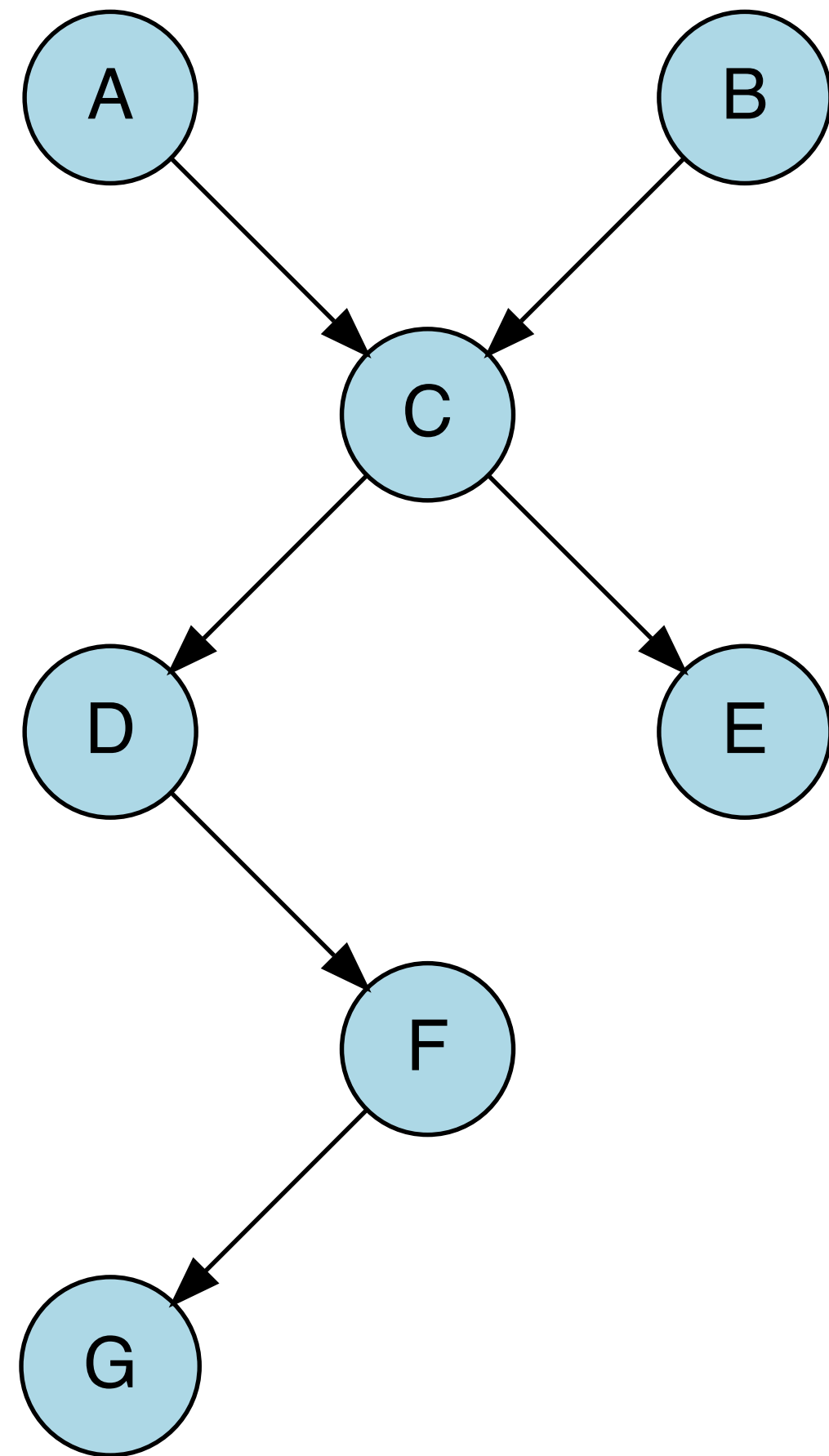
Replace arrows with lines



When should I control for variables?

Are D and E independent?

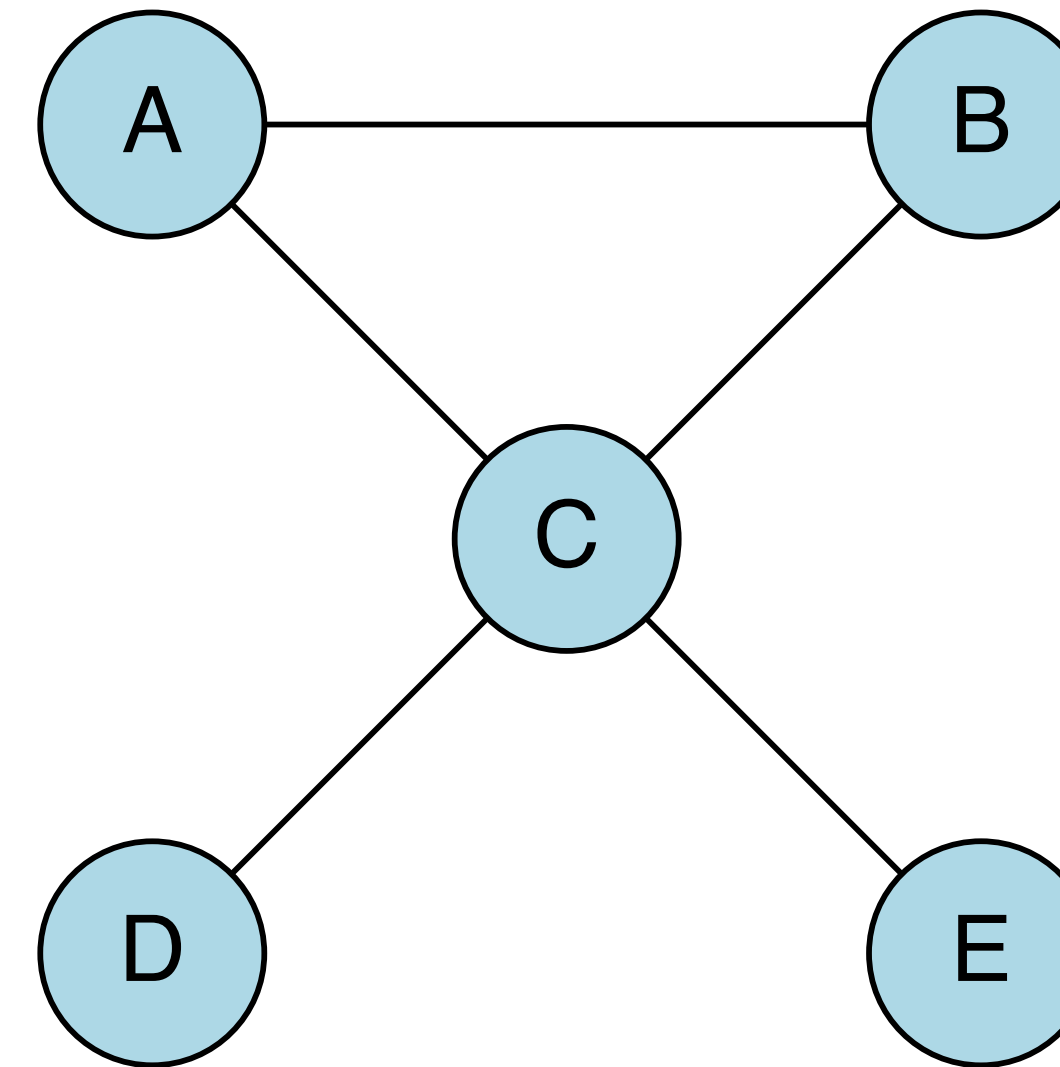
$$p(D | E) = p(D) \text{ ?}$$



4. Delete the givens

Remove the variables that we condition on, as well as their edges

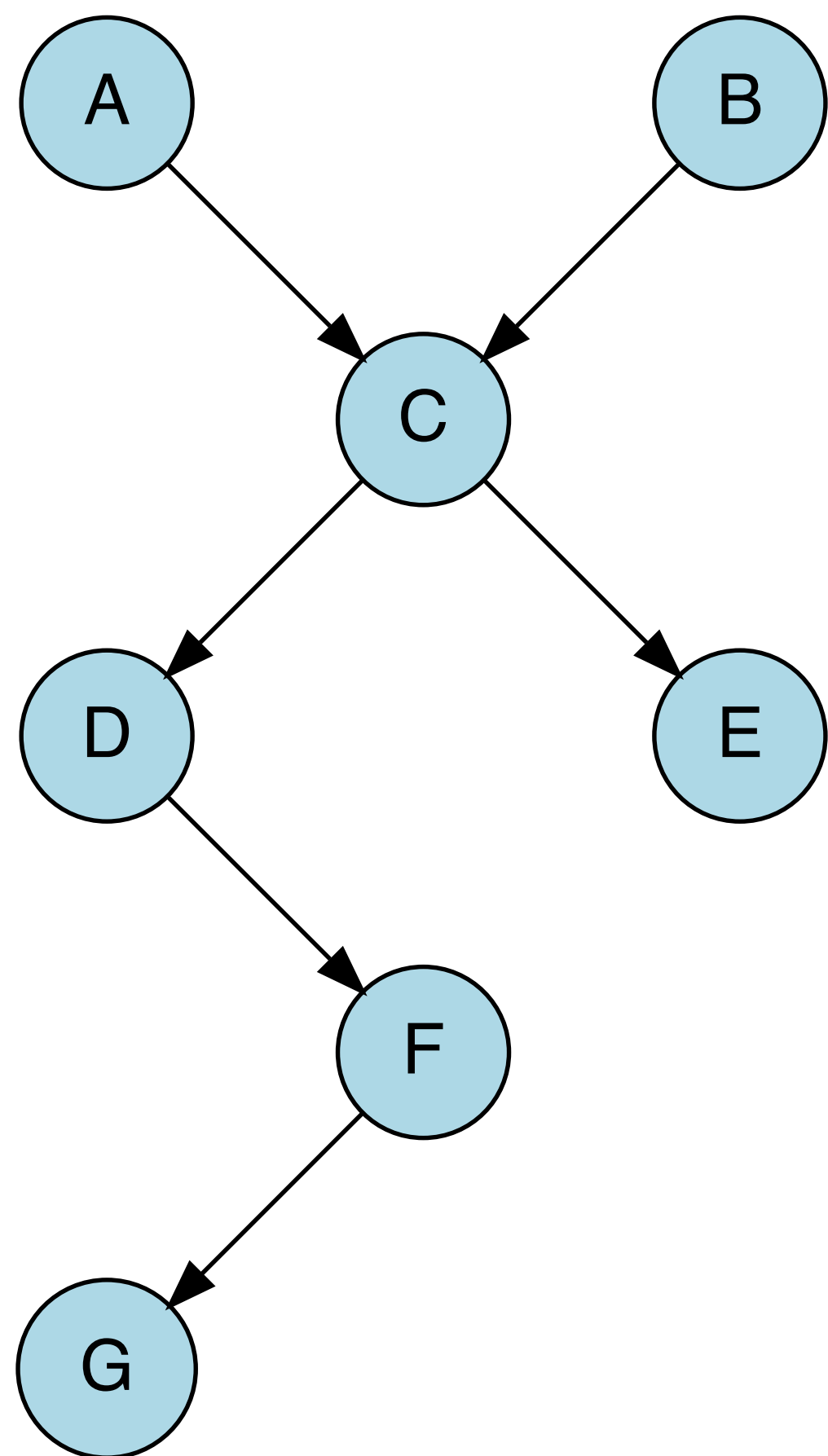
**we didn't condition on anything,
so there is nothing to delete**



When should I control for variables?

Are D and E independent?

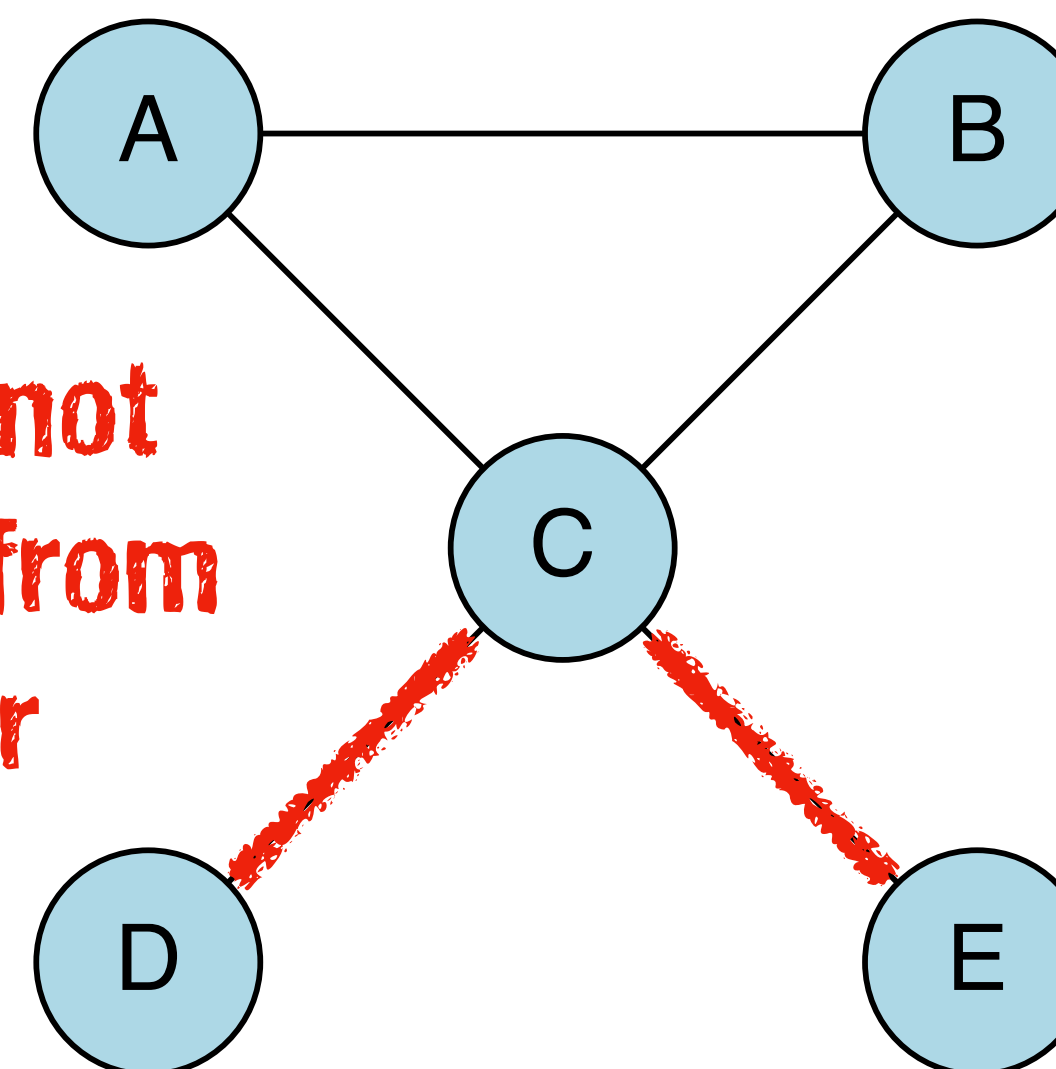
$$p(D | E) = p(D) \text{ ?}$$



5. Read answer off the graph

- if variables are **disconnected** they are independent
- if variables are connected (have a path between them) they are not guaranteed to be independent

D and E are not independent from each other

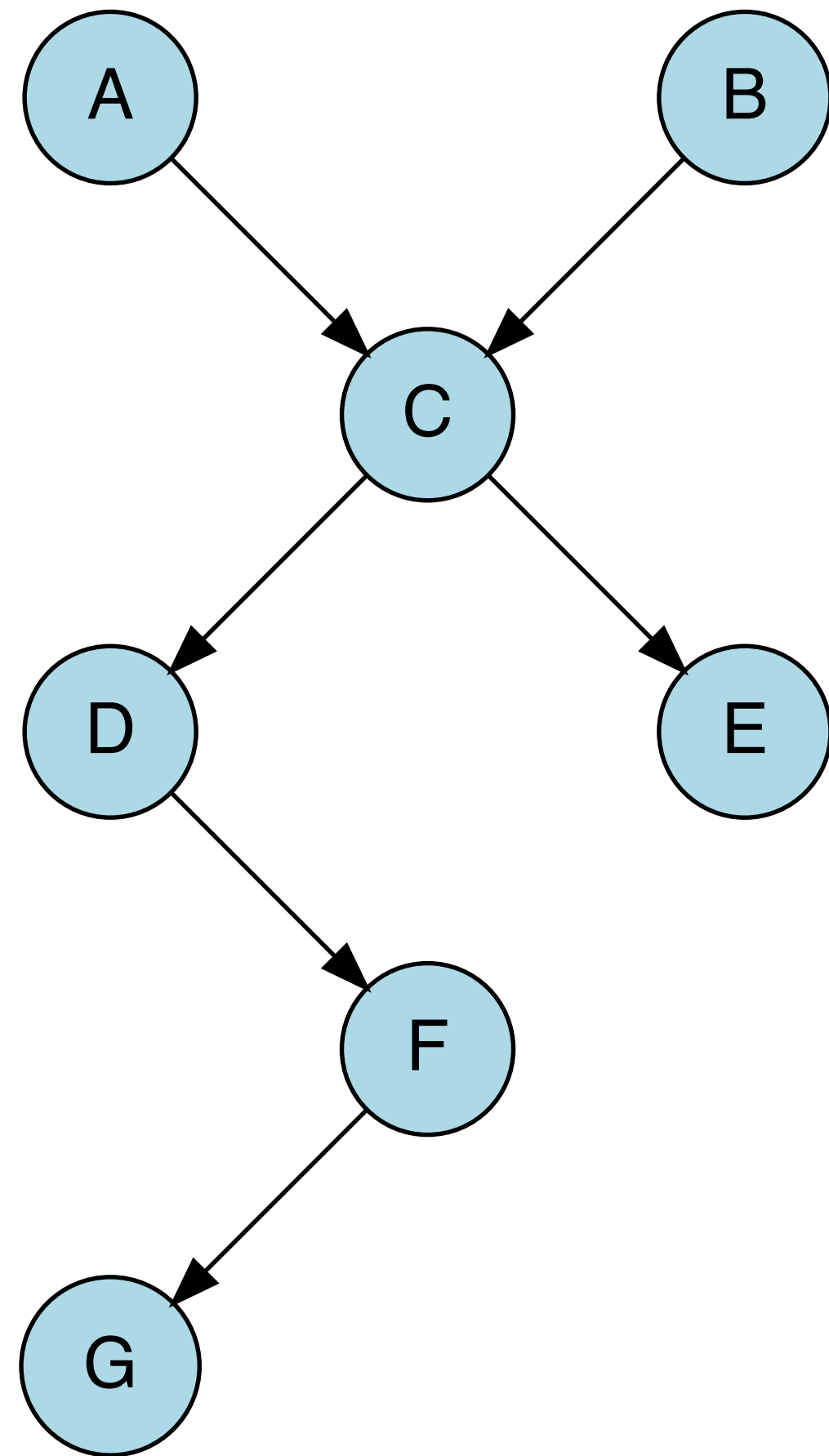


they are connected via at least one path

When should I control for variables?

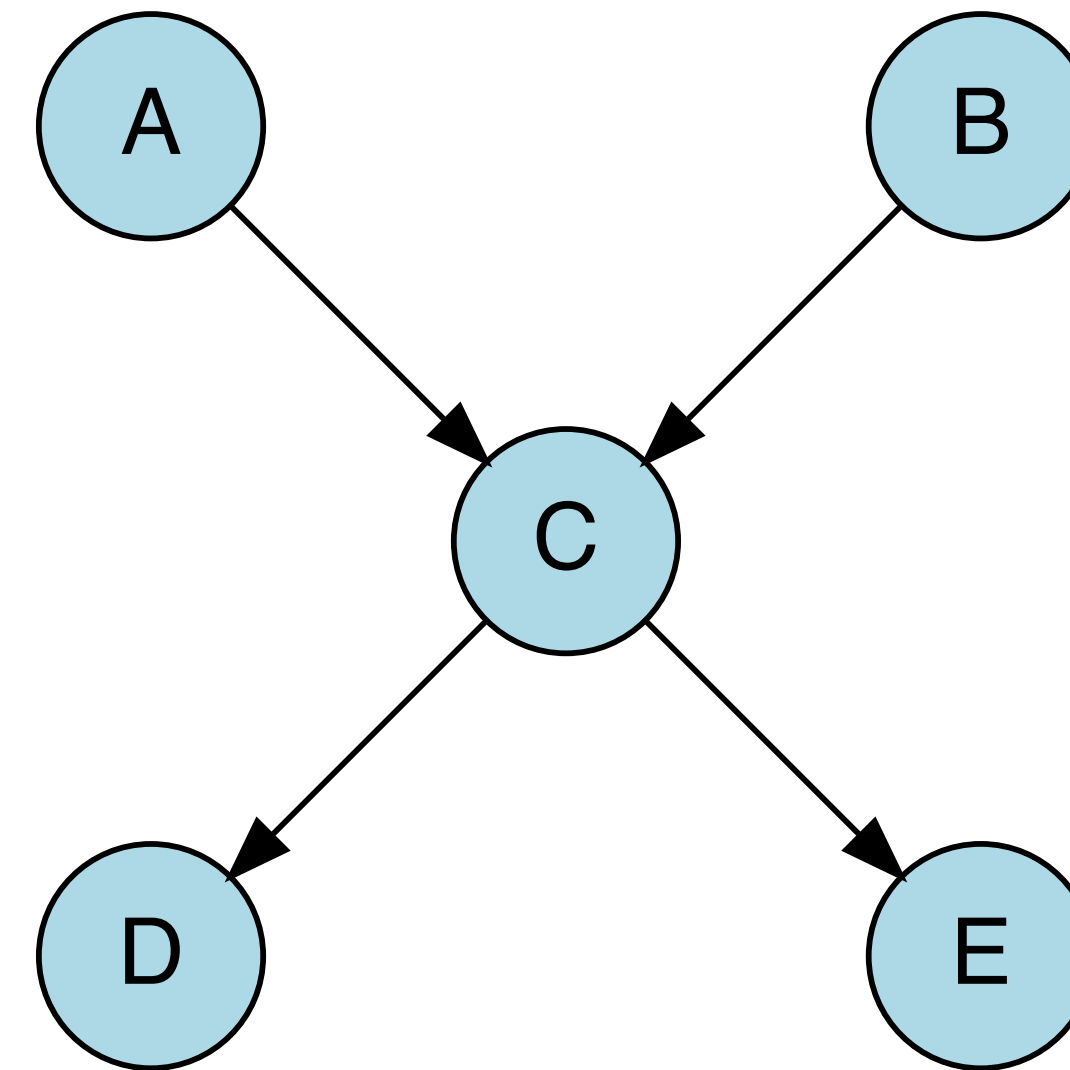
Are D and E independent, given C?

$$p(D | E, C) = p(D | C) ?$$



1. Draw the ancestral graph

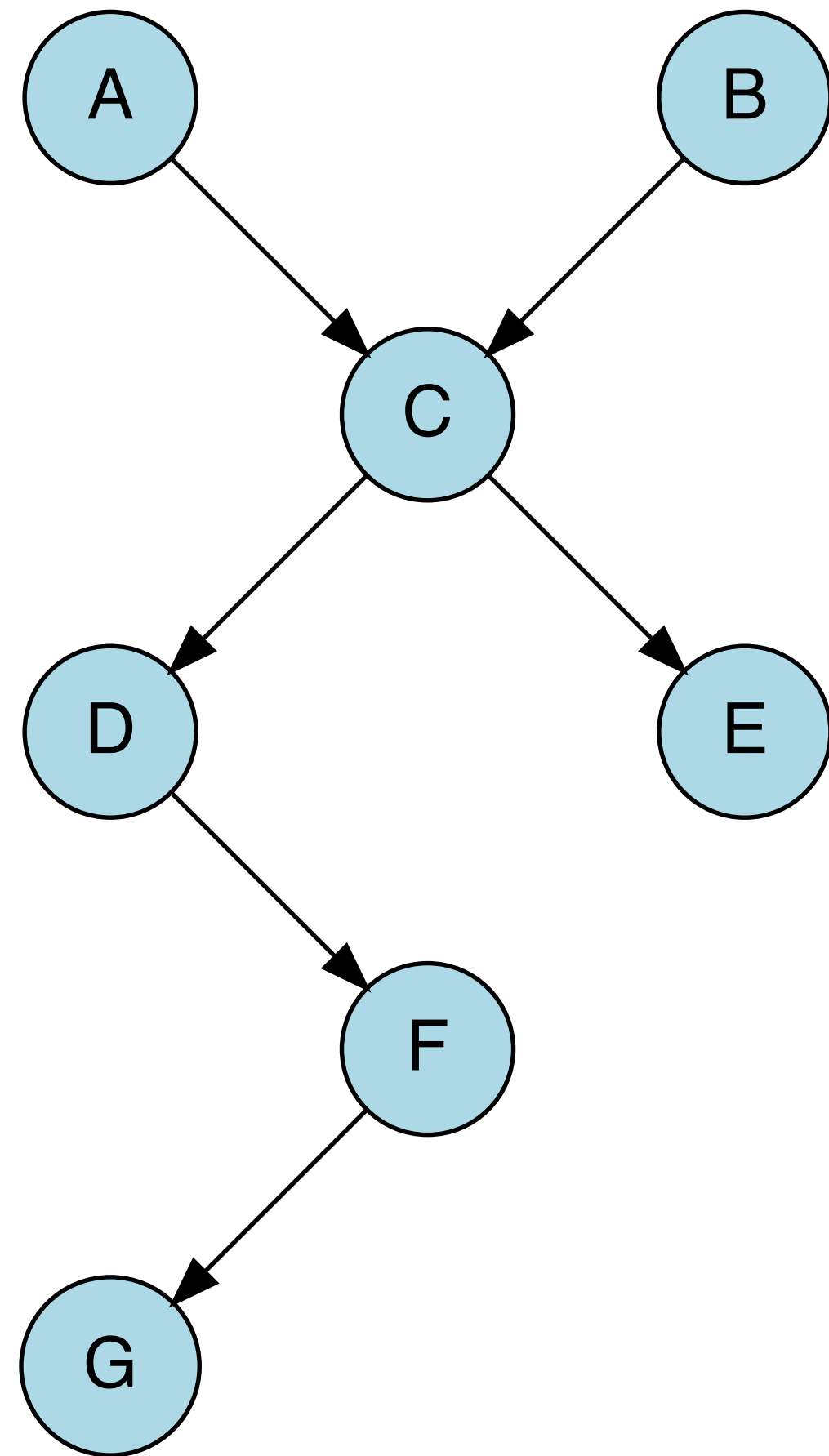
Construct the "ancestral graph" of all variables mentioned in the probability expression. This is a reduced version of the original net, consisting only of the variables mentioned and all of their ancestors (parents, parents' parents, etc.)



When should I control for variables?

Are D and E independent, given C?

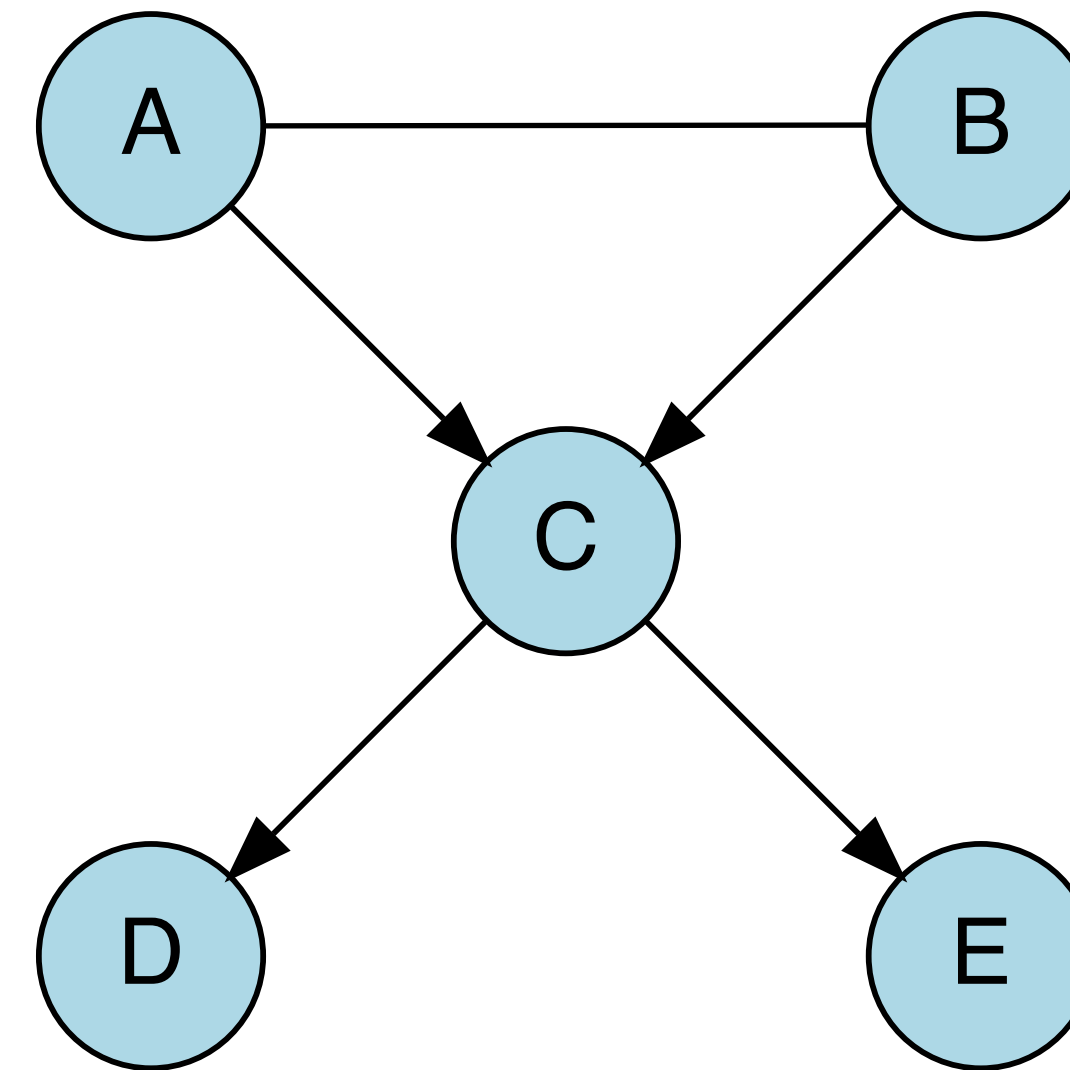
$$p(D | E, C) = p(D | C) ?$$



2. "Moralize" the graph

let's get married!

For each pair of variables with a common child, draw an undirected edge (line) between them. (If a variable has more than two parents, draw lines between every pair of parents.)

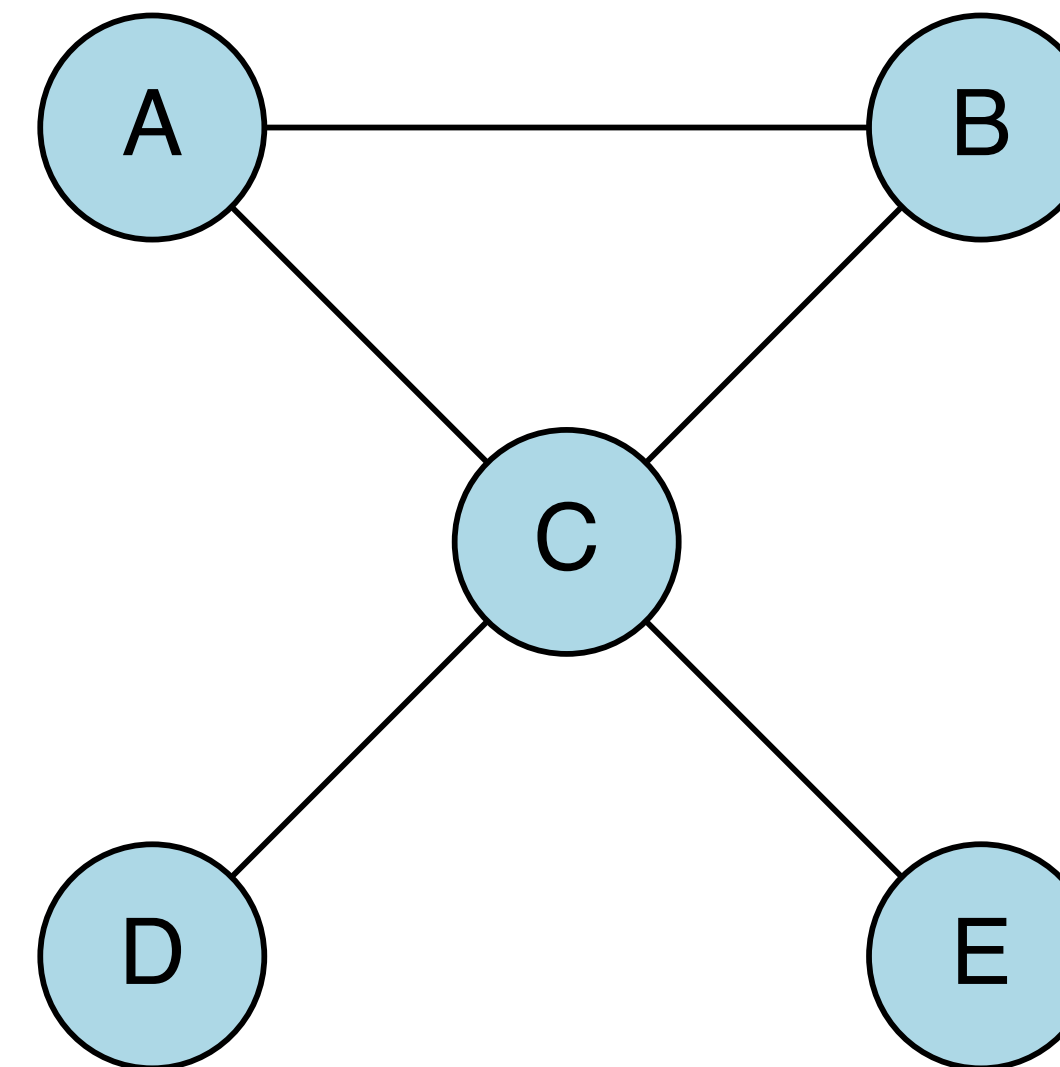
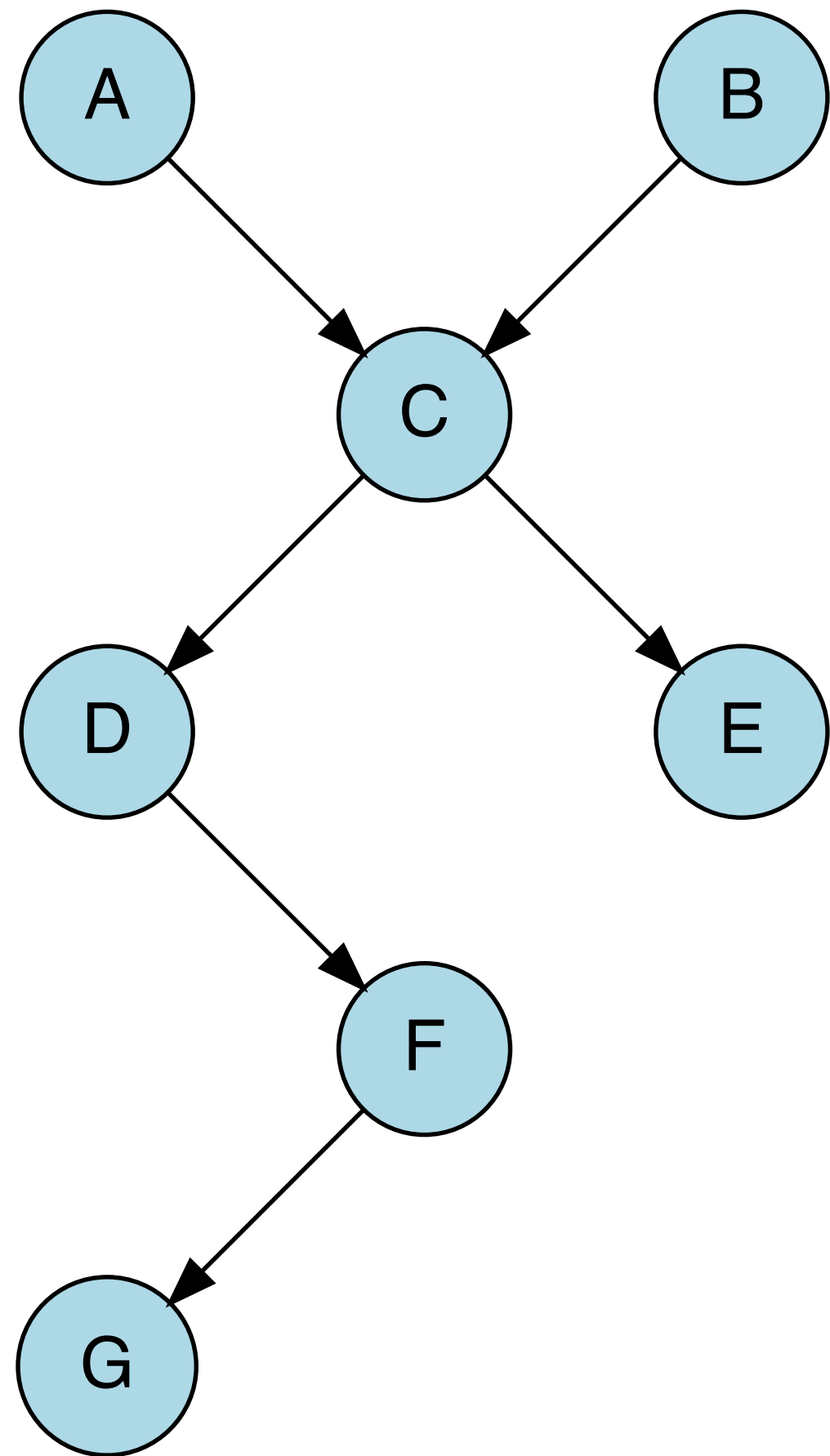


When should I control for variables?

Are D and E independent, given C? 3. "Disorient" the graph

$$p(D | E, C) = p(D | C) ?$$

Replace arrows with lines



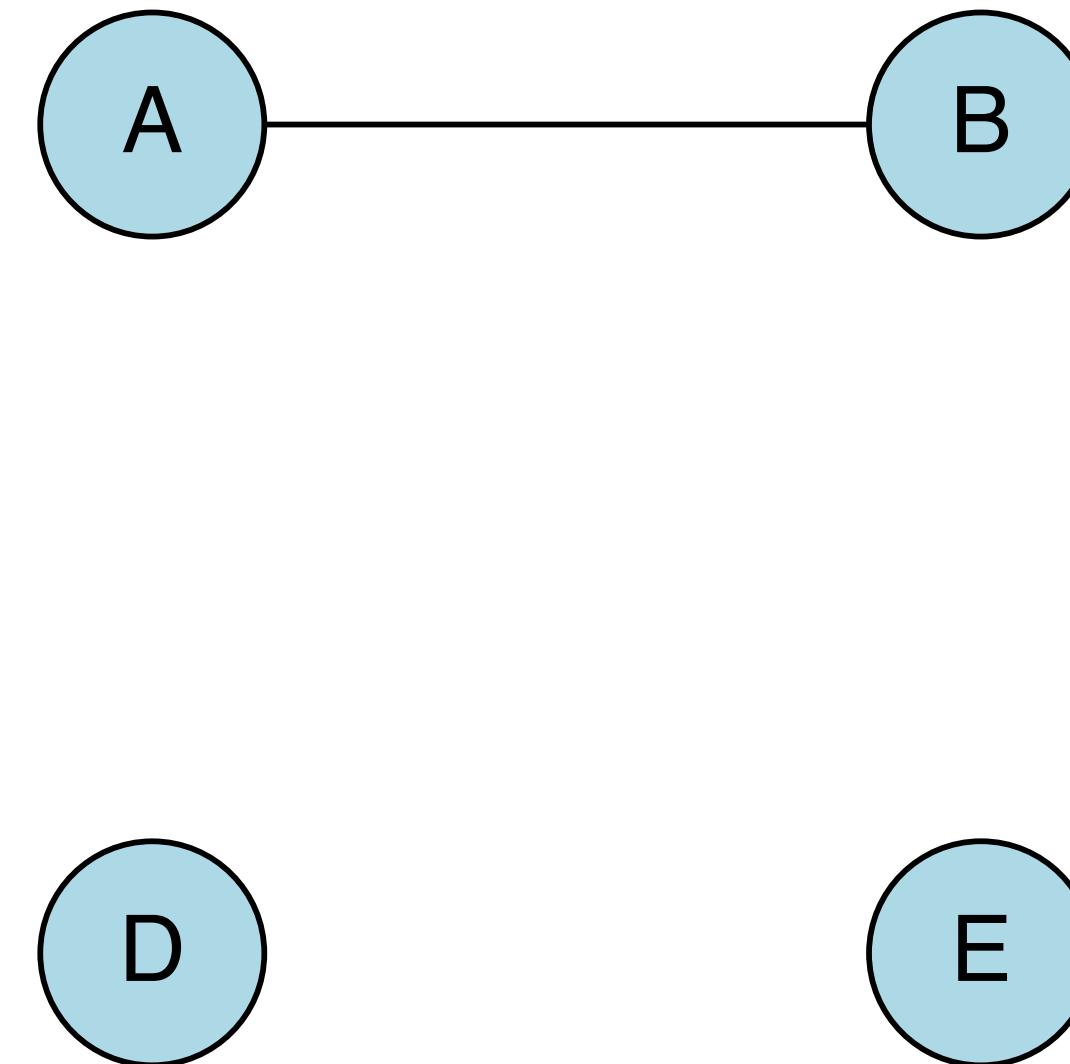
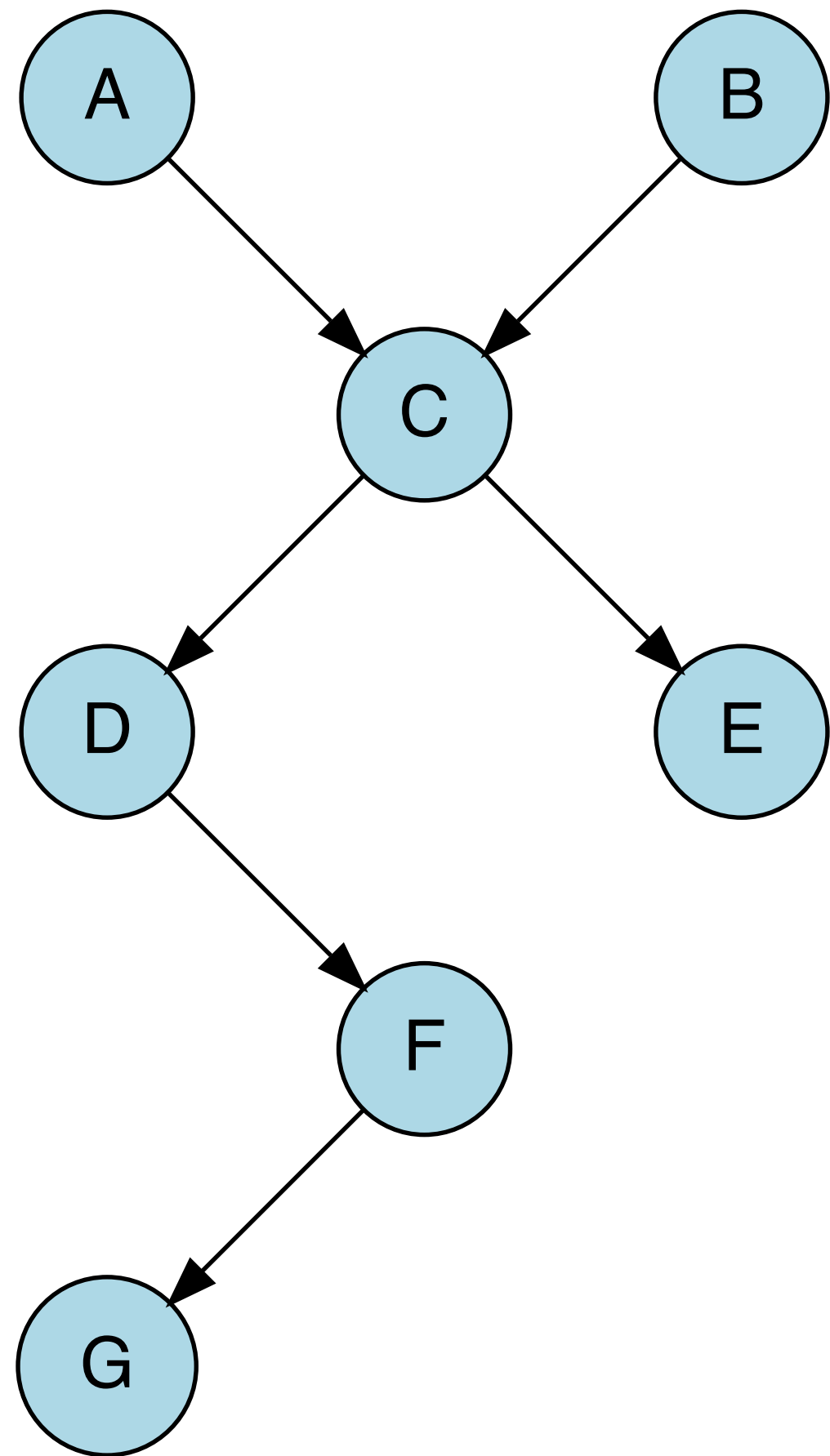
When should I control for variables?

Are D and E independent, given C? 4. Delete the givens

$$p(D | E, C) = p(D | C) \text{ ?}$$

Remove the variables that we condition on, as well as their edges

we conditioned on C!

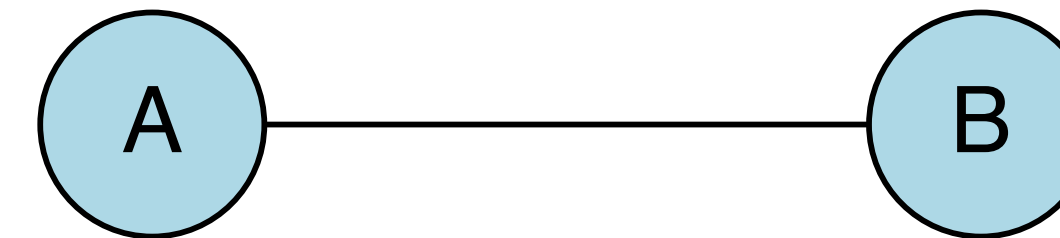
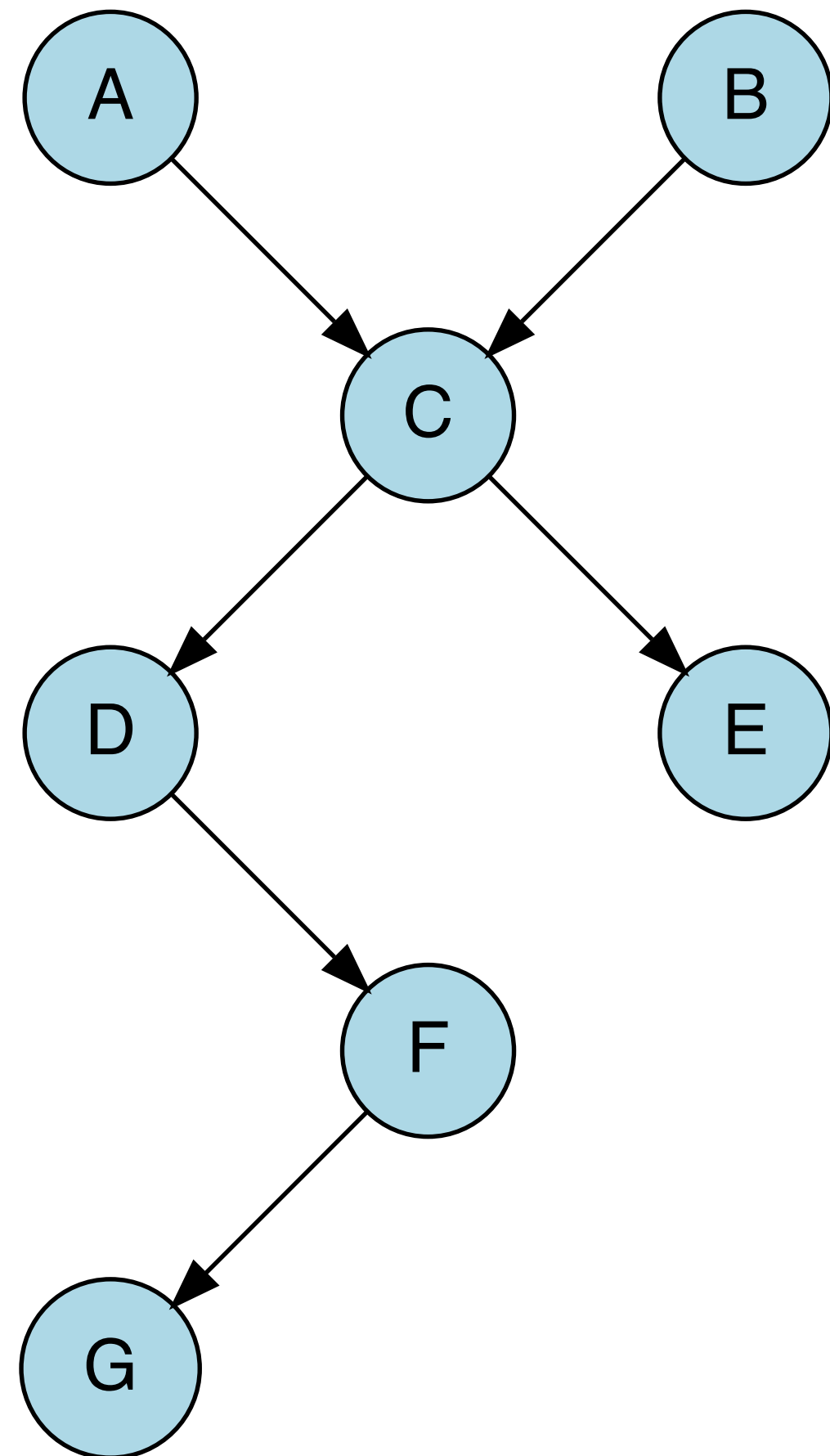


When should I control for variables?

Are D and E independent, given C? 5. Read answer off the graph

$$p(D | E, C) = p(D | C) ?$$

- if variables are **disconnected** they are independent
- if variables are connected (have a path between them) they are not guaranteed to be independent



D and E are independent from each other conditioned on C



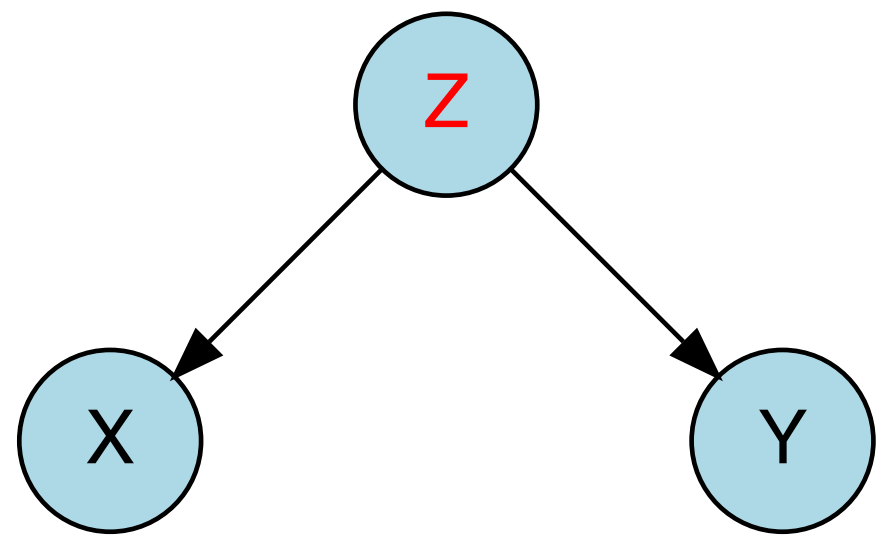
they aren't connected via a path

So what?

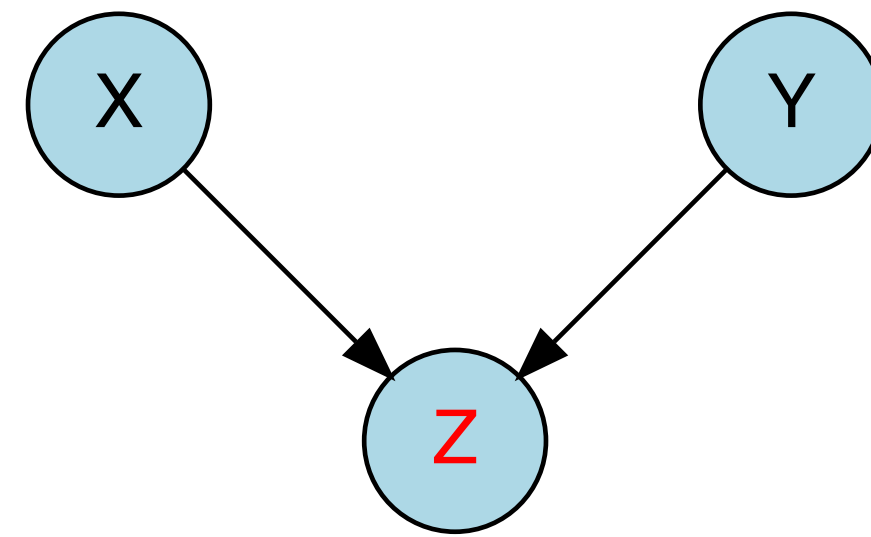


Patterns of inference

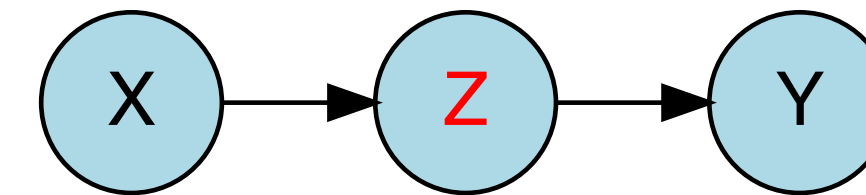
We want to estimate the (causal) relationship between X and Y



common cause



common effect



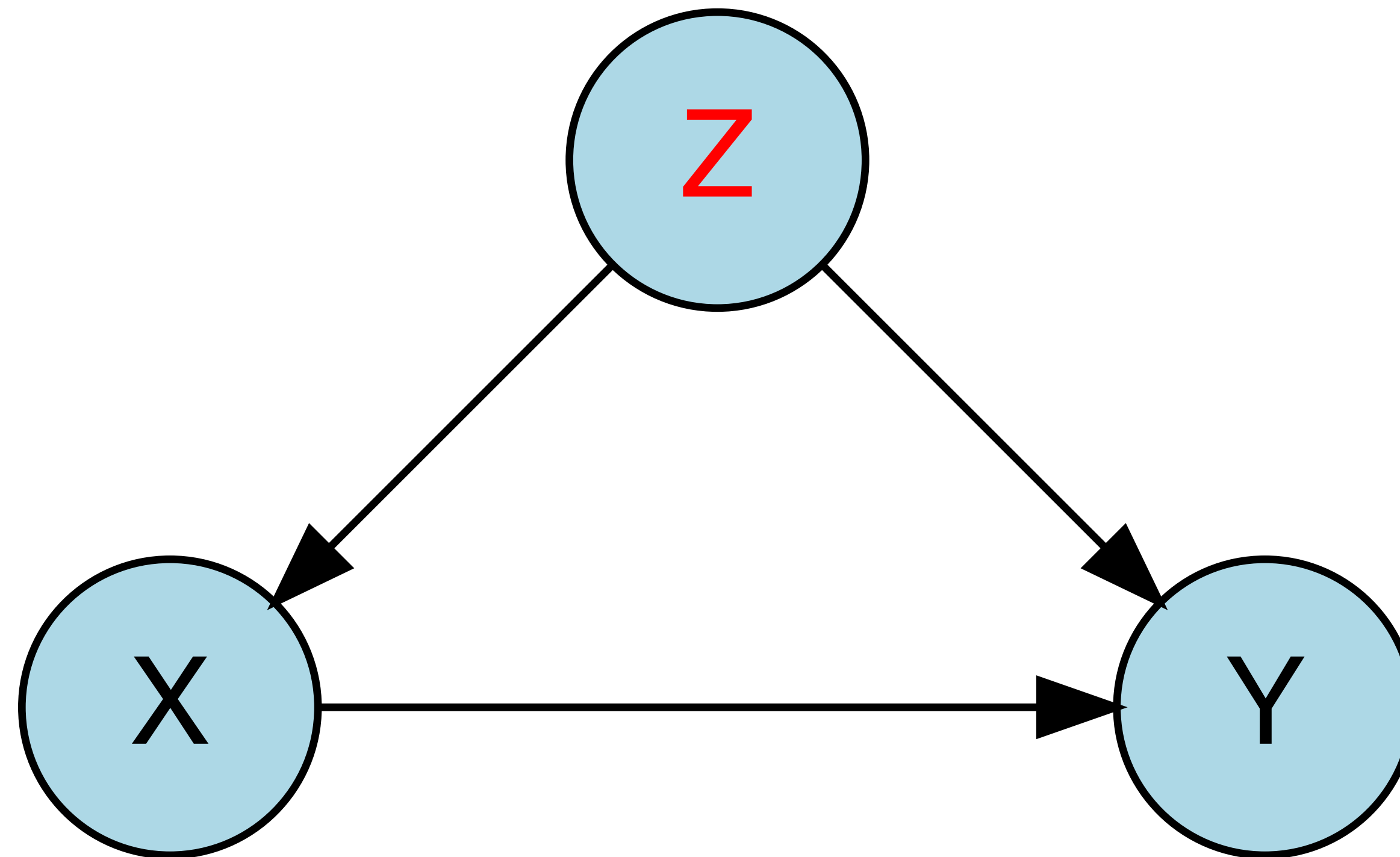
causal chain

by controlling for Z we hope to get a better estimate of the relationship between X and Y

d -separation helps us tell apart **good controls** from **bad controls**

When should I control for variables?

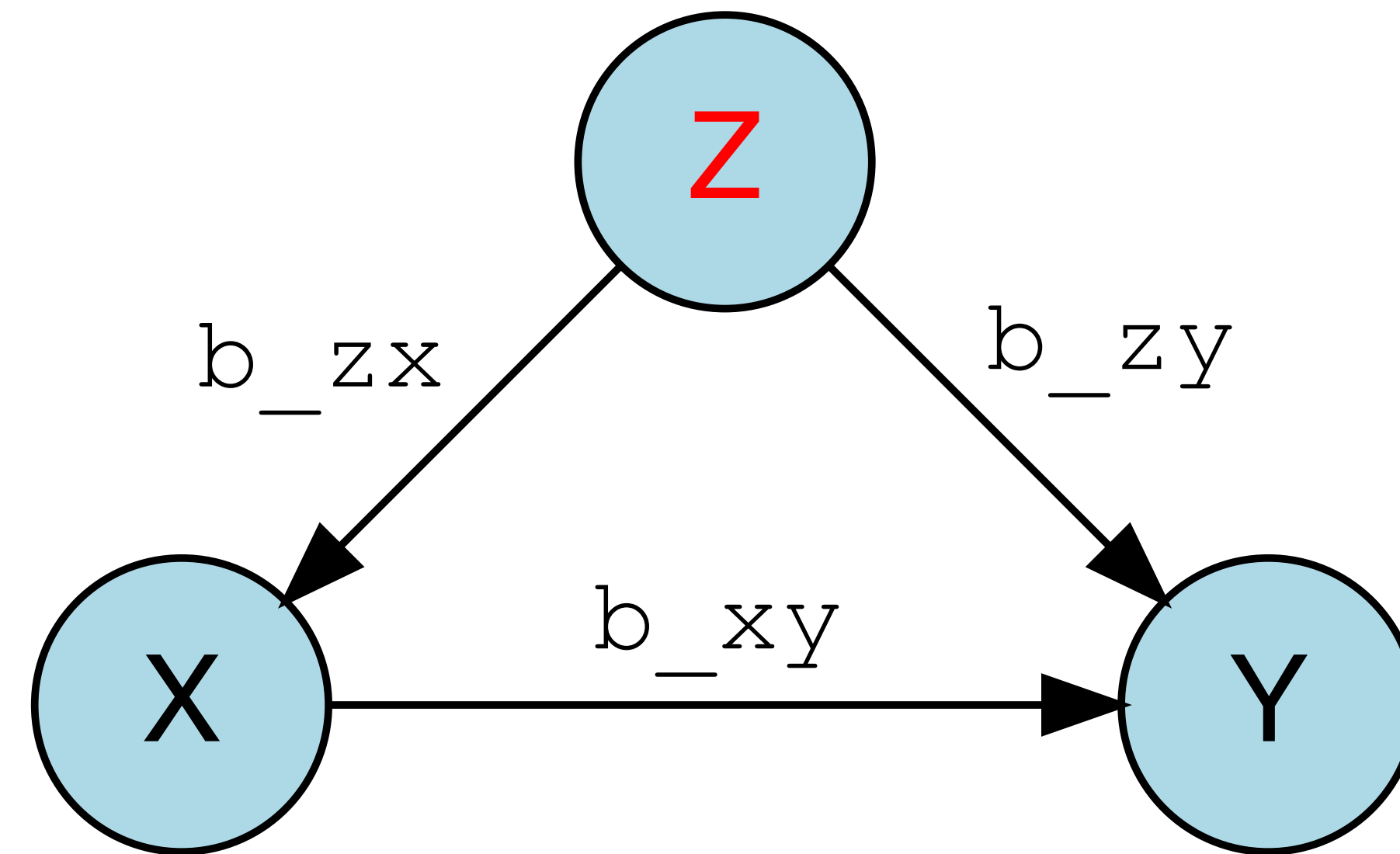
I want to estimate the effect that X has on Y



Is Z a **good** or a **bad** control here?

When should I control for variables?

```
1 set.seed(1)
2
3 n = 1000
4 b_zx = 2
5 b_xy = 2
6 b_zy = 2
7 sd = 1
8
9 df = tibble(z = rnorm(n = n, sd = sd),
10             x = b_zx * z + rnorm(n = n, sd = sd),
11             y = b_zy * z + b_xy * x + rnorm(n = n, sd = sd))
```



overestimating X's
effect on Y

$$Y = b_0 + b_1 \cdot X + e$$

```
1 # without control
2 lm(formula = y ~ 1 + x,
3     data = df) %>%
4 summary()
```

```
Call:
lm(formula = y ~ x, data = df)

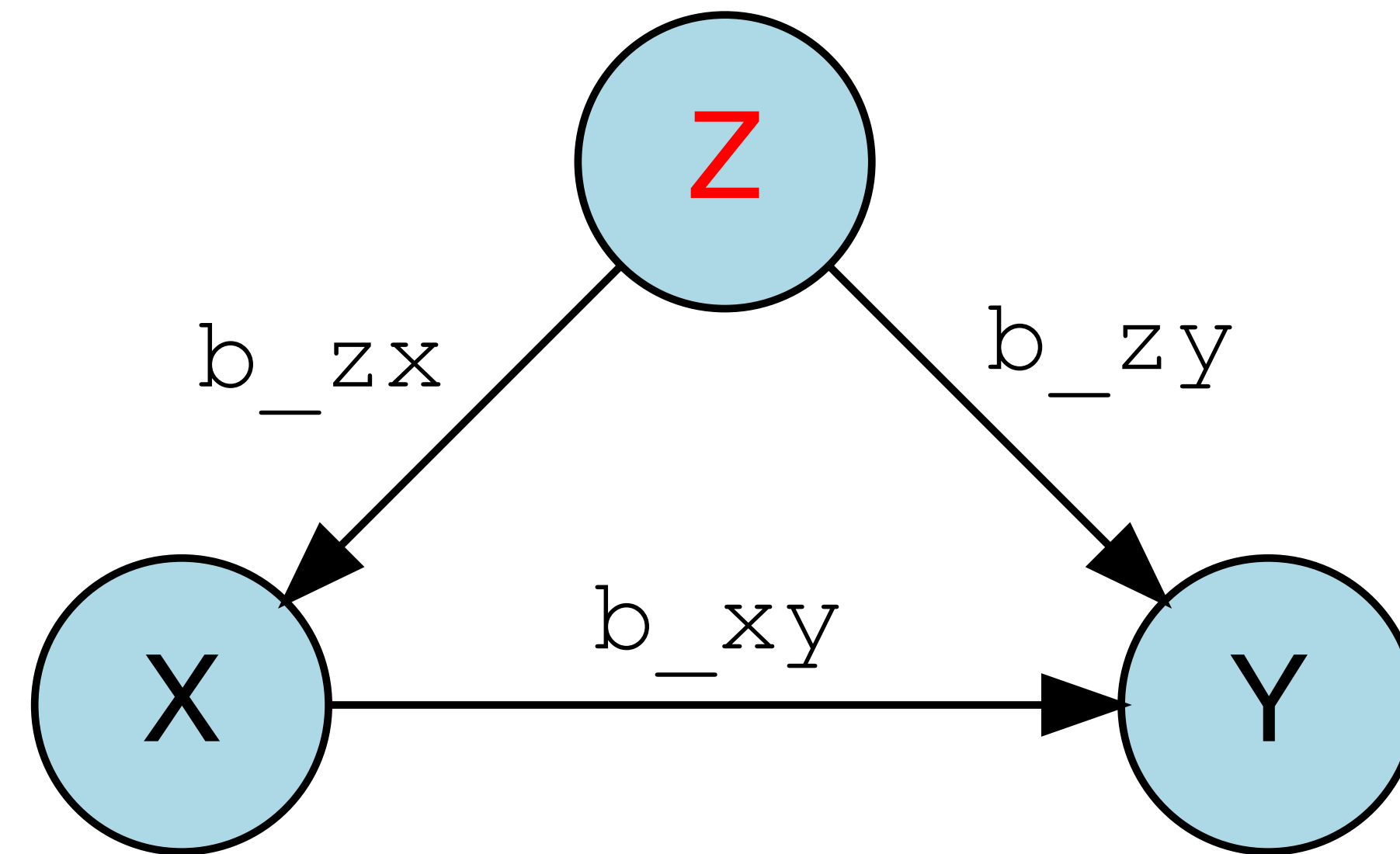
Residuals:
    Min       1Q   Median       3Q      Max
-4.6011 -0.9270 -0.0506  0.9711  4.0454

Coefficients:
(Intercept) 0.02449 0.04389 0.558 0.577
x          2.82092 0.01890 149.225 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.388 on 998 degrees of freedom
Multiple R-squared:  0.9571,    Adjusted R-squared:  0.9571 
F-statistic: 2.227e+04 on 1 and 998 DF,  p-value: < 2.2e-16
```

When should I control for variables?

```
1 set.seed(1)
2
3 n = 1000
4 b_zx = 2
5 b_xy = 2
6 b_zy = 2
7 sd = 1
8
9 df = tibble(z = rnorm(n = n, sd = sd),
10             x = b_zx * z + rnorm(n = n, sd = sd),
11             y = b_zy * z + b_xy * x + rnorm(n = n, sd = sd))
```



**accurate estimate
of X's effect on Y**

$$Y = b_0 + b_1 \cdot X + b_2 \cdot Z + e$$

```
1 # with control
2 lm(formula = y ~ 1 + x + z,
3     data = df) %>%
4 summary()
```

```
Call:
lm(formula = y ~ x + z, data = df)

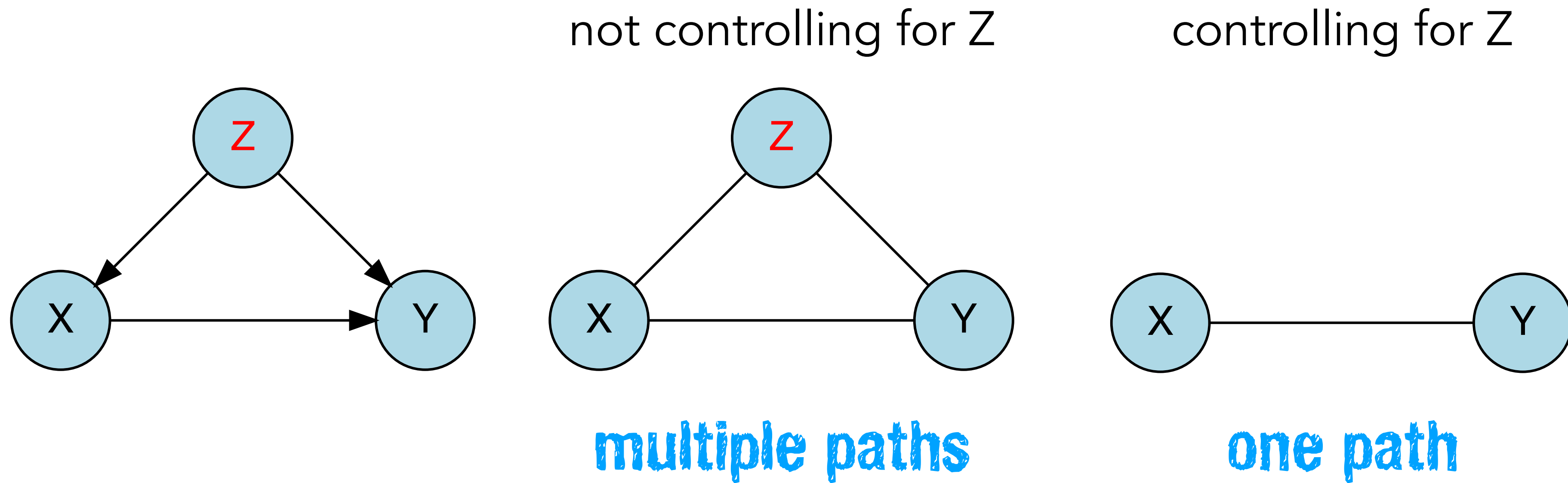
Residuals:
    Min       1Q   Median       3Q      Max
-3.6151 -0.6564 -0.0223  0.6815  2.8132

Coefficients:
(Intercept)  0.01624  0.03260  0.498  0.618
x           2.02202  0.03135 64.489  <2e-16 ***
z           2.00501  0.07036 28.497  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.031 on 997 degrees of freedom
Multiple R-squared:  0.9764,    Adjusted R-squared:  0.9763
F-statistic: 2.059e+04 on 2 and 997 DF,  p-value: < 2.2e-16
```


When should I control for variables?

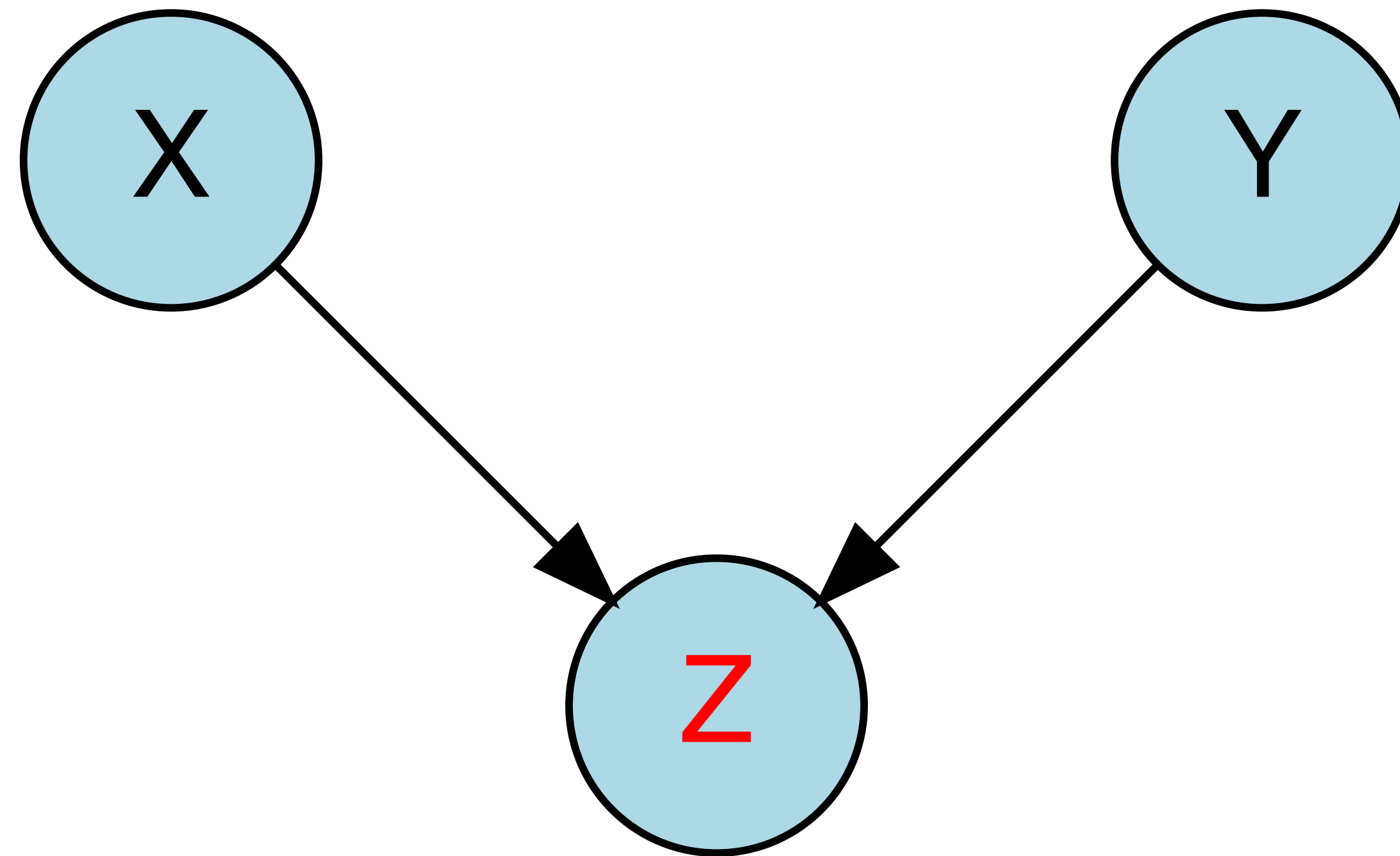
I want to estimate the effect that X has on Y



Z is a **good** control here!

When should I control for variables?

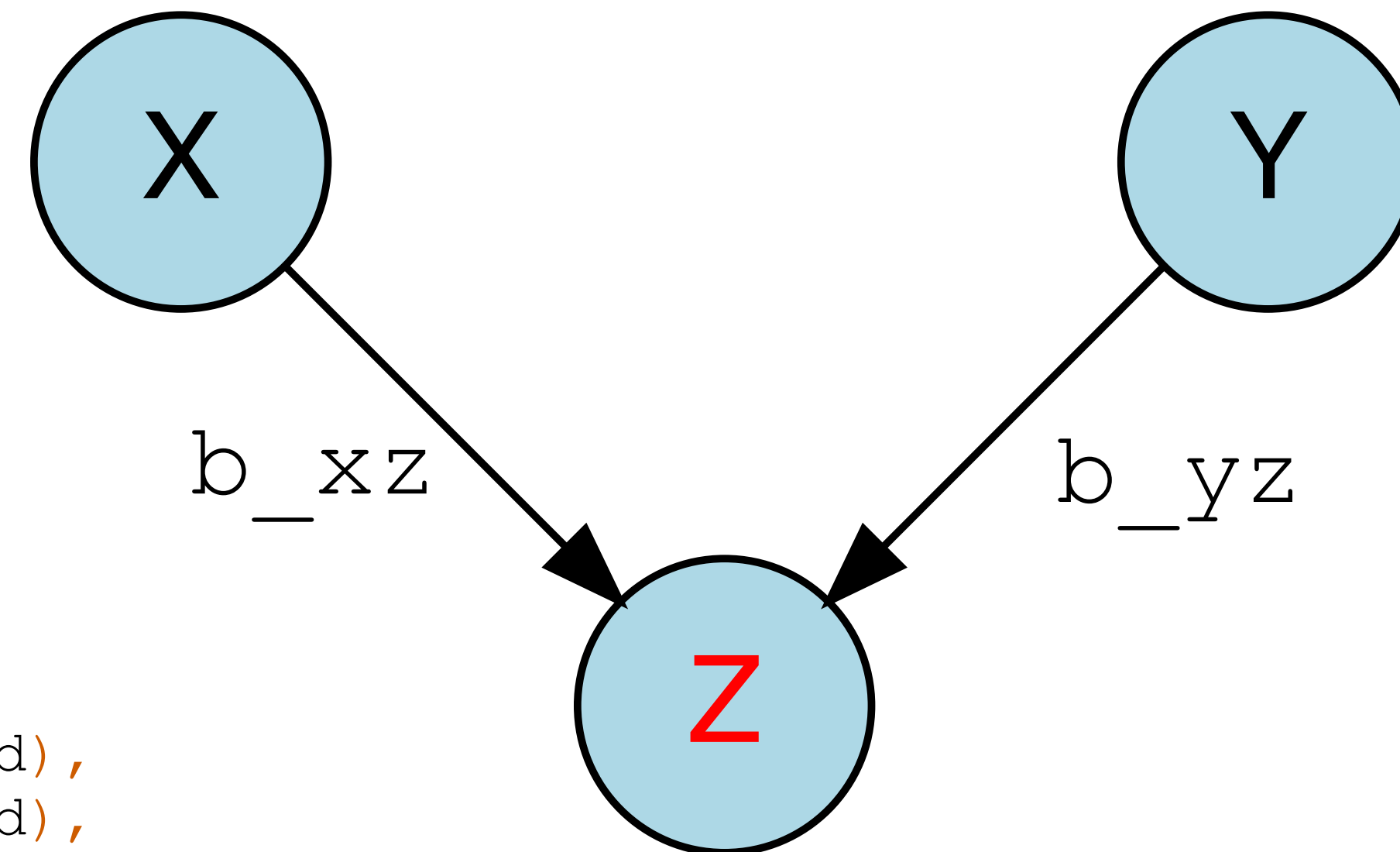
I want to estimate the effect that X has on Y



Is Z a **good** or a **bad** control here?

When should I control for variables?

```
1 set.seed(1)
2
3 n = 1000
4 b_xz = 2
5 b_yz = 2
6 sd = 1
7
8 df = tibble(x = rnorm(n = n, sd = sd),
9               y = rnorm(n = n, sd = sd),
10              z = x * b_xz + y * b_yz + rnorm(n = n, sd = sd))
```



$$Y = b_0 + b_1 \cdot X + e$$

```
1 # without control
2 lm(formula = y ~ 1 + x,
3     data = df) %>%
4 summary()
```

```
Call:
lm(formula = y ~ x, data = df)

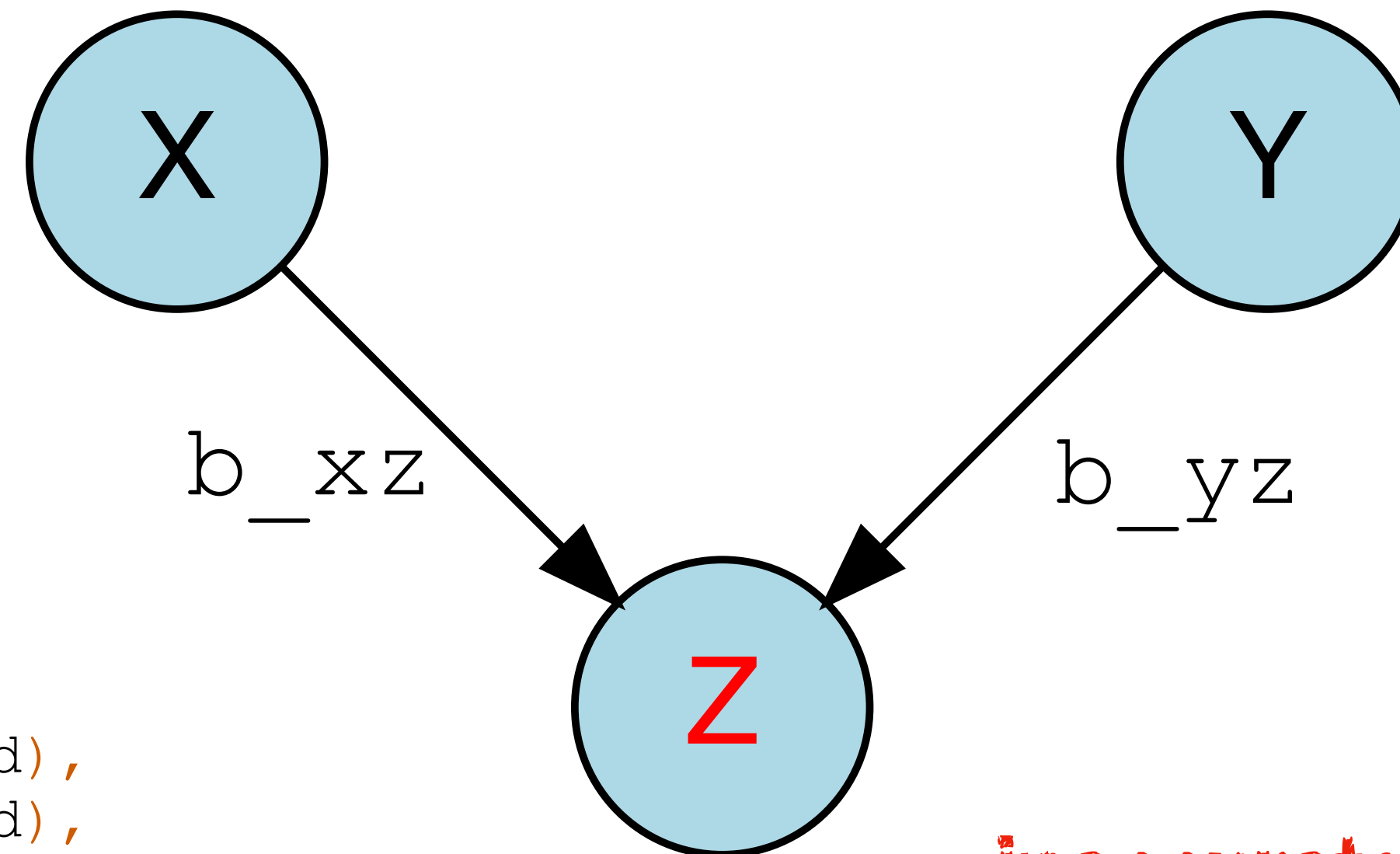
Residuals:
    Min       1Q   Median       3Q      Max
-3.2484 -0.6720 -0.0138  0.7554  3.6443

Coefficients:
(Intercept) -0.016187
x            0.006433

Residual standard error: 1.04 on 998 degrees of freedom
Multiple R-squared:  4.098e-05,    Adjusted R-squared:  -0.000961
F-statistic: 0.0409 on 1 and 998 DF,  p-value: 0.8398
```

accurate estimate
of X's effect on Y

When should I control for variables?



```
1 set.seed(1)
2
3 n = 1000
4 b_xz = 2
5 b_yz = 2
6 sd = 1
7
8 df = tibble(x = rnorm(n = n, sd = sd),
9              y = rnorm(n = n, sd = sd),
10             z = x * b_xz + y * b_yz + rnorm(n = n, sd = sd))
```

**inaccurate
estimate of X's
effect on Y**

$$Y = b_0 + b_1 \cdot X + b_2 \cdot Z + e$$

```
1 # with control
2 lm(formula = y ~ 1 + x + z,
3    data = df) %>%
4 summary()
```

```
Call:
lm(formula = y ~ x + z, data = df)

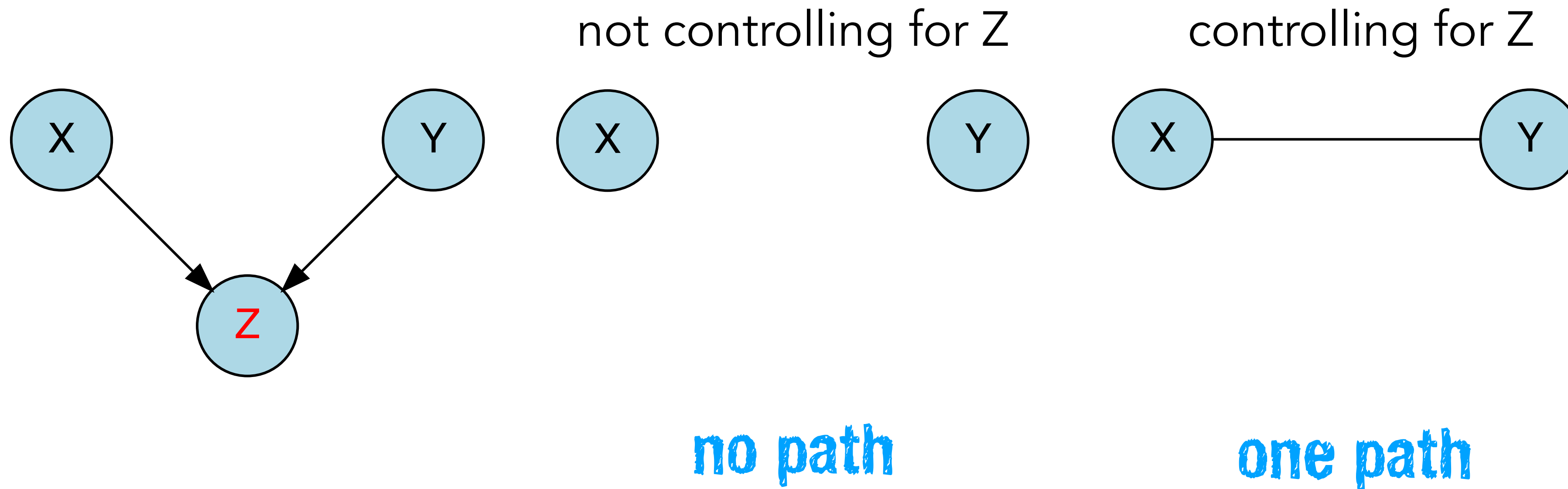
Residuals:
    Min       1Q   Median       3Q      Max
-1.35547 -0.30016  0.00298  0.31119  1.73408

Coefficients:
(Intercept) -0.009608  0.014477 -0.664    0.507
x           -0.816164  0.018936 -43.102   <2e-16 ***
z            0.398921  0.006186  64.489   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4578 on 997 degrees of freedom
Multiple R-squared:  0.8066,    Adjusted R-squared:  0.8062
F-statistic: 2079 on 2 and 997 DF,  p-value: < 2.2e-16
```


When should I control for variables?

I want to estimate the effect that X has on Y



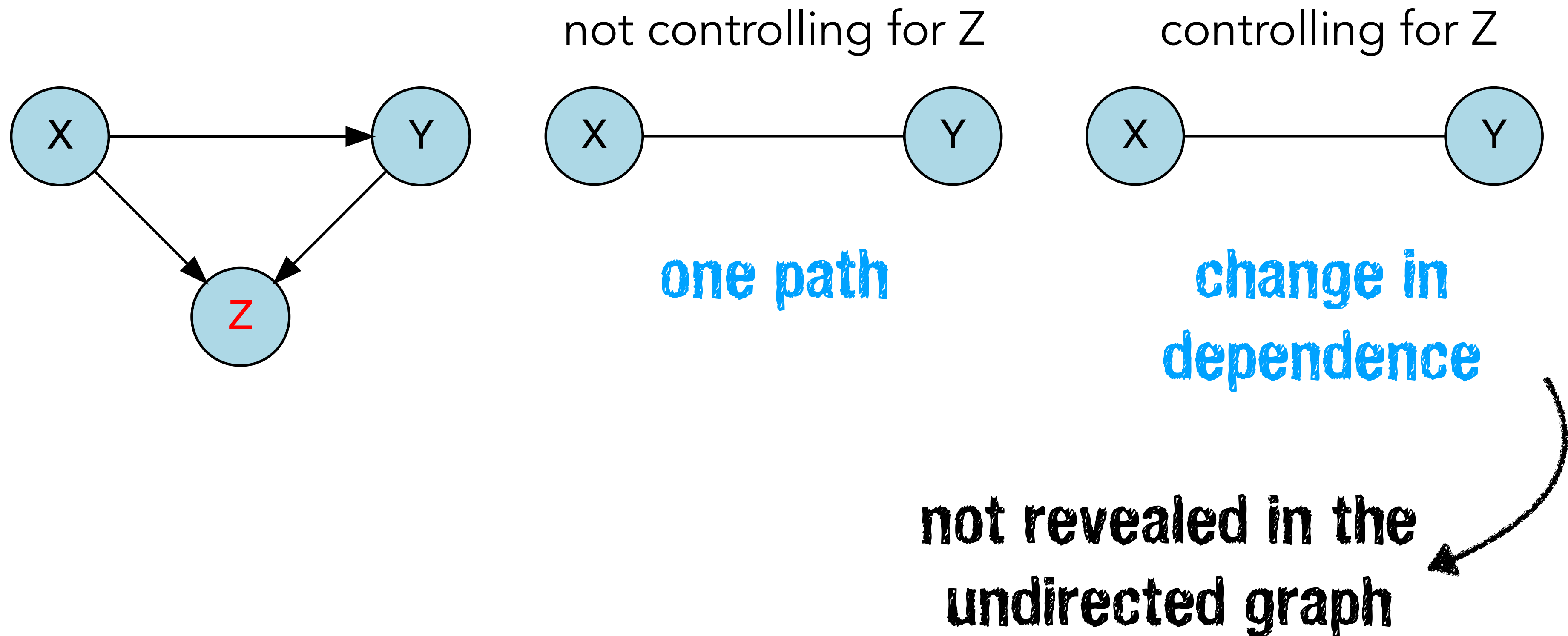
Z is a **bad** control here!

When should I control for variables?

- checking for **d-separation** tells us whether or not variables are (conditionally) independent
- it also tells us whether paths of dependence "open up", or get "closed down"
- the graphical procedure doesn't necessarily reveal whether the dependence between variables changes: it reveals the **structure** of dependence but not the **strength**
- you can always double check via running simulations in R

When should I control for variables?

I want to estimate the effect that X has on Y



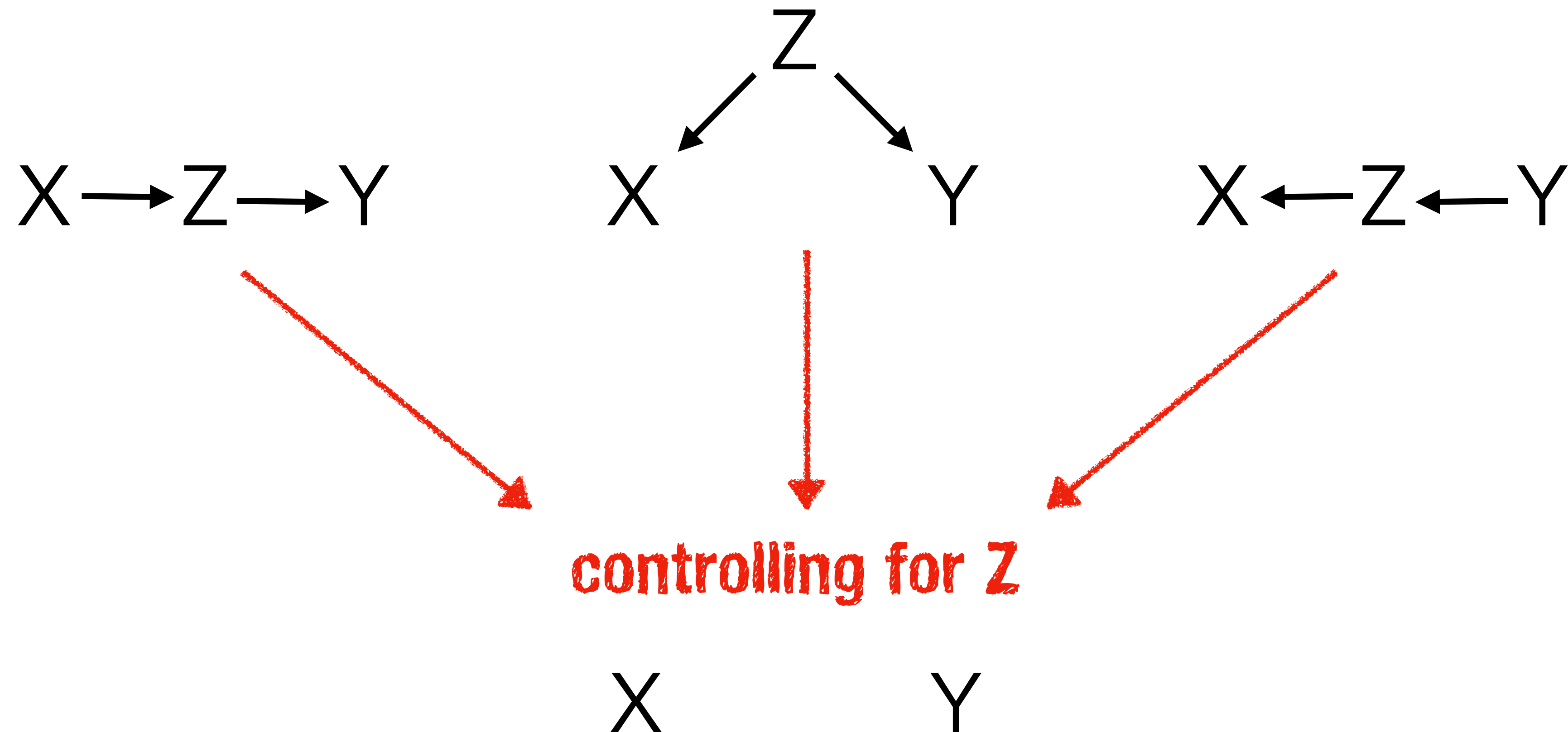
Z is a **bad** control here!

When should I control for variables?

- **good controls** reduce additional paths from X to Y apart from the direct path we are interested in estimating
- **bad controls** introduce additional paths (or change existing ones) that lead to a biased estimate of the direct path between X and Y

When should I control for variables?

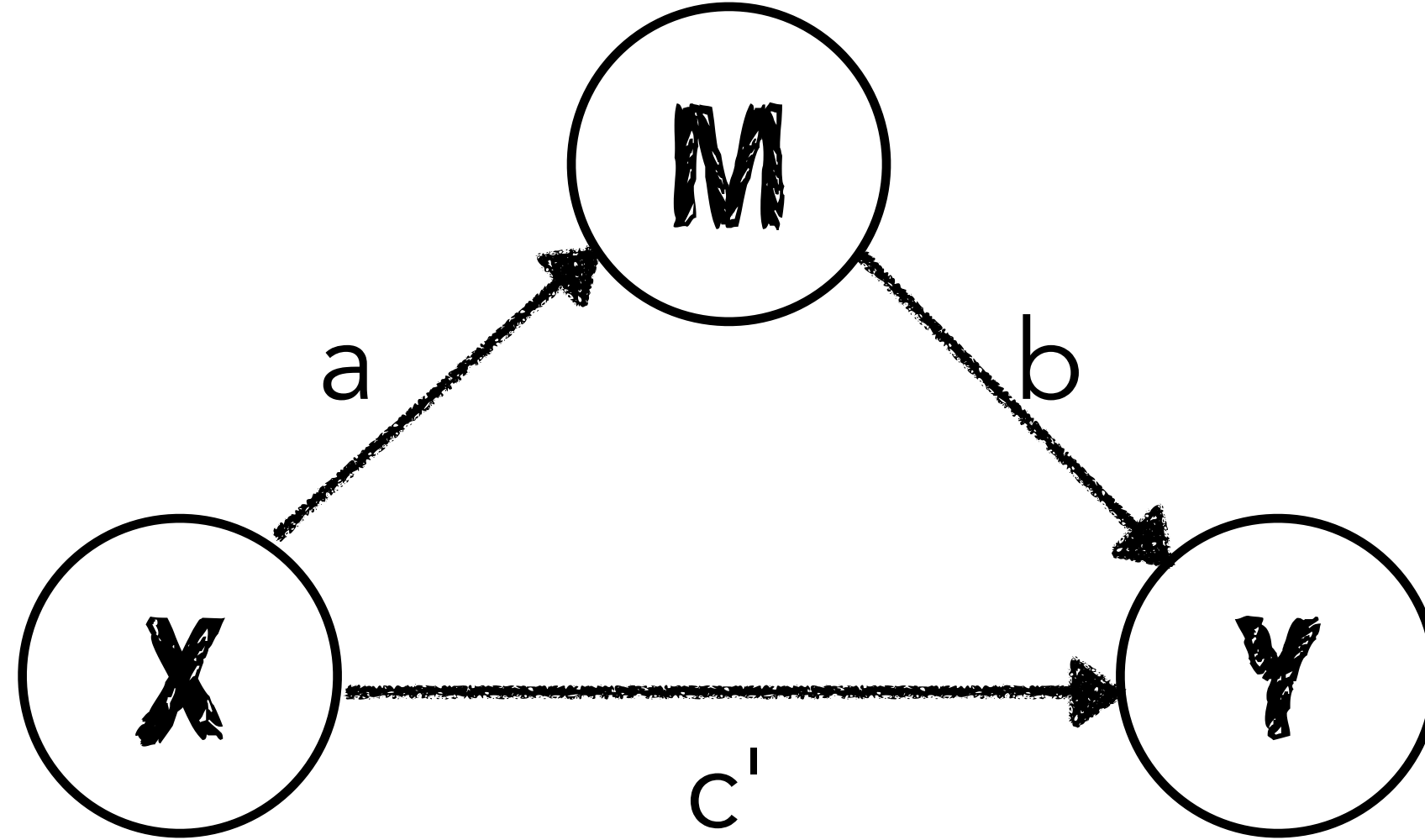
Problem: We don't know the ground truth ...



we need to manipulate X experimentally to tell these apart

Mediation

Definition

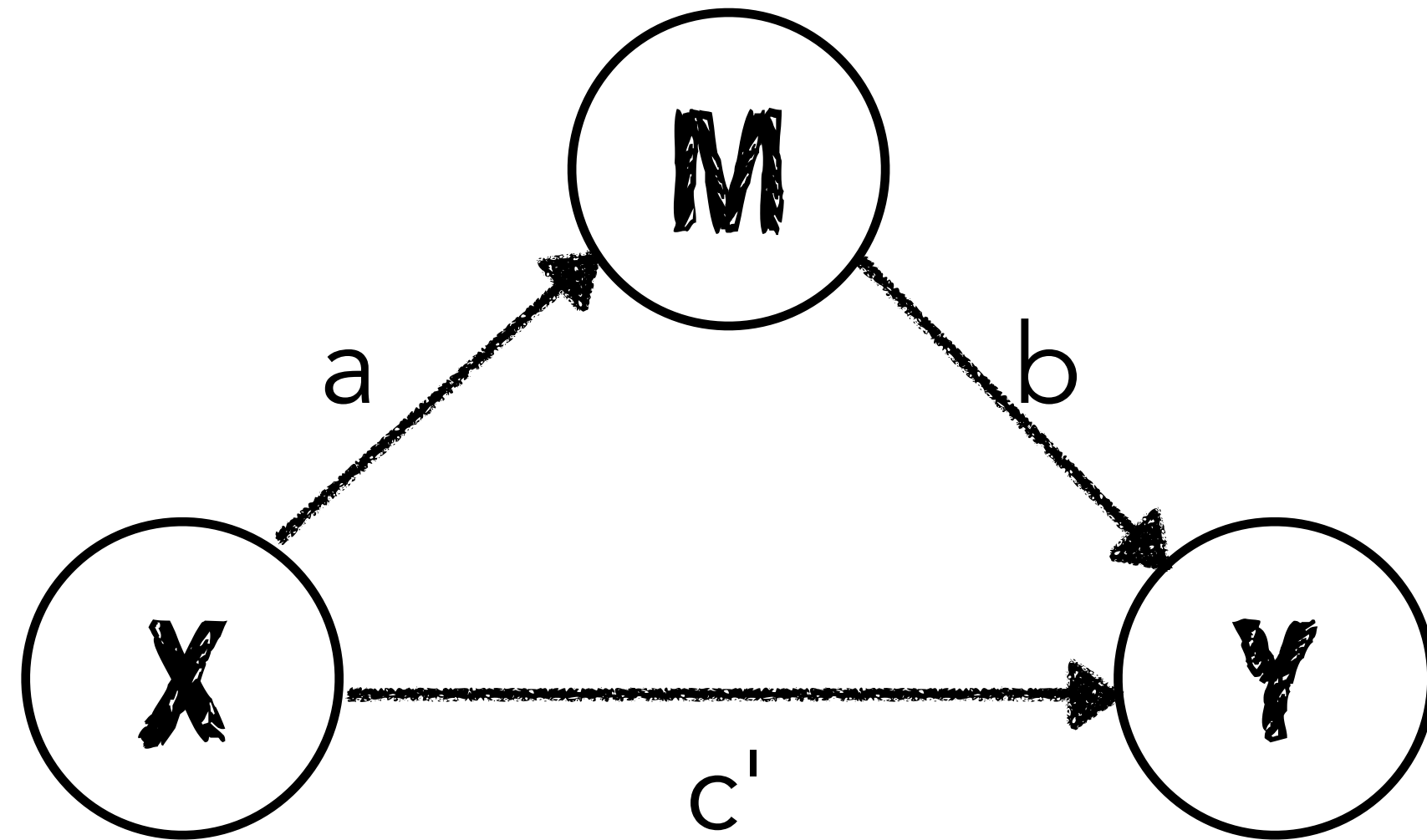


Rather than a direct causal relationship between **X** and **Y**, a mediation model proposes that **X** influences the mediator variable **M**, which in turn influences **Y**. Thus, the mediator variable serves to clarify the nature of the relationship between **X** and **Y**.

Adapted from Wikipedia

[https://en.wikipedia.org/wiki/Mediation_\(statistics\)](https://en.wikipedia.org/wiki/Mediation_(statistics))

Example



X = grades in Psych 252

M = feelings of self-esteem

Y = happiness

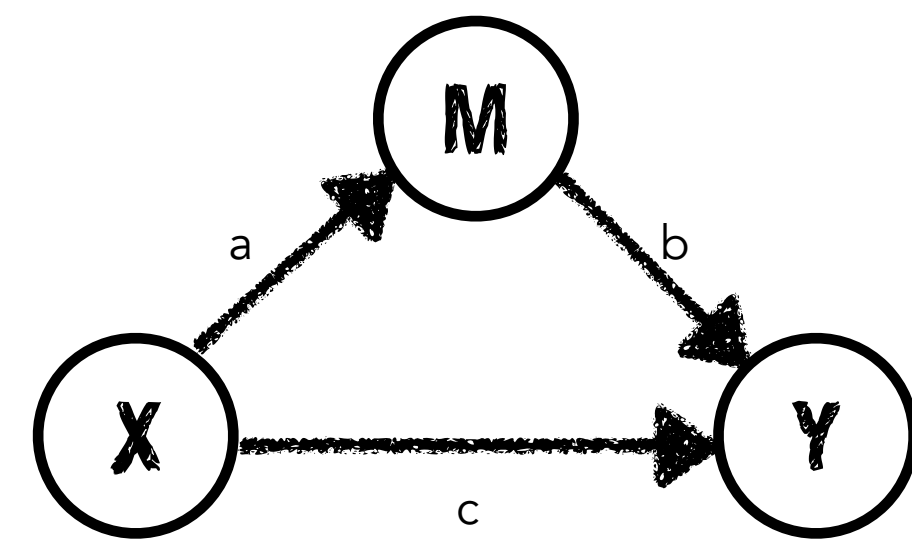
Is the relationship between grades in Psych 252 and happiness mediated by feelings of self-esteem?

Bootstrapping

```

1 library("mediation")
2
3 # bootstrapped mediation
4 fit.mediation = mediate(model.m = fit.m_x,
5                          model.y = fit.y_mx,
6                          treat = "x",
7                          mediator = "m",
8                          boot = T)
9
10 # summarize results
11 fit.mediation %>% summary()

```



$$\hat{m} = b_0 + b_1 \cdot x$$

$$\hat{y} = b_0 + b_1 \cdot m + b_2 \cdot x$$

Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	0.28078	0.14059	0.42	<2e-16 ***
ADE	-0.11179	-0.29276	0.10	0.272
Total Effect	0.16899	-0.00415	0.34	0.064 .
Prop. Mediated	1.66151	-3.22476	11.46	0.064 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 100

Simulations: 1000

2. Bootstrapping

Causal Mediation Analysis

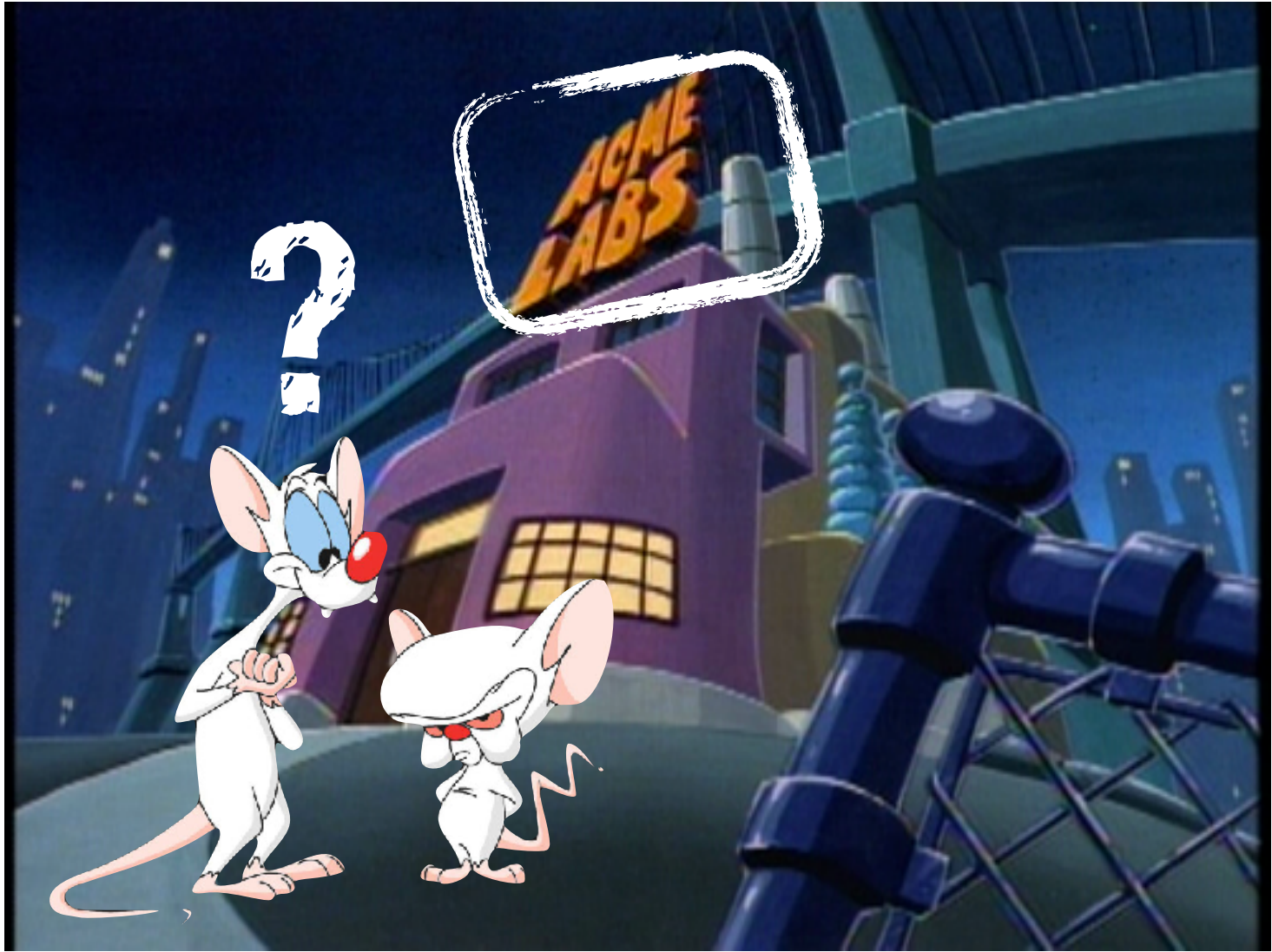
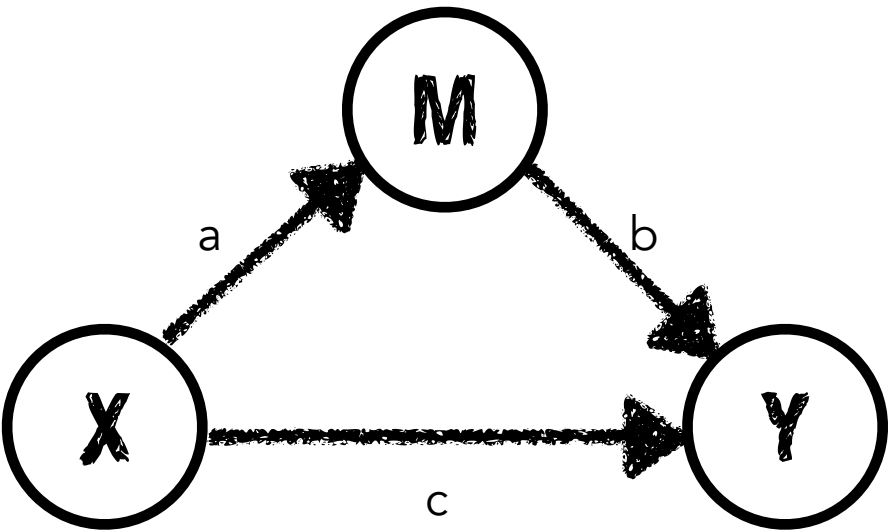
Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	0.28078	0.14059	0.42	<2e-16 ***
ADE	-0.11179	-0.29276	0.10	0.272
Total Effect	0.16899	-0.00415	0.34	0.064 .
Prop. Mediated	1.66151	-3.22476	11.46	0.064 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 100

Simulations: 1000



2. Bootstrapping

Causal Mediation Analysis

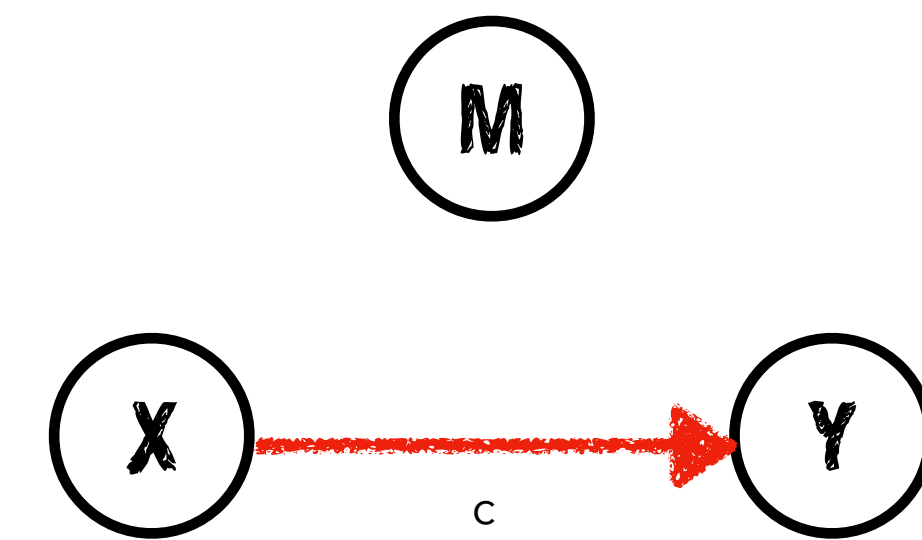
Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	0.28078	0.14059	0.42	<2e-16 ***
ADE	-0.11179	-0.29276	0.10	0.272
Total Effect	0.16899	-0.00415	0.34	0.064 .
Prop. Mediated	1.66151	-3.22476	11.46	0.064 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 100

Simulations: 1000



$$\hat{y} = b_0 + b_1 \cdot x$$

```
Call:
lm(formula = y ~ 1 + x, data = df.mediation)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.917	-3.738	-0.259	2.910	12.540

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.78300	6.16002	1.426	0.1571
x	0.16899	0.08116	2.082	0.0399 *

2. Bootstrapping

Causal Mediation Analysis

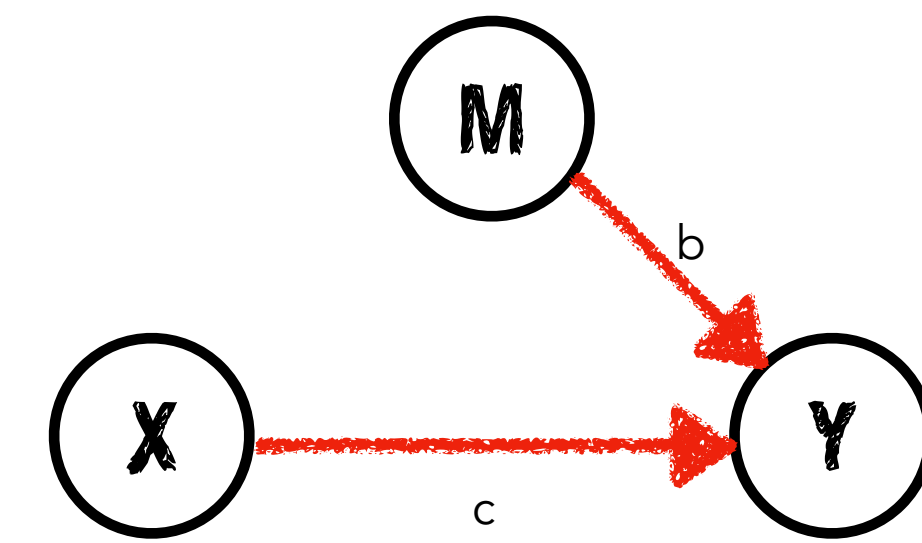
Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	0.28078	0.14059	0.42	<2e-16 ***
ADE	-0.11179	-0.29276	0.10	0.272
Total Effect	0.16899	-0.00415	0.34	0.064 .
Prop. Mediated	1.66151	-3.22476	11.46	0.064 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample Size Used: 100

Simulations: 1000



$$\hat{y} = b_0 + b_1 \cdot m + b_2 \cdot x \quad \textbf{ADE: Average direct effect}$$

Call:

```
lm(formula = y ~ 1 + m + x, data = df.mediation)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.3651	-3.3037	-0.6222	3.1068	10.3991

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.80952	5.68297	1.374	0.173
m	0.42381	0.09899	4.281	4.37e-05 ***
x	-0.11179	0.09949	-1.124	0.264

2. Bootstrapping

```
Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

      Estimate 95% CI Lower 95% CI Upper p-value
ACME      0.28078      0.14059      0.42  <2e-16 ***
ADE     -0.11179     -0.29276      0.10   0.272
Total Effect  0.16899     -0.00415      0.34   0.064 .
Prop. Mediated 1.66151     -3.22476     11.46   0.064 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

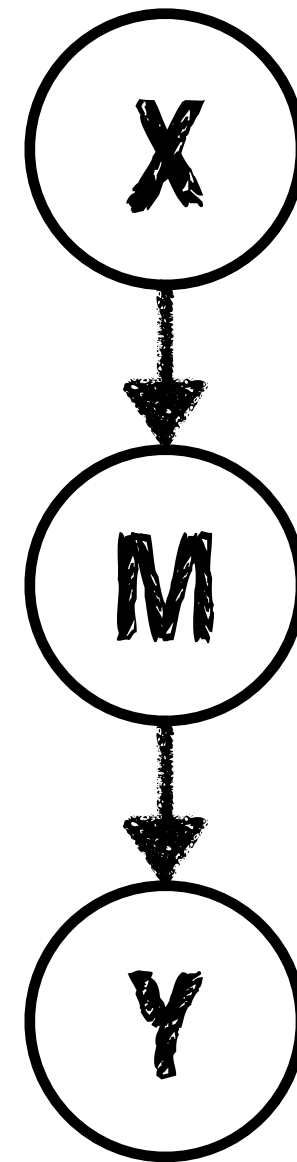
Sample Size Used: 100

Simulations: 1000
```

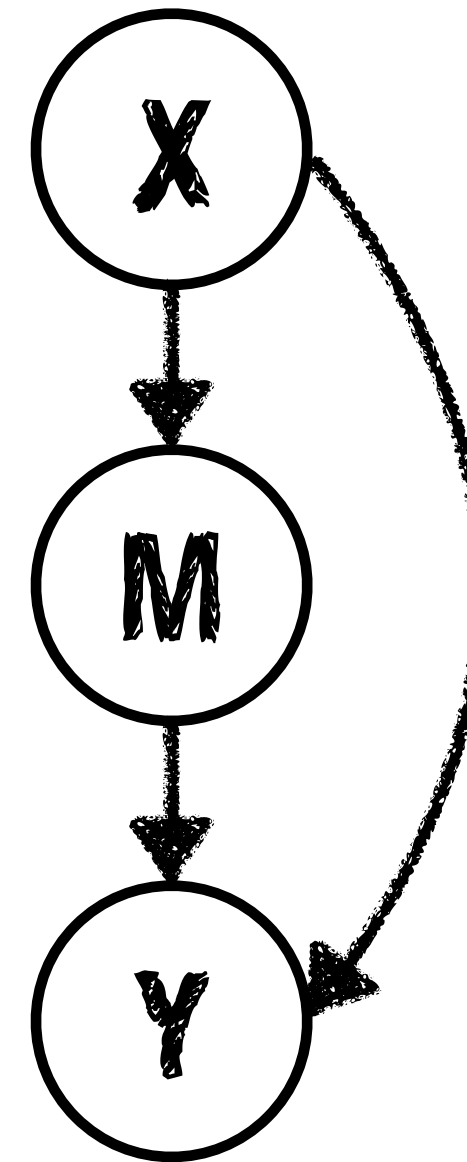
Average causal mediation effect (**ACME**) = Total effect - Average direct effect (**ADE**)

Underlying causal model

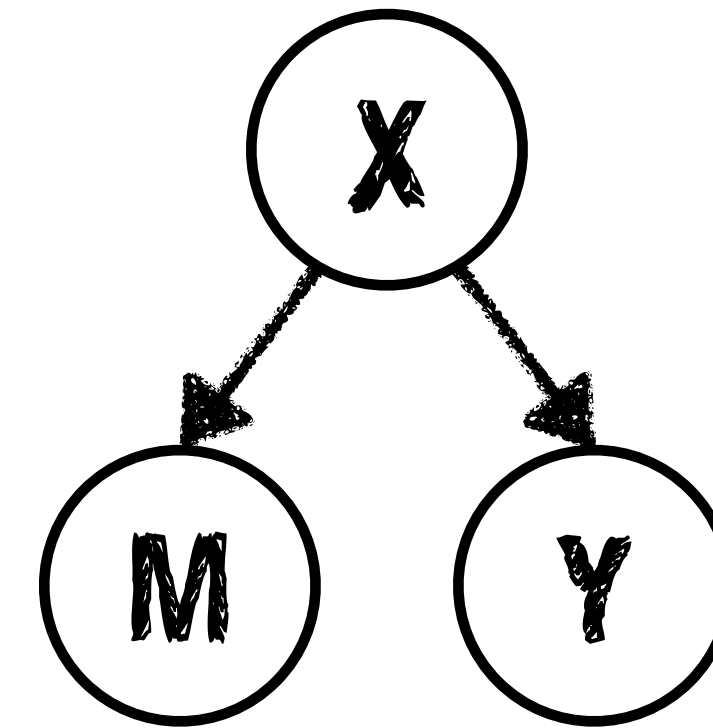
Full mediation



Partial mediation



No mediation

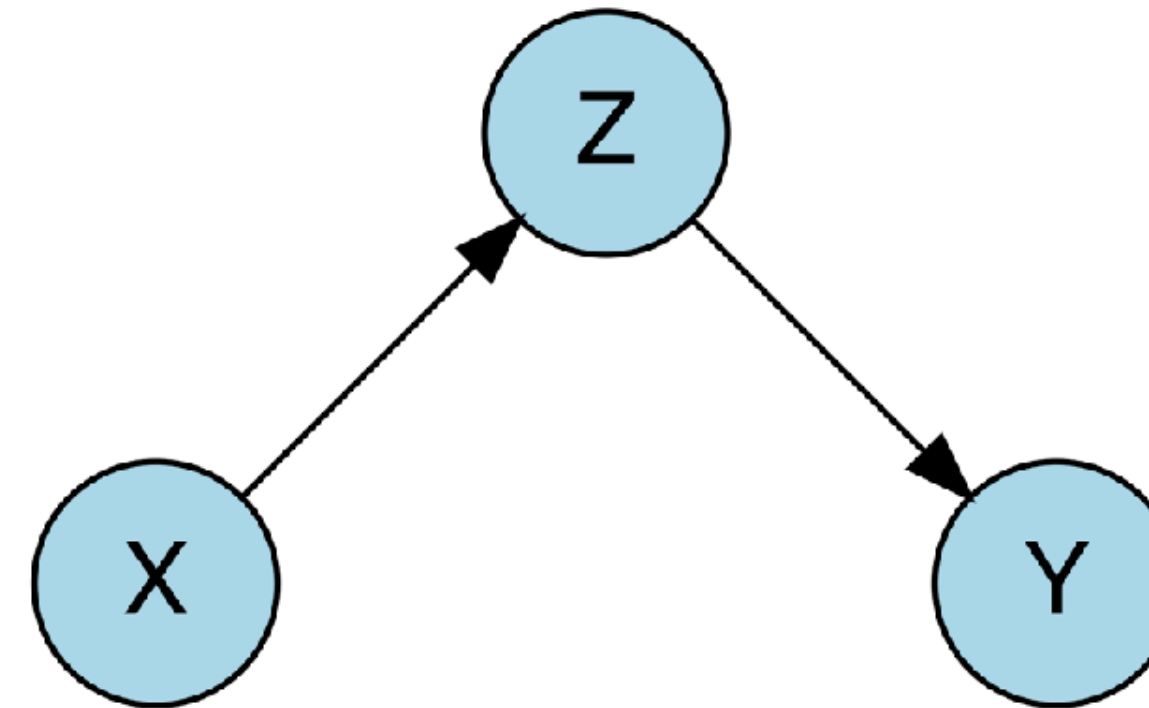


Full mediation: When the effect of **X** on **Y** completely disappears, **M** fully mediates between **X** and **Y**.

Partial mediation: When the effect of **X** on **Y** still exists, but in a smaller magnitude, **M** partially mediates between **X** and **Y**.

Beware of mediation analyses!!!

```
1 set.seed(1)
2
3 n = 100 # number of observations
4
5 # causal chain
6 df.causal_chain = tibble(x = rnorm(n, 0, 1),
7                           z = 2 * x + rnorm(n, 0, 1),
8                           y = 2 * z + rnorm(n, 0, 1))
```



Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	0.8287	0.6234	1.05	<2e-16 ***
ADE	-0.0535	-0.2548	0.15	0.55
Total Effect	0.7752	0.6391	0.90	<2e-16 ***
Prop. Mediated	1.0690	0.8131	1.35	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

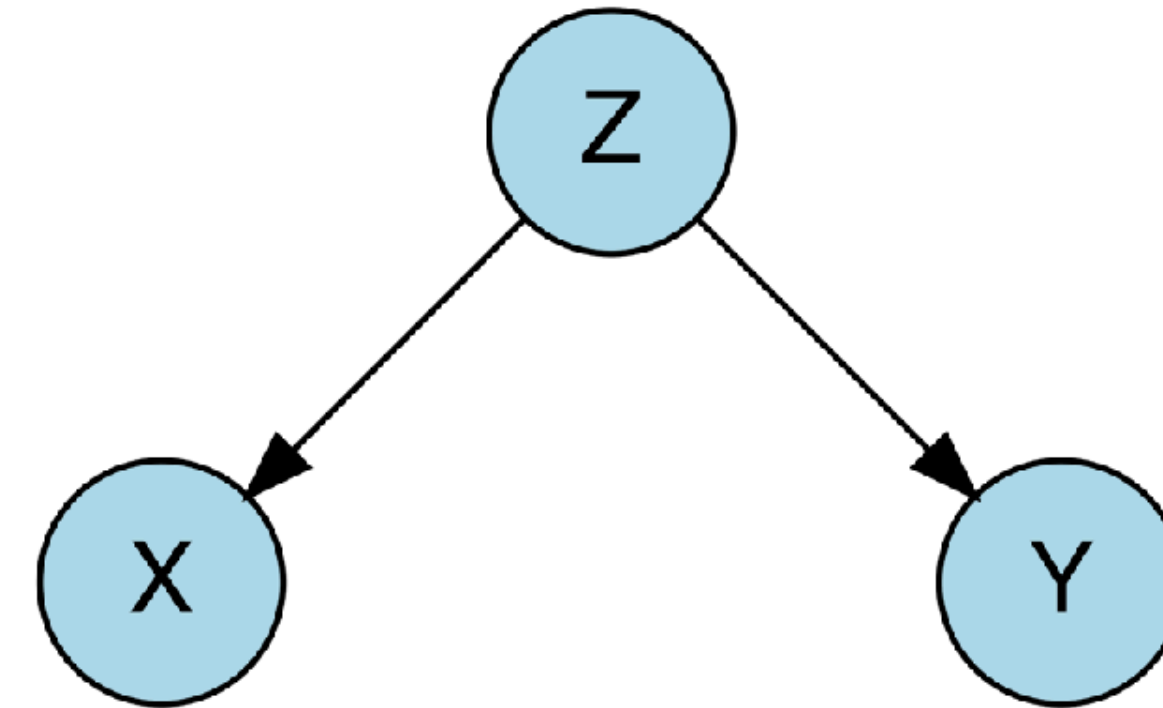
Sample Size Used: 100

Simulations: 1000

nice mediation result!

Beware of mediation analyses!!!

```
1 set.seed(1)
2
3 n = 100 # number of observations
4
5 # common cause
6 df.common_cause = tibble(z = rnorm(n, 0, 1),
7                             x = 2 * z + rnorm(n, 0, 1),
8                             y = 2 * z + rnorm(n, 0, 1))
```



Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the Percentile Method

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	0.8287	0.6065	1.04	<2e-16 ***
ADE	-0.0535	-0.2675	0.16	0.56
Total Effect	0.7752	0.6353	0.90	<2e-16 ***
Prop. Mediated	1.0690	0.8134	1.37	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

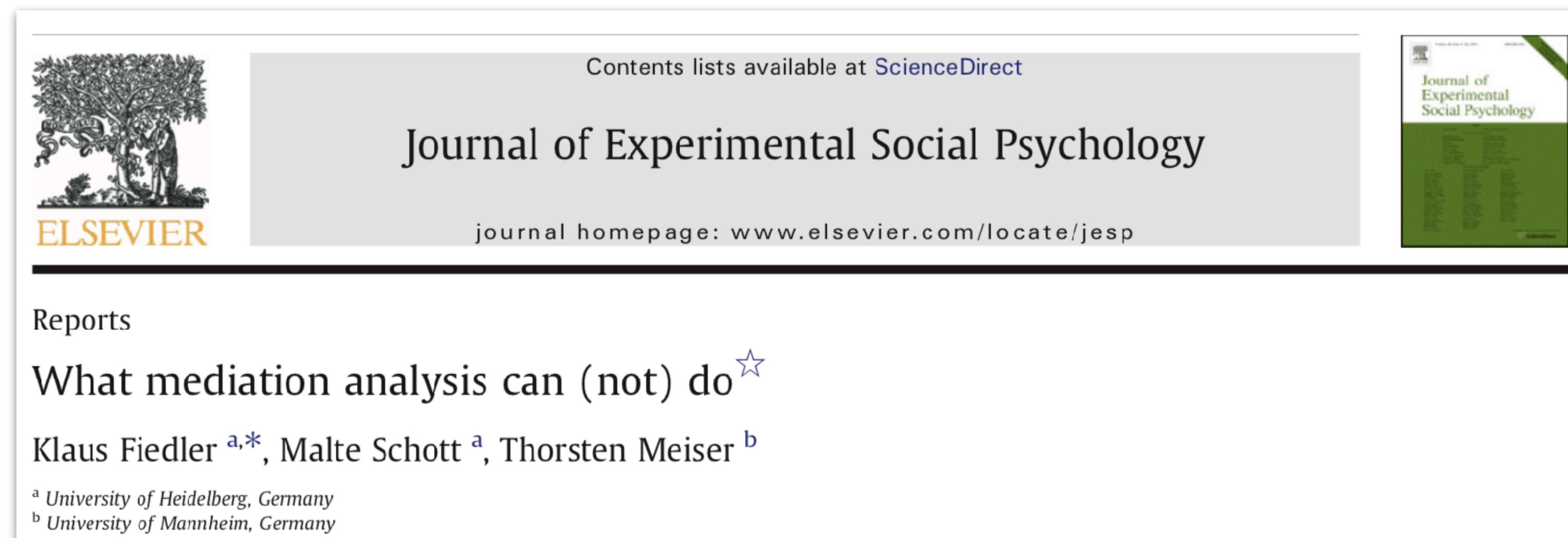
Sample Size Used: 100

Simulations: 1000

(not) nice mediation result!

Limitations

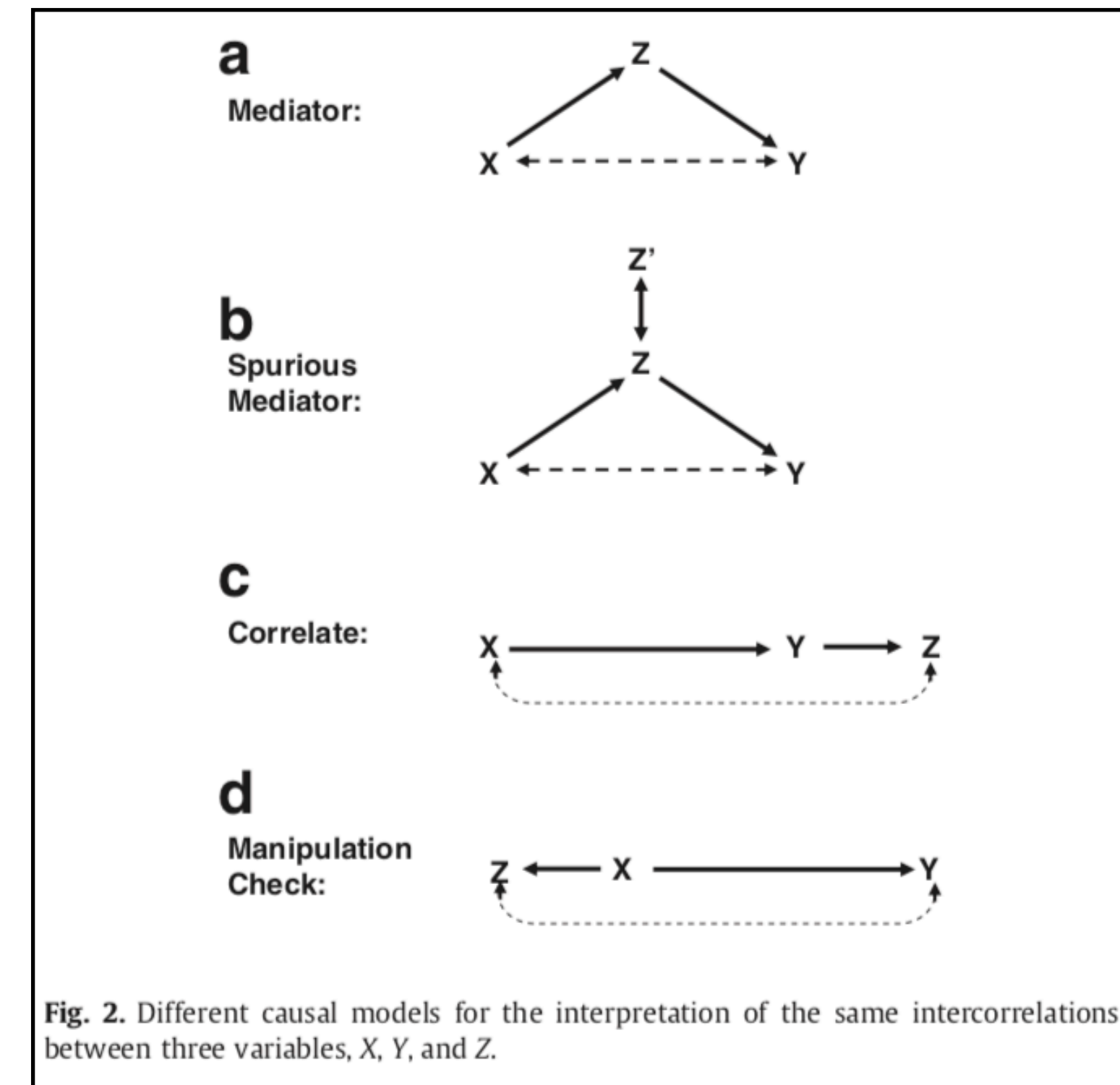
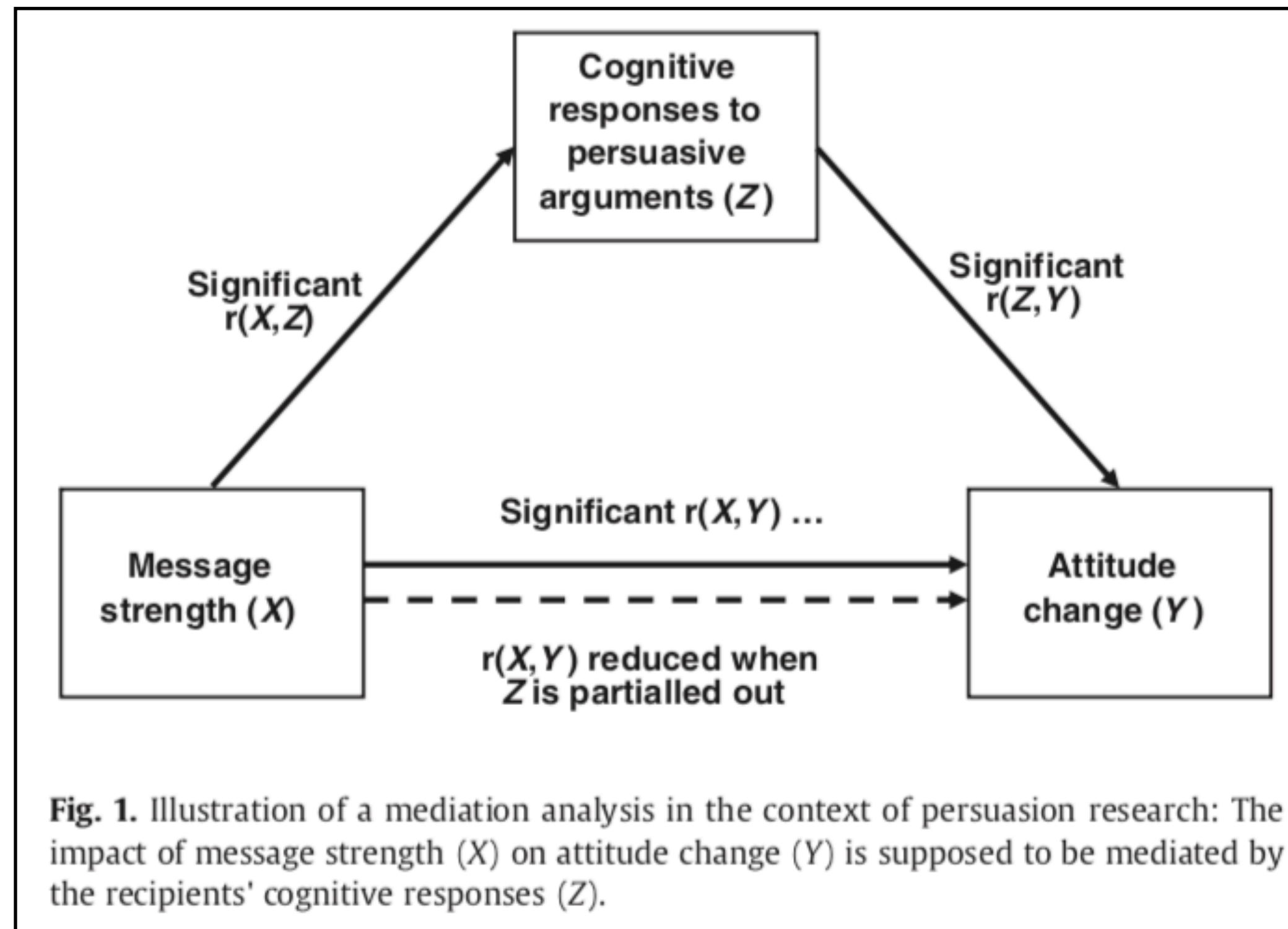
- correlational analysis
 - we need theories / experiments to tease apart causes and effects to properly map our variables onto the diagram



Fiedler, K., Schott, M., & Meiser, T. (2011). What mediation analysis can (not) do. *Journal of Experimental Social Psychology*, 47(6), 1231-1236.

Limitations

many-to-one mapping

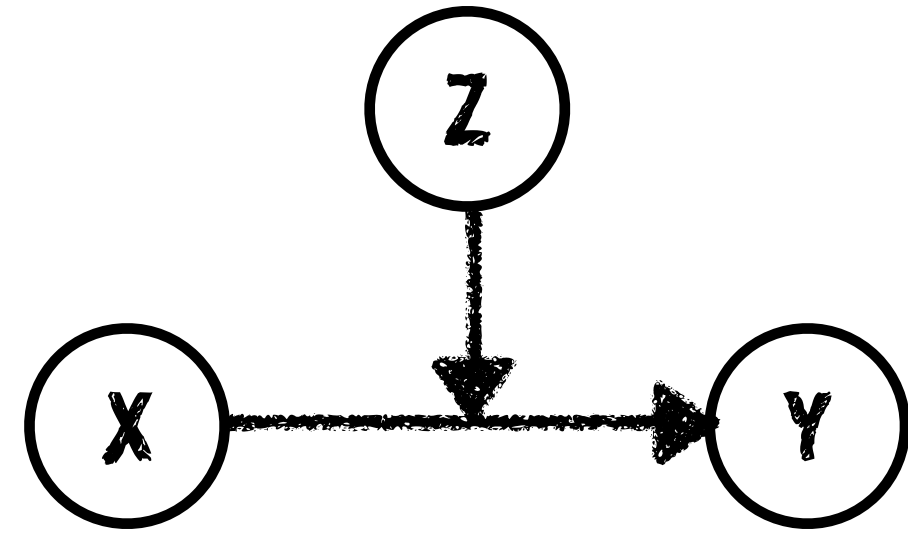


only experiments allow us to tell apart possible causal structures

Fiedler, K., Schott, M., & Meiser, T. (2011). What mediation analysis can (not) do. *Journal of Experimental Social Psychology*, 47(6), 1231-1236.

Moderation

Definition

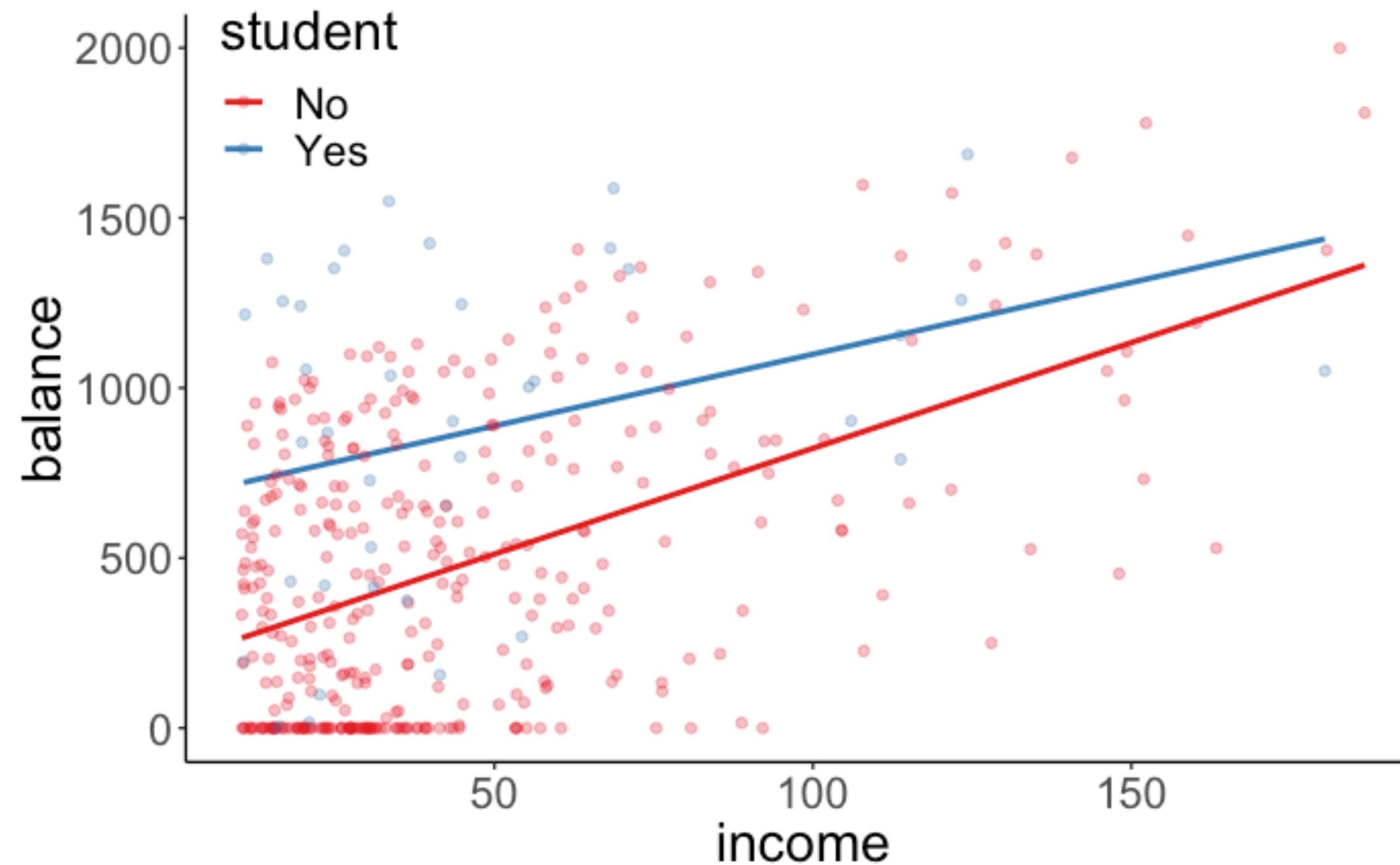


Moderation means that the effect of a predictor depends on the value of another.

Here, the nature of the relationship between **X** and **Y** depends on **Z**.

Have we come across moderation already?

Relationship
between credit card
balance, income,
and whether the
person is a student.



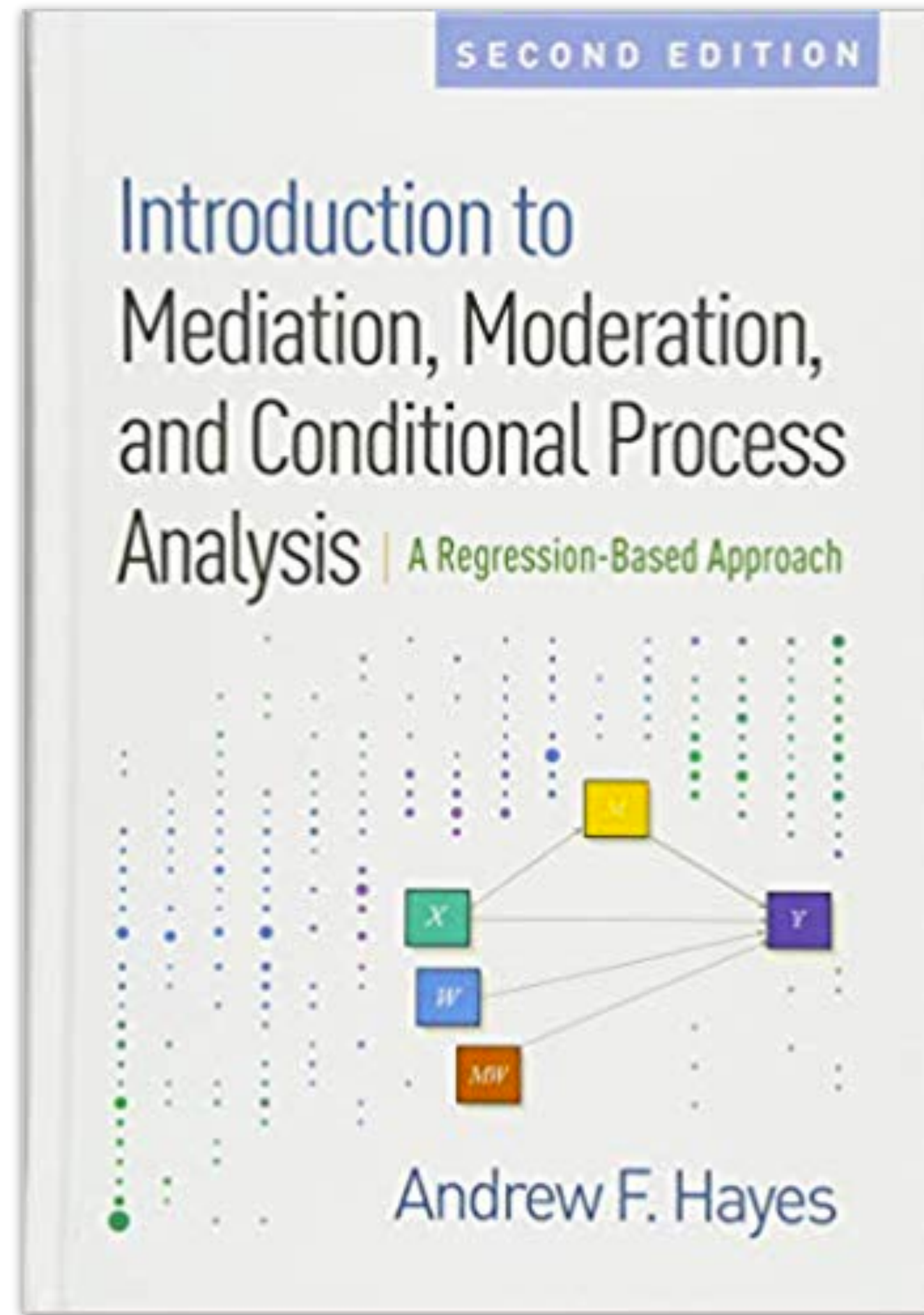
$$\widehat{\text{balance}}_i = 200.62 + 6.22 \cdot \text{income}_i + 476.68 \cdot \text{student}_i - 2.00 \cdot (\text{income}_i \times \text{student}_i)$$

if student = "No" $\widehat{\text{balance}}_i = 200.62 + 6.22 \cdot \text{income}_i$

if student = "Yes"

$$\begin{aligned}\widehat{\text{balance}}_i &= 200.62 + 6.22 \cdot \text{income}_i + 476.68 \cdot 1 - 2.00 \cdot (\text{income}_i \times 1) \\ &= 677.3 + 6.22 \cdot \text{income}_i - 2.00 \cdot \text{income}_i \\ &= 677.3 + 4.22 \cdot \text{income}_i\end{aligned}$$

Learn more about mediation and moderation



Recoded with `brms` by
Solomon Kurz here:
<https://solomon.quarto.pub/immcpa2/>

Plan for today

- Quick recap
- Causation
 - Causation vs. correlation
 - Controlling for variables
 - Mediation
 - Moderation

Feedback

How was the pace of today's class?

much too slow a little too slow just right a little too fast much too fast

How happy were you with today's class overall?



What did you like about today's class? What could be improved next time?

Thank you!